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Positron Emission Tomography (PET)

Clinical Policy Bulletins | Medical Clinical Policy Bulletins

Number: 0071

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Last Review
12/16/2025
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Next Review: 01/08/2026

Review History

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Scope of Policy

This Clinical Policy Bulletin addresses positron emission tomography (PET).

Note: Precertification may be required for select procedures and interventions. Use the CPT code search tool (https://www.aetna.com/health-care-professionals/precertification/precertification-lists.html) to see if precertification is required.

I. Medical Necessity

A. *Cardiac Indications*

Aetna considers positron emission tomography (PET) medically necessary for the following cardiac indications:

1. Evaluation of Coronary Artery Disease

   PET scans using rubidium-82 (Rb-82) or N-13 ammonia done at rest or with pharmacological stress are considered medically necessary for non-invasive imaging of the perfusion of the heart for the diagnosis and management of members with known or suspected coronary artery disease, provided such scans meet *either* one of the two following criteria:

   a. The PET scan is used in place of, but not in addition to, a single photon emission computed tomography (SPECT), in persons who meet medical necessity criteria for a cardiac SPECT (see CPB 0376 Single Photon Emission Computed Tomography (SPECT)) (../300_399/0376.html); *or*
   b. For use in assessment of coronary artery disease after cardiac transplant.

   Aetna considers absolute quantitation of myocardial blood flow (AQMBF) a medically necessary adjunct to cardiac PET when criteria for rest/stress perfusion for coronary artery disease are met.

2. Assessment of Myocardial Viability

   Fluorodeoxy-D-glucose (FDG)-PET scans are considered medically necessary for the determination of myocardial viability prior to re-vascularization, either as a primary or initial diagnostic study or following an inconclusive SPECT. The greater specificity of PET makes a SPECT following an inconclusive PET not medically necessary.

   The identification of members with partial loss of heart muscle movement or hibernating myocardium is important in selecting candidates with compromised ventricular function to determine appropriateness for re-vascularization. Diagnostic tests such as FDG-PET

distinguish between dysfunctional but viable myocardial tissue and scar tissue in order to affect the management decisions in members with ischemic cardiomyopathy and left ventricular dysfunction.

3. Cardiac Sarcoid

   FDG-PET is considered medically necessary to identify and monitor response to therapy for established or strongly suspected cardiac sarcoid.

B. *Oncologic indications*

Aetna considers FDG-PET medically necessary for the following oncologic indications, when medical necessity criteria are met in the following eviCore guidelines:

- eviCore Oncology Imaging Guidelines (https://www.evicore.com/provider/clinical-guidelines)
- eviCore Pediatric and Special Populations Oncology Imaging Guidelines (https://www.evicore.com/provider/clinical-guidelines)

Note: To access these clinical guidelines from landing page, select 'Cardiovascular & Radiology', enter 'EviCore by Evernorth' in search bar, accept terms and agreement, then scroll to find link to the guideline of interest.

Note: eviCore guidelines undergo a formal review annually; however, eviCore reserves the right to change and update the guidelines without prior notice. Draft guidelines are posted 90 days prior to implementation. Additional clinical guidelines may be developed as needed or may be withdrawn from use.

- Adrenal carcinoma
- Adrenocortical tumors
- Ampullary cancer
- Anal cancer
- Appendiceal cancer
- Brain tumors
- Breast cancer
- Burkitt's lymphoma

- Cancer of ampulla
- Castleman's disease
- Cervical cancer
- Chordoma
- Chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL) with suspected Richter's transformation
- Colorectal cancer
- Diffuse large B-cell lymphoma (DLBCL)
- Esophageal cancer
- Ewing sarcoma and osteosarcoma
- Fallopian tube cancer
- Follicular lymphoma
- Gallbladder cancer
- Gastric cancer
- Gastrointestinal stromal tumors
- Head and neck cancers (excluding cancers of the central nervous system)
- Hodgkin lymphoma
- Light-chain deposition disease
- Mantle cell lymphoma
- Marginal Zone and MALT lymphoma
- Melanoma
- Merkel cell carcinoma
- Mesothelioma
- Multiple myeloma and plasmacytomas
- Neuroblastoma
- Neuroendocrine tumors
- Non-Hodgkin's lymphoma
- Non-small cell lung carcinoma (NSCLC)
- Occult primary cancers
- Ovarian cancer
- Pancreatic cancer
- Paraneoplastic syndrome
- Penile cancer
- Post-transplant lymphoproliferative disorder
- Primary cutaneous B-cell lymphoma
- Primary peritoneal cancer
- Skin cancer
- Small cell lung carcinoma (SCLC)
- Small bowel adenocarcinoma

- Soft tissue sarcoma
- Solitary pulmonary nodules
- T-cell lymphoma (includes peripheral T-cell lymphoma, Mycosis Fungoides/Sezary Syndrome, primary cutaneous CD30+ T-cell lymphoproliferative disorders)
- Testicular cancer
- Thymic malignancies
- Thyroid cancer
- Urothelial carcinoma (bladder, renal pelvis, ureter)
- Uterine sarcoma
- Vaginal cancer
- Vulvar squamous cell cancer.

Upon individual case review, FDG-PET scanning may be considered medically necessary for other oncologic indications that are not listed as medically necessary above, when the conventional imaging that is indicated for that oncologic indication is equivocal, and general medical necessity criteria for oncologic indications (section B.1. below) are met.

PET-CT Fusion: The fusion of PET and CT imaging into a single system (PET/CT fusion) is considered medically necessary for any indication where PET scanning is considered medically necessary.

1. General Criteria

   a. Diagnosis:

      The PET results may assist in avoiding an invasive diagnostic procedure, or the PET results may assist in determining the optimal anatomic location to perform an invasive diagnostic procedure. In general, for most solid tumors, a tissue diagnosis is made prior to the performance of PET scanning. PET scans following a tissue diagnosis are performed for the purpose of staging, not diagnosis.

   b. Staging:

PET is considered medically necessary in situations in which clinical management of the member would differ depending on the stage of the cancer identified and *either:*

- The stage of the cancer remains in doubt after completion of a standard diagnostic work-up, including conventional imaging (computed tomography, magnetic resonance imaging, or ultrasound); *or*
- The use of PET would potentially replace one or more conventional imaging studies when it is expected that conventional study information is insufficient for the clinical management of the member.

c. Re-staging:

PET is considered medically necessary for re-staging after completion of treatment for the purpose of detecting residual disease, for detecting suspected recurrence in persons with signs or symptoms of recurrence, or to determine the extent of recurrence. Use of PET is also considered medically necessary if it could potentially replace one or more conventional imaging studies when it is expected that conventional study information is insufficient for the clinical management of the member.

d. Post-treatment surveillance

PET for post-treatment surveillance is considered experimental, investigational, or unproven where surveillance is defined as use of PET beyond the completion of treatment, in the absence of signs or symptoms of cancer recurrence or progression, for the purpose of detecting recurrence or progression or predicting outcome.

2. PET Radiopharmaceuticals / Radiotracers for Oncologic Indications

a. FDG-PET in Place of ^{99m}Tc Skeletal Scintigraphy:

There is no longer a temporary shortage of technetium 99-m (^{99m}Tc), which is used in nuclear medicine for skeletal scintigraphy (bone scans). Therefore, Aetna will no longer consider FDG-PET an acceptable alternative to bone scans for detecting skeletal abnormalities for medically necessary indications.

b. Cerianna (fluoroestradiol F-18) PET

Aetna considers Cerianna (fluoroestradiol F-18) PET imaging medically necessary for the detection of estrogen receptor (ER)-positive lesions as an adjunct to biopsy in recurrent or metastatic breast cancer when medical necessity criteria in eviCore oncology imaging guidelines are met.

c. Fluciclovine f-18 PET and Choline C-11 PET:

Aetna considers fluciclovine f-18 PET or choline c-11 PET medically necessary for restaging of men with a suspected recurrence of prostate cancer when medical necessity criteria in eviCore oncology imaging guidelines are met.

d. Gallium-68 PSMA PET, Piflufolastat F-18 (Pylarify) PET, and 18F-Flotufolastat (Posluma) PET:

Aetna considers Ga-68 PSMA-11 and piflufolastat F-18 (Pylarify) medically necessary for newly diagnosed and suspected recurrence of prostate cancer when medical necessity criteria in eviCore oncology imaging guidelines are met.

Aetna considers 18F-flotufolastat (Posluma) injection medically necessary for PET imaging of PSMA-positive lesions in individuals with prostate cancer when medical necessity criteria in eviCore oncology imaging guidelines are met.

e. Gallium Ga 68 dotatate PET, Gallium Ga 68 DOTA-TOC PET, and Copper Cu 64 dotatate PET:

Aetna considers gallium Ga 68 dotatate PET (Netspot), copper Cu 64 dotatate PET (DetectNet) and gallium Ga 68 -DOTA-TOC medically necessary for low grade neuroendocrine tumors when medical necessity criteria in eviCore oncology imaging guidelines are met.

Aetna considers Ga 68 dotatate PET or gallium Ga 68 -DOTA-TOC medically necessary for meningioma, for suspected recurrence with inconclusive findings on MRI when medical necessity criteria in eviCore oncology imaging guidelines are met.

Aetna considers Ga 68 dotatate PET or gallium Ga 68 -DOTA-TOC medically necessary for restaging medullary thyroid cancer, for inconclusive findings on conventional imaging when medical necessity criteria in eviCore oncology imaging guidelines are met.

Aetna considers ^{68}Ga-DOTATATE, ^{64}Cu-DOTATATE, and ^{68}Ga-DOTA-TOC experimental, investigational, or unproven for all other indications.

f. Gallium Ga 68 gozetotide (Gozellix, Illuccix, Locametz) PET:

Aetna considers the kit for the preparation of gallium Ga 68 gozetotide (Gozellix, Illuccix, Locametz) PET of prostate-specific membrane antigen (PSMA)-positive lesions medically necessary in men with prostate cancer when medical necessity criteria in eviCore oncology imaging guidelines are met.

C. *Neurologic Indications*

Aetna considers FDG-PET medically necessary only for pre-surgical evaluation for the purpose of localization of a focus of refractory seizure activity.

Aetna considers an amyloid PET scan (including florbetapir F18 [Amyvid], florbetaben F18 [Neuraceq] flortaucipir F 18 injection [Tauvid], flutemetamol F18 [Vizamyl]) medically necessary for persons with mild cognitive impairment due to

Alzheimer disease who are being considered for aducanumab (Aduhelm), lecanemab-irmb (Leqembi), or donanemab-azbt (Kisunla) therapy; see CPB 0996 - Aducanumab-avwa (Aduhelm) (../900_999/0996.html), CPB 1026 - Lecanemab-irmb (Leqembi) (../1000_1099/1026.html), or CPB 1066 - Donanemab-azbt (Kisunla) (../1000_1099/1066.html).

Aetna considers PET scans experimental, investigational, or unproven for other neurologic indications not listed as medically necessary in this policy because of insufficient evidence of its effectiveness.

D. *Other Indications*

Aetna considers FDG-PET medically necessary for the following non-oncologic indications:

1. Diagnosis, staging and re-staging of Langerhans cell histiocytosis;
2. Evaluation of large-vessel vasculitis (including giant cell arteritis and Takayasu arteritis) when diagnosis is uncertain following negative temporal artery biopsy results;
3. Staging and restaging of Erdheim-Chester disease;
4. Staging and restaging of hemangiopericytoma;
5. Staging, restaging, and surveillance of Rosai-Dorfman disease;
6. Workup of monoclonal gammopathy of undefined significance (MGUS) if computed tomography/magnetic resonance imaging (CT/MRI) are negative.

E. **Experimental, Investigational, or Unproven**

The following procedures are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

1. Amyloid-PET for the diagnosis of sporadic cerebral amyloid angiopathy
2. Carbon-11 labeled 5-HTP PET for carcinoid and all other indications

3. FDG-PET scans for the following non-oncological indications:

    - Adrenoleukodystrophy
    - Chronic osteomyelitis
    - Coccidioidomycosis (also known as valley fever, San Joaquin Valley fever, California valley fever, and desert fever)
    - Cortico-basal syndrome
    - Eosinophilia
    - Fever of unknown origin
    - Hepatic encephalopathy
    - Infection of knee replacement prostheses
    - Infection of hip arthroplasty
    - Joint infection
    - Joint inflammation associated with osteoarthritis
    - Lymphangiomatosis
    - Mesenteric lymphadenopathy
    - Moyamoya disease
    - Neurodegenerative progression in Parkinson disease
    - Opsoclonus (opsillopsia) myoclonus syndrome
    - Pigmented villonodular synovitis
    - Pleural effusion
    - Rheumatoid arthritis
    - Sarcoid/sarcoidosis (other than cardiac sarcoidosis as noted above)
    - Screening in Li-Fraumeni syndrome
    - Splenomegaly
    - Takayasu's arteritis
    - Xanthogranuloma
    - Other indications not listed as medically necessary in this policy

4. FDG-PET scans for the following oncological indications:

    - Amyloidosis in bone marrow transplant recipients, follow-up
    - Atypical teratoid/rhabdoid tumor
    - Chondrosarcoma
    - Clear cell carcinoma of the uterus
    - Desmoid tumors/fibromatosis

- Endometrial cancer
- Intraductal papillary mucinous neoplasm of the pancreas
- Gestational trophoblastic neoplasia
- Giant cell tumor of the bone
- Hemangioendothelioma
- Hepatic sarcoma
- Hepatocellular carcinoma
- Hypercalcemia of malignancy
- Initial staging of primary clear eccrine hidradenocarcinoma
- Kidney cancer
- Leukemia
- Malignant degeneration of neurofibromas
- Neurofibromatosis
- Neuroganglioma of the psoas muscle
- Osteoblastoma
- Paget's disease (including extra-mammary Paget's disease)
- Peri-ampullary cancer
- Pilar tumor
- Pituitary adenoma
- Placental cancer
- Plasmacytoid dendritic cell neoplasm
- Pleomorphic adenoma
- Schwannoma
- Salivary gland tumors
- Serous papillary endometrial carcinoma
- Solitary fibrous tumor of the pelvis (initial workup and post-treatment surveillance)
- Solitary fibrous tumor of the pleura, staging
- Spindle cell sarcoma
- Surgical planning for perivascular epithelioid cell tumors (PEComas)
- Sweat gland tumor, treatment planning
- Uterine papillary mesothelioma
- Waldenstrom macroglobulinemia, restaging
- Wilms tumor
- Other oncologic indications not listed as medically necessary in this policy

5. 11C-choline PET/CT for hyperparathyroidism

6. 11C-metomidate PET for prediction of treatment response in primary aldosteronism
7. 18F-FDOPA-PET/CT scan for prognosis of brain cancer
8. Florbetapi-PET imaging for the diagnosis of cerebral amyloid angiopathy-related hemorrhages
9. Fluorodopa F-18 PET for evaluation of adults with suspected Parkinsonian syndromes/Parkinson disease
10. Gallium-68 dotatate PET/CT for monitoring SDHB mutation (pathogenic) consistent with hereditary paraganglioma/pheochromocytoma syndrome
11. NaF-18 PET for identifying bone metastasis of cancer and all other indications
12. PET imaging for disseminated histoplasmosis, or hepatoblastoma
13. PET imaging as part of selective internal radiation therapy (SIRT) for hepatocellular carcinoma, or for evaluation of abnormal CT findings (e.g., possible carcinomatosis) status post resection of dermatofibrosarcoma protuberans
14. PET-probe guided surgical resection for recurrent ovarian cancer and other indications
15. PET scans for Huntington disease
16. PET scans for screening of asymptomatic members for breast cancer, lung cancer and other indications
17. PET/CT for monitoring recurrence of littoral cell splenic angioma
18. PET/MRI for all indications. (Exceptions to cover PET/MRI may be made where PET/CT is unavailable and medical necessity criteria for FDG-PET are met)
19. PET scanning with a gamma camera for all indications
20. Positron emission mammography (PEM)
21. SCINTIX biology-guided radiotherapy for the treatment of liver metastases
22. Whole body PET for evaluation of unspecified periodic fever syndrome.

F. **Related Policies**

- CPB 0376 - Single Photon Emission Computed Tomography (SPECT) (../300_399/0376.html)
- CPB 0996 - Aducanumab-avwa (Aduhelm) (../900_999/0996.html)

- CPB 1026 - Lecanemab-irmb (Leqembi) (../1000_1099/1026.html)
- CPB 1066 - Donanemab-azbt (Kisunla) (../1000_1099/1066.html)

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Cardiac indications:	
CPT codes covered if selection criteria are met:	
78429	Myocardial imaging, positron emission tomography (PET), metabolic evaluation study (including ventricular wall motion[s] and/or ejection fraction[s], when performed), single study; with concurrently acquired computed tomography transmission scan
78430	Myocardial imaging, positron emission tomography (PET), perfusion study (including ventricular wall motion[s] and/or ejection fraction[s], when performed); single study, at rest or stress (exercise or pharmacologic), with concurrently acquired computed tomography transmission scan
78431	multiple studies at rest and stress (exercise or pharmacologic), with concurrently acquired computed tomography transmission scan
78432	Myocardial imaging, positron emission tomography (PET), combined perfusion with metabolic evaluation study (including ventricular wall motion[s] and/or ejection fraction[s], when performed), dual radiotracer (eg, myocardial viability);
78433	with concurrently acquired computed tomography transmission scan
+78434	Absolute quantitation of myocardial blood flow (AQMBF), positron emission tomography (PET), rest and pharmacologic stress (List separately in addition to code for primary procedure)
78459	Myocardial imaging, positron emission tomography (PET), metabolic evaluation

Code	Code Description
78491	Myocardial imaging, positron emission tomography (PET), perfusion; single study at rest or stress
78492	multiple studies at rest and/or stress
HCPCS codes covered if selection criteria are met:	
A9526	Nitrogen N-13 ammonia, diagnostic, per study dose, up to 40 millicuries
A9552	Fluorodeoxyglucose F-18 FDG, diagnostic, per study dose, up to 45 millicuries
A9555	Rubidium Rb-82, diagnostic, per study dose, up to 60 millicuries
ICD-10 codes covered if selection criteria are met (not all-inclusive):	
D86.85	Sarcoid myocarditis
I20.0 - I20.9	Angina pectoris
I21.01 - I22.2	ST elevation (STEMI) myocardial infarction involoving left main or left anterior descending coronary artery
I21.A1	Myocardial infarction type 2
I21.A9	Other myocardial infarction type
I24.0 - I24.9	Other acute and subacute forms of ischemic heart disease
I25.10	Atherosclerotic heart disease of native coronary artery
I25.110 – I25.119	Atherosclerotic heart disease of native coronary artery with angina pectoris
I25.2	Old myocardial infarction
I25.700 – I25.799	Atherosclerosis of coronary artery bypass graft(s) and coronary artery of transplanted heart with angina pectoris
I25.83	Coronary atherosclerosis due to lipid rich plaque
I25.84	Coronary atherosclerosis due to calcified coronary lesion
I25.810	Atherosclerosis of coronary artery bypass graft(s) without angina pectoris
I25.811	Atherosclerosis of native coronary artery of transplanted heart without angina pectoris
I25.812	Atherosclerosis of bypass graft of coronary artery of transplanted heart without angina pectoris
I44.4 - I45.5	Atrioventricular, bundle branch, and other heart block
I48.0, I48.20 - I48.21, I48.91	Atrial fibrillation

Code	Code Description
I50.1 - I50.9	Heart failure
Oncologic indications and conditions other than cardiac and neurologic for PET and PET-CT Fusion:	
CPT codes covered if selection criteria are met:	
78811	Positron emission tomography (PET) imaging; limited area (e.g., chest, head/neck)
78812	skull base to mid-thigh
78813	whole body
78814	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; limited area (e.g., chest, head/neck)
78815	skull base to mid-thigh
78816	whole body
Other CPT codes related to the CPB:	
32097	Thoracotomy, with diagnostic biopsy(ies) of lung nodule(s) or mass(es) (eg, wedge, incisional), unilateral
32100	Thoracotomy; with exploration
32408	Core needle biopsy, lung or mediastinum, percutaneous, including imaging guidance, when performed
38500 - 38530	Biopsy or excision of lymph node(s); open, superficial, by needle, superficial (e.g., cervical, inguinal, axillary), open, deep cervical node(s), with or without excision scalene fat pad, open deep axillary node(s) or open, internal mammary node(s)
61534	Craniotomy with elevation of bone flap; for excision of epileptogenic focus without electrocorticography during surgery
61536	for excision of cerebral epileptogenic focus, with electrocorticography during surgery (includes removal of electrode array)
82378	Carcinoembryonic antigen (CEA)
HCPCS codes covered if selection criteria are met:	
A9515	Choline c-11, diagnostic, per study dose up to 20 millicuries
A9587	Gallium ga-68, dotatate, diagnostic, 0.1 millicurie
A9588	Fluciclovine f-18, diagnostic, 1 millicurie

Code	Code Description
A9593	Gallium ga-68 psma-11, diagnostic, (ucsf), 1 millicurie
A9594	Gallium ga-68 psma-11, diagnostic, (ucla), 1 millicurie
A9596	Gallium ga-68 gozetotide, diagnostic, (illuccix), 1 millicurie [68Ga-PSMA-11]
A9608	Flotufolastat f 18, diagnostic, 1 millicurie
A9616	Gallium ga-68 gozetotide (gozellix), diagnostic, 1 millicurie
A9800	Gallium ga-68 gozetotide, diagnostic, (locametz), 1 millicurie
C9067	Gallium Ga-68, Dotatoc, diagnostic, 0.01 mci
G0235	PET imaging, any site, not otherwise specified
HCPCS codes not covered for indications listed in the CPB:	
A9586	Florbetapir f18, diagnostic, per study dose, up to 10 millicuries
A9591	Fluoroestradiol F 18, diagnostic, 1 millicurie
A9597	Positron emission tomography radiopharmaceutical, diagnostic, for tumor identification, not otherwise classified
A9598	Positron emission tomography radiopharmaceutical, diagnostic, for non-tumor identification, not otherwise classified
A9602	Fluorodopa f-18, diagnostic, per millicurie
G0219	PET imaging whole body; melanoma for non-covered indications
G0252	PET imaging, full and partial-ring PET scanners only, for initial diagnosis of breast cancer and/ or surgical planning for breast cancer (e.g., initial staging of axillary lymph nodes)
S8085	Fluorine-18 fluorodeoxyglucose (F-18 FDG) imaging using dual-head coincidence detection system (non-dedicated PET scan) [when described as an FDG-SPECT scan]
Other HCPCS codes related to the CPB:	
A4641	Radiopharmaceutical, diagnostic, not otherwise classified
A9513	Lutetium lu 177, dotatate, therapeutic, 1 millicurie
A9580	Sodium fluoride F-18, diagnostic, per study dose, up to 30 millicuries [not covered when with NaF-18 PET for identifying bone metastasis of cancer]
A9607	Lutetium lu 177 vipivotide tetraxetan, therapeutic, 1 millicurie
J0175	Injection, donanemab-azbt, 2 mg

Code	Code Description
Q9951	Low osmolar contrast material, 400 or greater mg/ml iodine concentration, per ml
Q9958 - Q9967	High osmolar contrast material
ICD-10 codes covered if selection criteria are met:	
C00.0 - C21.8, C24.1, C25.0 - C25.9, C30.0 - C34.92, C38.0 - C43.9, C48.0 - C52, C53.0 - C53.9, C56.1 - C57.9, C60.0 - C60.9, C61, C62.00 - C62.92, C71.0 - C71.9, C73, C76.0	Malignant neoplasm of lip, oral cavity, and pharynx, esophagus, stomach, penis, prostate, testis, small intestine, colon, rectum, rectosigmoid junction, and anus [PET not covered for re-staging and surveillance], ampulla of vater, pancreas, peritoneum, nasal cavities, middle ear, and accessory sinuses, mediastinum, mesothelioma, respiratory system [PET not covered for screening of asymptomatic members for lung cancer and staging of biopsy for solitary fibrous tumor of pleura] and other intrathoracic organs, bones of skull and face, mandible, connective tissue and other soft tissue, [PET not covered for schwannoma], melanoma of skin, skin, breast, vulva, vagina [PET not covered for staging and restaging of vulvar or vaginal cancer], cervix uteri, ovary, fallopian tube, testis, eye, brain, [PET not covered for atypical teratoid/rhabdoid tumor], thyroid gland, thymus, head, face, and neck, other endocrine glands and related structures [including paragangliomas], other malignant neoplasm without specification of site [occult primary cancers] [including Angiomatoid fibrous histiocytoma] [not covered for Planning of radiotherapy for glioblastoma]
C7A.00 - C7A.090, C7A.092 - C7A.8	Malignant neuroendocrine tumors [not covered for Surgical planning for perivascular epithelioid cell tumors (PEComas)]
C45.0 - C45.9	Mesothelioma
C70.0 - C70.9	Malignant neoplasm of meninges
C80.1	Malignant (primary) neoplasm, unspecified [unknown primary]
C81.00 - C90.02, C96.0 - C96.9	Malignant neoplasm of lymphatic and hematopoietic tissue [except xanthogranuloma]
C90.20 - C90.22	Extramedullary plasmacytoma

Code	Code Description
C90.30 - C90.32	Solitary plasmacytoma
C91.10 - C91.12	Chronic lymphocytic leukemia of B-cell type
D00.00 - D01.3, D02.0 - D02.4, D05.00 - D05.92, D06.0 - D06.9, D09.0 - D09.19, D09.3 - D09.9	Carcinoma in situ lip, oral cavity, and pharynx, esophagus, stomach, colon, rectum, respiratory system, breast, and eye [major salivary glands not covered]
D37.01 - D37.09 D38.0 - D38.6	Neoplasm of uncertain behavior of major salivary glands, lip, oral cavity, and pharynx, larynx, trachea, bronchus, and lung, pleura, thymus, and mediastinum, and other and unspecified respiratory organs
D3a.00 - D3a.8	Neuroendocrine tumors
D32.0-D32.9	Benign neoplasm of meninges
D43.0 - D43.9	Neoplasm of uncertain behavior of brain and central nervous system
D47.2	Monoclonal gammopathy
D47.Z1	Post-transplant lymphoproliferative disorder (PTLD)
D47.Z2	Castleman disease [not covered for re-staging of multi-centric Castleman's disease]
D48.110 - D48.2	Neoplasm of uncertain behavior of connective and other soft tissue [hemangiopericytoma][not covered for giant cell tumor of the bone]
D76.3	Other histiocytosis with massive lymphadenopathy [Roasi-Dorfman disease][covered for staging, restaging, and surveillance]
E88.89	Other specified metabolic disorders [Erdheim-Chester disease]
G30.0 – G30.9	Alzheimer's disease
G31.84	Mild cognitive impairment of uncertain or unknown etiology

Code	Code Description
G40.001 - G40.919	Epilepsy and recurrent seizures [pre-surgical evaluation for localization of seizure focus]
M31.4	Aortic arch syndrome [Takayasu]
M31.6	Other giant cell arteritis
R22.0 - R22.1, R90.0	Swelling, mass, or lump in head and neck
R56.1	Post traumatic seizures
R56.9	Unspecified convulsions [pre-surgical evaluation for localization of seizure focus only]
R91.1	Solitary pulmonary nodule
R97.20	Elevated prostate specific antigen [PSA]
R97.21	Rising PSA following treatment for malignant neoplasm of prostate
T66.xxxA - T66.xxxS	Radiation sickness, unspecified [radiation necrosis]
Z85.01, Z85.028, Z85.030 - Z85.038 Z85.040 - Z85.048, Z85.110 - Z85.12 Z85.21 - Z85.22, Z85.3 Z85.41, Z85.43, Z85.46, Z85.47 Z85.71 - Z85.79, Z85.820, Z85.830 Z85.840 - Z85.841, Z85.850 - Z85.858	Personal history of malignant neoplasm of esophagus, large intestine, prostate, rectum, rectosigmoid junction, and anus, trachea, bronchus and lung, larynx, nasal cavities, middle ear, and accessory sinuses, breast, stomach, cervix uteri, ovary, testis, lymphosarcoma and reticulosarcoma, Hodgkin's disease, bone, melanoma of skin, eye, brain, thyroid and neuroendocrine tumor

Code	Code Description
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
B38.0 - B38.9	Coccidioidomycosis
B39.0 – B39.9	Histoplasmosis
C23	Malignant neoplasm of gallbladder
C24.0	Malignant neoplasm of extrahepatic bile duct
C24.8 - C24.9, C48.0	Gallbladder and other and unspecified parts of biliary tract and retro peritoneum
C46.0 - C46.9, C51.0 - C51.9, C55	Malignant neoplasm of Kaposi's sarcoma, vulva and uterus, part unspecified
C54.0 - C54.9	Malignant neoplasm of corpus uteri
C58	Malignant neoplasm of placenta
C63.0-C63.9	Malignant neoplasm of other and unspecified male genital organs
C64.0-C64.9	Malignant neoplasm of kidney, except renal pelvis
C65.0-C65.9	Malignant neoplasm of renal pelvis
C67.0-C67.9	Malignant neoplasm of bladder
C70.0 - C70.9, C72.0 - C72.1 C72.50 - C72.9	Malignant neoplasm of other and unspecified parts of nervous system
C77.9	Secondary and unspecified malignant neoplasm of lymph nodes, site unspecified
C78.2 - C78.39	Secondary malignant neoplasm of pleura and other respiratory organs
C78.6 - C78.7	Secondary malignant neoplasm of retroperitoneum and peritoneum, and liver, specified as secondary
C79.00 - C79.52	Secondary malignant neoplasm of kidney, other urinary organs, skin, brain and spinal cord, other parts of nervous system, and bone and bone marrow
C79.70 - C79.72	Secondary malignant neoplasm of adrenal gland
C79.82	Secondary malignant neoplasm of genital organs
C79.89	Secondary malignant neoplasm of other specified sites
C7A.091	Malignant carcinoid tumor of the thymus

Code	Code Description
C86.30 - C86.31	Blastic NK-cell lymphoma [plasmacytoid dendritic cell neoplasm]
C88.80 - C88.81	Other malignant immunoproliferative diseases
C90.10, C91, C91.40 - C91.42, C93.00 - C93.Z2, C94.00 - C94.82, C95.00 - C95.92	Plasma cell leukemia, lymphoid leukemia, monocytic leukemia, other leukemias of specified cell type, Leukemia of unspecified cell type, Hairy cell leukemia
D01.40 - D01.9	Carcinoma in situ of anal canal, anus, unspecified, other and unspecified parts of intestine, liver and biliary system, and other and unspecified digestive organs
D04.0 - D04.9	Carcinoma in situ of skin
D07.1	Carcinoma in situ of vulva
D10.0 - D36.9	Benign neoplasms [including paraganglioma][including littoral cell splenic angioma]
D21.5	Benign neoplasm of connective and other soft tissue of pelvis [solitary fibrous tumor of pelvis]
D37.6, D48.3 - D8.4	Neoplasm of uncertain behavior of liver and biliary passages, and retroperitoneum and peritoneum
D39.0 - D41.9, D44.0 - D44.9	Neoplasm of uncertain behavior of other and unspecified respiratory organs, genitourinary organs, and endocrine glands
D45 - D47.1, D47.3 – D47.Z9, D48.60 - D48.62	Neoplasm of uncertain behavior of breast and other lymphatic and hematopoietic tissues
D49.3 - D49.7	Neoplasm of unspecified behavior of breast, bladder, other genitourinary organs, brain, and endocrine glands and other parts of nervous system
D49.9	Neoplasm of unspecified behavior, of unspecified site

Code	Code Description
D72.10 - D72.19	Eosinophilia
D73.2	Chronic congestive splenomegaly
D86.0 - D86.9	Sarcoidosis
E21.0 - E21.5	Hyperparathyroidism and other disorders of parathyroid gland
E26.09	Other primary hyperaldosteronism
E71.520 - E71.529	X-linked adrenoleukodystrophy
E83.52	Hypercalcemia
E85.0 - E85.9	Amyloidosis
F01.50 - F99	Mental disorders
G00.0 - G37.9, G43.001 - G99.8 H00, 011 - H59.89, H60.00 - H95.89	Diseases of the nervous system and sense organs [except pre-surgical evaluation for localization of seizure focus] [Monitoring of cortico-basal syndrome] [Neurodegenerative progression in Parkinson disease]
I00 - I52	Heart disease
I67.5	Moyamoya disease
I88.0	Nonspecific mesenteric lymphadenitis
J84.82	Adult pulmonary Langerhans cell histiocytosis
J90 - J91.8	Pleural effusion
K72.01, K72.11, K72.91	Hepatic encephalopathy
L72.11	Pilar cysts [Pilar tumor]
M01.X0 - M01.X79	Direct infections of joint in infectious and parasitic diseases classified elsewhere [joint infection]
M04.1	Periodic fever syndromes [evaluation of unspecified periodic fever syndrome]
M05.00 - M14.89	Inflammatory polyarthropathies

Code	Code Description
M12.20 - M12.29	Villonodular synovitis (pigmented)
M15.0 - M19.93	Osteoarthritis [Joint inflammation associated with osteoarthritis]
M86.30 - M86.9	Chronic multifocal osteomyelitis
M88.0 - M88.9	Osteitis deformans [Paget's disease of bone]
O01.0 - O01.9	Hydatidiform mole [gestational trophoblastic neoplasia]
Q89.09	Congenital malformations of spleen [congenital splenomegaly]
R16.1	Splenomegaly, not elsewhere classified
R22.2	Localized swelling, mass and lump, trunk [retroperitoneal mass]
R25.0 - R29.91	Symptoms and signs involving nervous and musculoskeletal systems
R40.0 - R40.4	Somnolence, stupor and coma
R41.1 - R41.3	Amnesia
R41.9, R68.89	Other general symptoms
R42	Dizziness and giddiness
R44.0, R44.2 - R44.3 R55	Hallucinations and syncope and collapse
R50.9	Fever, unspecified [fever of unknown origin (FUO)]
R93.0	Abnormal findings on diagnostic imaging of skull and head, not elsewhere classified
R93.1 - R93.5	Abnormal findings on radiological and other examination of other intrathoracic organ
R94.01 - R94.138	Abnormal results of function studies of brain and central nervous system and peripheral nervous system and special senses
R94.30 - R94.39	Abnormal results of cardiovascular function studies,
T81.40xA - T81.49xS	Infection following a procedure [infection of hip arthroplasty]

Code	Code Description
T84.51xA - T84.54xS	Infection and inflammatory reaction due to internal joint prosthesis [infection of hip arthroplasty or knee replacement prostheses]
Z00.00 - Z13.9	Persons encountering health services for examination
Z15.01	Genetic susceptibility to malignant neoplasm of breast [Li-Fraumeni syndrome]
Z85.00, Z85.810 - Z85.819	Personal history of malignant neoplasm of gastrointestinal tract, unspecified
Z85.05 - Z85.09	Personal history of malignant neoplasm of liver and of other gastrointestinal tract
Z85.20 - Z85.29	Personal history of malignant neoplasm of other respiratory and intrathoracic organs
Z85.42	Personal history of malignant neoplasm of other parts of uterus
Z85.48 - Z85.59	Personal history of malignant neoplasm of epididymis, other male genital organs, urinary organs, unspecified
Z85.6	Personal history of leukemia
Z85.71 - Z85.79	Personal history of other malignant neoplasms of lymphoid, hematopoietic and related tissues
Z85.820 - Z85.828	Personal history of other malignant neoplasm of skin
Z85.848	Personal history of malignant neoplasm of other parts of nervous system
Z85.858	Personal history of malignant neoplasm of other endocrine glands and related structures, other, and unspecified sites
Z96.641 - Z96.649	Presence of artificial hip
FDG-PET :	
HCPCS codes not covered for indications listed in the CPB:	
A9552	Fluorodeoxyglucose f-18 fdg, diagnostic, per study dose, up to 45 millicuries
ICD-10 codes covered if selection criteria are met:	
C23	Malignant neoplasm of gallbladder
C43.0-C43.9	Malignant melanoma of skin

Code	Code Description
C44.00-C44.99	Other and unspecified malignant neoplasm of skin
C4A.0-C4A.9	Merkel cell carcinoma
C47.0	Malignant neoplasm of peripheral nerves of head, face and neck [Neuroblastoma]
C65.1-C65.9	Malignant neoplasm of renal pelvis
C66.1-C66.9	Malignant neoplasm of ureter
C67.0-C67.9	Malignant neoplasm of bladder
C74.00-C74.92	Malignant neoplasm of adrenal gland
D47.2	Monoclonal gammopathy [Light chain deposition disease]
Cerianna (fluoroestradiol F-18) PET:	
HCPCS codes covered for indications listed in the CPB:	
A9591	Fluoroestradiol f 18, diagnostic, 1 millicurie [Cerianna]
ICD-10 codes covered if selection criteria are met:	
C50.011-C50.929	Malignant neoplasm of breast [recurrent or metastatic breast cancer]
Ga 68 dotatate Pet or Ga 68-DOTA-TOC:	
HCPCS codes covered if selection criteria are met:	
A9587	Gallium ga-68, dotatate, diagnostic, 0.1 millicurie
C9067	Gallium ga-68, dotatoc, diagnostic, 0.01 mci
ICD-10 codes covered if selection criteria are met:	
C73	Malignant neoplasm of thyroid gland
D32.0-D32.9	Benign neoplasm of meninges
Non Oncologic Rheumatologic indications for FDG-PET:	
CPT codes covered for indications listed in the CPB:	
78811	Positron emission tomography (PET) imaging; limited area (e.g., chest, head/neck)
78812	skull base to mid-thigh
78813	whole body
78814	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; limited area (e.g., chest, head/neck)
78815	skull base to mid-thigh
78816	whole body

Code	Code Description
Other CPT codes related to the CPB:	
37609	Ligation or biopsy, temporal artery
77084	Magnetic resonance (eg, proton) imaging, bone marrow blood supply
ICD-10 codes covered if selection criteria are met:	
D47.2	Monoclonal gammopathy
M31.4	Aortic arch syndrome [Takayasu]
M31.6	Other giant cell arteritis
Neurologic indications for PET:	
CPT codes covered for indications listed in the CPB:	
78608	Brain imaging, positron emission tomography (PET); metabolic evaluation
78609	perfusion evaluation
HCPCS codes not covered for indications listed in the CPB:	
A9587	Gallium ga-68, dotatate, diagnostic, 0.1 millicurie
A9602	Fluorodopa f-18, diagnostic, per millicurie
ICD-10 codes covered if selection criteria are met:	
G40.001 - G40.919	Epilepsy and recurrent seizures [pre-surgical evaluation for localization of seizure focus]
R56.1	Post traumatic seizures
R56.9	Unspecified convulsions [pre-surgical evaluation for localization of seizure focus only]
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
F01.50 - F01.C4 F03.90 - F03.C4	Dementias
F02.80 - F02.C4	Dementia in other diseases classified elsewhere with or without behavioral disturbance
F07.0	Personality change due to known physiological condition
G10	Huntington's disease
G20.A1 - G21.9	Parkinson's disease
G23.1	Progressive supranuclear ophthalmoplegia [Steele-Richardson-Olszewski]
G31.83	Neurocognitive disorder with Lewy bodies

Code	Code Description
G31.85	Corticobasal degeneration
G90.3	Multi-system degeneration of the autonomic nervous system
I68.0	Cerebral amyloid angiopathy [sporadic]
R41.1 - R41.3	Amnesia
R43.0 - R43.9	Disturbances of smell and taste
Z13.858	Encounter for screening for other nervous system disorders
Z82.0	Family history of epilepsy and other diseases of the nervous system
Amyloid PET scan:	
CPT codes covered for indications listed in the CPB:	
78811	Positron emission tomography (PET) imaging; limited area (eg, chest, head/neck)
78814	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; limited area (eg, chest, head/neck)
HCPCS codes covered if selection criteria are met:	
A9586	Florbetapir f18, diagnostic, per study dose, up to 10 millicuries
A9601	Flortaucipir f 18 injection, diagnostic, 1 millicurie
Q9982	Flutemetamol F18, diagnostic, per study dose, up to 5 millicuries
Q9983	Florbetaben f18, diagnostic, per study dose, up to 8.1 millicuries
Other HCPCS codes related to the CPB:	
J0172	Injection, aducanumab-avwa, 2 mg
J0174	Injection, lecanemab-irmb, 1 mg
ICD-10 codes covered if selection criteria are met:	
G30.0 - G30.9	Alzheimer's disease
G31.84	Mild cognitive impairment of uncertain or unknown etiology
Piflufolastat F-18 (Pylarify) PET:	
HCPCS codes covered if selection criteria are met:	
A9595	Piflufolastat f-18, diagnostic, 1 millicurie
ICD-10 codes covered if selection criteria are met:	
C61	Malignant neoplasm of prostate
R97.20	Elevated prostate specific antigen [PSA]

Code	Code Description
R97.21	Rising PSA following treatment for malignant neoplasm of prostate
Z85.46	Personal history of malignant neoplasm of prostate
Copper Cu 64 dotatate PET:	
CPT codes covered for indications listed in the CPB:	
78811	Positron emission tomography (PET) imaging; limited area (eg, chest, head/neck)
78812	Positron emission tomography (PET) imaging; skull base to mid-thigh
78813	Positron emission tomography (PET) imaging; whole body
78814	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; limited area (eg, chest, head/neck)
78815	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; skull base to mid-thigh
78816	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; whole body
HCPCS codes covered if selection criteria are met:	
A9592	Copper cu-64, dotatate, diagnostic, 1 millicurie
ICD-10 codes covered if selection criteria are met:	
C74.00 – C74.92	Malignant neoplasm of medulla of adrenal gland [Pheochromocytoma]
C7A.00 – C7A.8	Malignant neuroendocrine tumors
D09.3	Carcinoma in situ of thyroid and other endocrine glands
D35.00 – D35.02	Benign neoplasm of adrenal gland [Pheochromocytoma]
D3A.00 – D3A.8	Neuroendocrine tumors
D44.10 – D44.12	Neoplasm of uncertain behavior of adrenal gland [Pheochromocytoma]

Code	Code Description
D44.7	Neoplasm of uncertain behavior of aortic body and other paraganglia
D49.7	Neoplasm of unspecified behavior of endocrine glands and other parts of nervous system [Pheochromocytoma]
SCINTIX biology-guided radiotherapy:	
HCPCS codes not covered if selection criteria are met:	
G0562	Therapeutic radiology simulation-aided field setting; complex, including acquisition of pet and ct imaging data required for radiopharmaceutical-directed radiation therapy treatment planning (i.e., modeling)
G0563	Stereotactic body radiation therapy, treatment delivery, per fraction to 1 or more lesions, including image guidance and real-time positron emissions-based delivery adjustments to 1 or more lesions, entire course not to exceed 5 fractions
ICD-10 codes not covered if selection criteria are met:	
C78.7	Secondary malignant neoplasm of liver and intrahepatic bile duct

Background

Policy above is adapted from eviCore imaging guidelines.

Positron emission tomography (PET) also known as positron emission transverse tomography (PETT), or positron emission coincident imaging (PECI), is a non-invasive diagnostic imaging procedure that assesses the level of metabolic activity and perfusion in various organ systems of the human body. A positron camera (tomograph) is used to produce cross-sectional tomographic images, which are obtained from positron emitting radioactive tracer substances (radiopharmaceuticals) such as 2-[F-18] fluoro-d-glucose (FDG) that are administered intravenously to the member.

PET assesses the function of tissues and organs by monitoring the metabolic or biochemical activity while tracking the movement and concentration of a radioactive contrast agent. The technique uses

special computerized imaging equipment and rings of detectors surrounding the individual to record gamma radiation produced when positrons (positively charged particles) emitted by the radioactive agent collide with electrons. Integrated PET/CT imaging is a technique in which both PET and CT are performed sequentially during a single visit on a hybrid PET/CT scanner. The CT and PET images are then co-registered using fusion software, enabling the physiologic data obtained on PET to be localized according to the anatomic CT images. When PET/CT is performed, a low radiation dose CT without contrast is typically used to keep the radiation dose as low as possible and to limit adverse events. A higher resolution CT requires a higher dose of radiation and intravenous (IV) contrast. The Biograph mCT-X and mCT-S (Biograph HD) are examples of PET/CT devices.

A BlueCross BlueShield Association's special report on PET for post-treatment surveillance of cancer (2009) found that there is simply inadequate direct and indirect evidence supporting the effectiveness of PET scanning for the purpose of surveillance. Reflecting this lack of evidence, current practice guidelines appear unanimously to recommend against the use of PET for surveillance. No strong support of the use of PET for surveillance was found in editorials, case reports, or other studies. The report concluded that given such problems such as lead time bias, length bias, and the uncertain diagnostic characteristics of PET in the surveillance setting, it would be difficult to determine if the effectiveness of PET for surveillance could be determined with observational data. Clinical trials may be necessary to determine whether PET surveillance is effective in improving health outcomes. (Note: surveillance is defined as use of PET beyond the completion of treatment, in the absence of signs or symptoms of cancer recurrence or progression, for the purpose of detecting recurrence or progression or predicting outcome).

Guidelines on screening for tumors in paraneoplastic syndromes from the European Federation of Neurological Societies (Titulaer et al, 2011) state that, for screening of the thoracic region, a CT-thorax is recommended, which if negative is followed by fluorodeoxyglucose-positron emission tomography (FDG-PET). The guidelines recommend mammography for breast cancer screening, followed by MRI. Ultrasound is the investigation of first choice for the pelvic region, followed by CT. The guidelines state that dermatomyositis patients should have CT of the thorax and abdomen, ultrasound of the pelvic

region and mammography in women, ultrasound of testes in men under 50 years and colonoscopy in men and women over 50 years. The guidelines recommend, if primary screening is negative, repeat screening after 3 to 6 months and screening every 6 months up until four years. In Lambert-Eaton myasthenic syndrome, screening for 2 years is sufficient.

PET/MRI is a hybrid imaging technology that incorporates MRI soft tissue morphological imaging and PET functional imaging. PET/MRI systems use software to fuse image data from two separately performed imaging examinations. Performing the PET and MRI scans simultaneously in the same imaging session purportedly prevents issues associated with image data mismatching that may occur with image fusion software. A combined PET/MRI approach is suggested for imaging anatomical, biochemical and functional characteristics of disease. The Biograph mMR is an example of a PET/MRI device. There are few studies that have focused on PET/MRI technology and its advantages over PET/CT fusion, which is the current standard of care. An assessment by the Australian Health Policy Advisory Committee on Technology (Mundy, 2012) concluded that there are few clinical studies in the literature reporting on the use of hybrid PET-MRI systems. The report noted that initial, small scale studies indicate that PET-MRI hybrid scanning systems are as effective at imaging regions of interest in certain brain cancers and head and neck cancer as PET-CT hybrid scanners; however these imaging studies do not indicate the effect on clinical outcomes for these patients or a change in patient management. The review stated that, based on the small number of published studies it appears that hybrid PET-MRI may be a promising imaging modality, especially for pediatric patients, with the added benefit of reduced exposure to radiation compared to a PET-CT scan. The report noted, however, that recent developments in CT design have resulted in scanners that deliver a reduced radiation dose. In addition, combined PET-MRI systems are not capable of producing as high-quality images as stand-alone imaging systems. The report concluded that "combined PET-MRI systems are currently of benefit in the research, rather than clinical setting." The report stated that larger studies with clinical outcomes are required to demonstrate the effectiveness of the modality. The report noted concerns regarding the paucity of evidence in respect to the clinical effectiveness of hybrid PET-MRI scanners and the potential for increased costs due to

workforce issues including training requirements, time taken for interpretation of images, increasing capacity for image storage and the impact on patient flow.

Absolute quantitation of myocardial blood flow (AQMBF) imaging is an additional physiological assessment during a pharmacologic stress/rest PET or PET/CT myocardial perfusion imaging. Following stress/rest PET or PET/CT myocardial perfusion imaging, images for PET myocardial perfusion imaging are acquired to allow quantitation of AQMBF. The report quantifies in ml/g/min for rest, stress, and indexed/reserve flow for each coronary bed and for the global left ventricular. Performance of quantitation of myocardial blood flow by cardiac PET is currently non-standardized between different vendor products.

Ngo et al (2022) provided an overview of the role of PET myocardial perfusion imaging (MPI) in the detection of CAD, focusing on the added value of myocardial blood flow (MBF) for diagnosis and prognostication. These investigators stated that PET MPI is increasingly used for the risk stratification of patients with suspected or established coronary artery disease (CAD). PET MPI provides accurate and reproducible non-invasive quantification of MBF at rest and during hyperemia, providing incremental information over conventional myocardial perfusion alone. Inclusion of MBF in PET MPI interpretation improves both its sensitivity and specificity. Furthermore, quantitative MBF measurements have repeatedly been shown to offer incremental and independent prognostic information over conventional clinical markers in a broad range of conditions, including in CAD. Quantitative MBF measurement is now an established and powerful tool enabling accurate risk stratification and guiding patients' management.

Anal Cancer

The National Comprehensive Cancer Network (NCCN) on "Anal carcinoma" (v 2.2018) states that PET/CT scanning can be considered to verify staging before treatment, and that it has been reported to be useful in the evaluation of pelvic nodes, even for those who have "normal-sized" lymph nodes on CT imaging and have anal canal cancer. However, the NCCN panel does not consider PET/CT to be a replacement for a diagnostic CT.

PET for Orthopedic Prostheses

Manthey and co-workers (2002) described 18F-FDG-PET findings in patients referred for evaluation of painful hip or knee prostheses. These investigators studied 23 patients with 28 prostheses, 14 hip and 14 knee prostheses, who had a complete operative or clinical follow-up. 18F-FDG-PET scans were obtained with an ECAT EXACT HR+ PET scanner. High glucose uptake in the bone prostheses interface was considered as positive for infection, an intermediate uptake as suspect for loosening, and uptake only in the synovia was considered as synovitis. The imaging results were compared with operative findings or clinical outcome. FDG-PET correctly identified 3 hip and 1 knee prostheses as infected, 2 hip and 2 knee prostheses as loosening, 4 hip and 9 knee prostheses as synovitis, and 2 hip and 1 knee prostheses as unsuspected for loosening or infection. In 3 patients covered with an expander after explantation of an infected prosthesis, FDG-PET revealed no further evidence of infection in concordance with the clinical follow-up. FDG-PET was false-negative for loosening in 1 case. The authors concluded that these preliminary findings suggested that FDG-PET could be a useful tool for differentiating between infected and loose orthopedic prostheses as well as for detecting only inflammatory tissue such as synovitis.

Germ Cell Tumors (CGT)

Karapetis and colleagues (2003) stated that FDG-PET may detect residual or recurrent malignancy in patients with germ cell tumors (GCT) following chemotherapy. The objective of the present study was to evaluate the use of FDG-PET in the setting of advanced GCT, and to determine the influence of FDG-PET on subsequent patient management. A computerized search of the patient database of the Department of Medical Oncology, Guy's Hospital, London, United Kingdom, and a manual search of medical records, were conducted. All male patients with metastatic or extra-gonadal GCT treated with chemotherapy between July 1996 and June 1999 inclusive were identified. Data from patients that had a PET scan following chemotherapy were analyzed. Reported PET scan findings were compared with subsequent clinical management and patient outcome. A total of 30 patients with metastatic testicular GCT and 3 patients with extra-gonadal GCT were treated with chemotherapy. Of these, 15 patients (12 testicular; 3 extra-gonadal; 10 non-seminoma; and 5 seminoma) were investigated following chemo-therapy with at least 1

FDG-PET scan. Seven patients had 2 or more PET scans, and a total of 26 FDG-PET scans was performed. The most frequent indication for PET scan was evaluation of a residual mass (11 patients). Three patients had an FDG-PET to evaluate thymic prominence. Minimum follow-up from first PET scan was 18 months. Three of 26 PET scans had false-positive findings. Four PET scans yielded findings of equivocal significance with repeat PET scan recommended. Relapse of disease occurred in 3 patients; 2 of whom had normal previous PET scans and 1 had a previous equivocal result. Moreover, PET had an impact on patient management in only 1 case where it "prompted" surgical excision of a residual mass. Normal PET scans provided reassurance in patients with residual small masses but did not alter their subsequent management. The authors concluded that a residual mass was the most common indication for PET. For the majority of patients PET did not have a discernible influence on clinical management. They stated that (i) oncologists should exercise caution in their interpretation of PET scan findings and guidelines for the appropriate use of PET in testicular cancer management need to be developed, and (ii) prospective trials are required to define the clinical role of PET in this setting.

Hepatocellular Carcinoma

Wolfort et al (2010) the effectiveness of FDG-PET for the detection and staging of hepatocellular carcinoma (HCC). In addition, these researchers also assessed the correlation between FDG-PET positivity, tumor size, AFP, and histological grade. All patients on the hepatobiliary and liver transplant service with biopsy proven HCC that underwent FDG-PET between January 2000 and December 2004 were selected for a retrospective review. Results of the FDG-PET scans were compared with other imaging studies (CT, MRI, ultrasonography), intra-operative findings, tumor size, AFP levels, and histological grade. Of the 20 patients who underwent 18F-FDG PET, increased FDG uptake was noted in 14 scans (70 %). These 20 patients fell into 2 groups: (i) for detecting HCC (group A) and (ii) for staging HCC (group B). There were 7 patients in group A; only 2 scans (28.6 %) showed increased uptake. There were 13 patients in group B; 12 scans (92.3 %) showed increased uptake. In group B, 11 of the 13 scans (84.6 %) provided an accurate representation of the disease process. Two scans failed to accurately portray the disease; 1 scan failed to show any increase in uptake, and the other scan failed to

detect positive nodes that were found during surgery. FDG-PET detected only 2 of 8 tumors (25 %) less than or equal to 5 cm in size. All 12 PET scans (100 %) in tumors greater than or equal to 5 cm and/or multiple in number were detected by FDG-PET. FDG-PET scans with AFP levels less than 100 ng/ml were positive in 5 of 9 patients (55.6 %). In patients with levels greater than 100 ng/ml, 6 of 7 patients (85.7 %) had positive scans. Histologically, there were 6 well-differentiated, 6 moderately differentiated, and 2 poorly differentiated HCCs. FDG-PET detected 4 of 6 for both well- and moderately-differentiated HCCs. Both poorly- differentiated HCCs were detected. The intensity was evenly distributed between the different histological grades. There was a strong correlation of FDG uptake with tumor size. There were 5 HCCs with primary tumors greater than 10 cm in size; 4 showed intense uptake on the scan. In contrast, of the 8 tumors less than or equal to 5 cm in size, 6 were negative for uptake. The sensitivity of FDG-PET in detecting HCC less than or equal to 5 cm in size is low and therefore may not be helpful in detecting all of these tumors. For larger tumors, there is a strong correlation of sensitivity and uptake intensity with tumor size and elevated AFP levels. FDG-PET sensitivity and uptake intensity did not correlate with histological grade. In the setting of extra-hepatic disease, FDG-PET seems to be an effective accurate method for HCC staging; however, whether PET offers any benefit over traditional imaging has yet to be determined.

An UpToDate review on "Staging and prognostic factors in hepatocellular carcinoma" (Curley et al, 2013) states that "Positron emission tomography with fluorodeoxyglucose (FDG-PET) is being investigated as a complementary staging tool that may help to define prognosis in some patients …. Positron-emitting radionuclides other than FDG (such as 11C-acetate) are under investigation as potentially more useful agents for imaging and staging HCC". Furthermore, National Comprehensive Cancer Network (NCCN)’s clinical guideline on "Hepatobiliary cancers" (Version 4.2018) notes that PET/CT is not adequate for screening and diagnosis of hepatocellular carcinoma. In addition, PET/CT is not recommended for detection of HCC because of limited sensitivity. There is no mentioning of its use for staging.

Pediatric Abdominal Tumors

Murphy et al (2008) stated that PET/CT scan provides both functional and anatomical information in a single diagnostic test. It has the

potential to be a valuable tool in the evaluation of pediatric abdominal tumors. The goal of this study was to report the authors' early experience with this technology. Children who underwent PET/CT scan in the work-up for abdominal neoplasms between July 2005 and January 2008 were identified. Retrospective reviews of all radiological studies, operative notes, and pathological reports were undertaken. A total of 36 patients were collected. These included Burkitt's lymphoma (n = 8), neuroblastoma (n = 7), rhabdomyosarcoma (n = 6), ovarian tumor (n = 3), Wilms tumor (n = 2), HCC (n = 2), paraganglioma (n = 1), germ cell tumor (n = 1), undifferentiated sarcoma (n = 1), renal primitive neuroectodermal tumor (n = 1), gastro-intestinal stromal tumor (n = 1), adrenocortical carcinoma (n = 1), inflammatory pseudotumor (n = 1), and adrenal adenoma (n = 1). All neoplasms were FDG were avid. These investigators identified several potential uses for PET/CT scan in this group of patients. These included (i) pre-operative staging, (ii) selection of appropriate site for biopsy, (iii) identification of occult metastatic disease, (iv) follow-up for residual or recurrent disease, and (v) assessment of response to chemotherapy. It can also be valuable when the standard diagnostic studies are equivocal or conflicting. The authors concluded that these preliminary data indicated that PET/CT is a promising tool in the evaluation of pediatric abdominal malignancies. The delineation of the exact role of this diagnostic modality will require additional experience. It should also be noted that the National Comprehensive Cancer Network's clinical practice guideline on "Kidney cancer" (Version 3.2025) states "PET alone is not a tool that is standardly used to diagnose kidney cancer or follow for evidence of relapse after nephrectomy."

Mody and colleagues (2006) described the use of FDG-PET in a series of 7 children (11 scans) with primary hepatic malignancies (5 patients with hepatoblastoma, 2 patients with hepatic embryonal rhabdomyosarcoma), together with other imaging (CT and MRI), serum tumor markers, and tumor pathology. These patients with pathologically proven hepatic malignancies underwent 11 FDG-PET scans for staging (1 patient) or restaging (6 patients). Tumor uptake of FDG was assessed qualitatively and compared with biochemical and radiological findings. Abnormal uptake was demonstrated in 6 of 7 patients (10 of 11 scans). Three patients subsequently underwent partial hepatic resection, and 1 underwent brain biopsy, confirming in each that the abnormal uptake of FDG indicated viable tumor. In 1 patient, intense uptake was due to necrotizing granulomas. In 1

patient, images were suboptimal due to non-compliance with fasting. The authors concluded that primary hepatic tumors of childhood usually demonstrate increased glycolytic activity, which allows them to be imaged using PET and the tracer 18F-FDG. The technique is probably most useful for assessing response to therapy, in following alfa-fetoprotein (AFP)-negative cases and for detecting metastatic disease although a large series of patients will need to be studied to confirm these initial findings. Non-neoplastic inflammation may also accumulate FDG and could be confused with malignancy. As these tumors are rare, prospective multi-center studies are needed to determine the true clinical utility of FDG-PET imaging in the management of children with primary hepatic malignancies.

Langerhans Cell Histiocytosis (LCH)

Phillips and colleagues (2009) assessed the effectiveness of FDG-PET scans in identifying sites of active disease and assessing response to therapy in patients with Langerhans cell histiocytosis (LCH). Changes in standardized uptake value (SUV) indicated increased or decreased disease activity before changes are evident by plain films or bone scans. A total of 102 PET scans for 44 patients (3 adults and 41 children) with biopsy-proven LCH were compared with 83 corollary imaging modalities and were rated for overall clinical utility: false positive or negative ("inferior"), confirming lesions identified by another imaging modality ("confirmatory"), or showing additional lesions, response to therapy or recurrence of disease activity ("superior"), in comparison to bone scans, MRI, CT or plain films. FDG-PET was rated superior in that 90/256 (35 %) new, recurrent, or lesions responding to therapy were identified via change in SUV before other radiographical changes. Positron emission tomography scans confirmed active LCH in 146/256 (57 %). FDG-PET was superior to bone scans in that 8/23 (34 %) lesions, 11/53 (21 %) comparisons to lesions found by MRI, 13/64 (20 %) CT, and 58/116 (50 %) plain films. Positron emission tomography scans confirmed lesions found by: 14/23 (61 %) bone scans, 33/53 (62 %) MRI, 45/64 (65 %) CT, and 54/116 (46 %) of plain films. The authors concluded that whole body FDG-PET scans can detect LCH activity and early response to therapy with greater accuracy than other imaging modalities in patients with LCH lesions in the bones and soft tissues. Whole-body FDG-PET scanning is an important and informative study at diagnosis and for following disease course in patients with LCH.

Krajicek et al (2009) stated that pulmonary Langerhans cell histiocytosis (PLCH) is an inflammatory lung disease strongly associated with cigarette smoking and an increased risk of malignant neoplasms. Although the chest CT scan characteristics of PLCH are well-recognized, the PET scan characteristics of adults with PLCH are unknown. These researchers identified 11 patients with PLCH who underwent PET scanning over a 6-year period from July 2001 to June 2007. The presenting clinic-radiologic features including PET scan and chest CT scan findings were analyzed. Five of 11 patients had positive PET scan findings. Of the 5 PET scan-positive patients, 4 (80 %) were women, 4 (80 %) were current smokers, and the median age was 45 years (age range of 31 to 52 years). PET scan-positive findings were more likely to be present if the scan was performed early in the clinical course. Three PET scan-positive patients (60 %) had multi-organ involvement. PET scan-positive patients had predominantly nodular inflammatory lung disease (greater than 100 nodules) with most nodules measuring less than 8 mm, whereas all PET scan-negative patients had predominantly cystic lung disease with fewer nodules (less than 25 nodules). Notable abnormal PET scan findings included foci of increased uptake in nodular lung lesions, thick-walled cysts, bone, and liver lesions. The mean maximum standardized uptake value of the PET scan-positive lesions ranged from 2.0 to 18.2. The authors concluded that PLCH may be associated with abnormal thoracic and extra-thoracic PET scan results. Patients with nodular disease seen on chest CT scans appear more likely to have abnormal PET scan findings. They stated that these findings suggested that PET scan imaging cannot reliably distinguish between the benign inflammatory nodular lesions of PLCH and malignant lesions.

Adam et al (2010) noted that PLCH manifests with dyspnea and a cough with no significant expectoration, with spontaneous pneumothorax being the first symptom in some patients. The disease is caused by multiple granulomas in terminal bronchioles, visible on high resolution CT (HRCT) as nodules. During the further course of the disease, these nodules progress through cavitating nodules into thick-walled and, subsequently, thin-walled cysts. LCH may affect the lungs only or multiple organs simultaneously. Pulmonary LCH may continually progress or remit spontaneously. Treatment is indicated in patients in whom pulmonary involvement is associated with multi-system involvement or when a progression of the pulmonary lesions has been confirmed. To document the disease progression,

examination of the lungs using HRCT is routinely applied. Increasing number of nodules suggests disease progression. However, determining the number of nodules is extremely difficult. Measuring radioactivity of the individual small pulmonary loci (nodules) using PET is not possible due to the high number and small size of the nodules. The authors' center has a register of 23 patients with LCH; the pulmonary form had been diagnosed in 7 patients. A total of 19 PET and PET-CT examinations were performed in 6 of these patients. PET-CT was performed using the technique of maximum fluorodeoxyglucose accumulation in a defined volume of the right lung --SUV(max) Pulmo. In order to compare the results of examinations performed using the same and different machines over time as well as in order to evaluate pulmonary activity, the maximum fluorodeoxyglucose accumulation in a defined volume of the right lung (SUV(max) Pulmo) to maximum fluorodeoxyglucose accumulation in a defined volume of the liver tissue (SUV(max) Hepar) ratio (index) was used. The disease progression was evaluated using the SUV(max) Pulmo/SUV(max) Hepar index in the six patients with pulmonary LCH. The index value was compared to other parameters characterizing the disease activity (HRCT of the lungs, examination of pulmonary function and clinical picture). The SUV(max) Pulmo/SUV(max) Hepar index correlated closely with other disease activity parameters. The traditional PET-CT examination is useful in detecting the LCH loci in the bone, nodes and other tissue but not in the presence of diffuse involvement of pulmonary parenchyma. Measuring the maximum fluorodeoxyglucose accumulation in a defined volume of the right lung and expressing this activity as the SUV(max) Pulmo/SUV(max) Hepar index appears to be a promising approach. The authors concluded that their initial experience suggested that the results obtained using this method correlate well with other parameters that characterize activity of P LCH. However, they noted that this was a pilot study and further verification is required.

An UpToDate review on "Pulmonary Langerhans cell histiocytosis" (King, 2012) states that "Fluorodeoxyglucose-PET (FDG-PET) scans may show increased uptake in patients with PLCH, particularly when obtained early in the course of disease. This was evaluated in a series of 11 patients with PLCH, five of whom had abnormal FDG uptake in the lungs. The patients with FDG-PET positivity were more likely to have nodular radiographic pattern, suggesting earlier disease; those

with negative FDG-PET scans were more likely to have a cystic pattern and fewer nodules, suggesting later disease". The study cited was that by Krajicek et al (2009).

Large-Vessel Vasculitis

Blockmans et al (2009) stated that ultrasonography, MRI, and PET are increasingly studied in large-vessel vasculitis. These imaging modalities have broadened the knowledge on these disorders and have a place in the diagnostic approach of these patients. Temporal artery ultrasonography can be used to guide the surgeon to that artery segment with the clearest "halo" sign to perform a biopsy, or in experienced hands can even replace biopsy. The distal subclavian, axillary, and brachial arteries can also be examined. High-resolution MRI depicts superficial cranial and extra-cranial involvement patterns in giant cell arteritis (GCA). Contrast enhancement is prominent in active inflammation and decreases under successful steroid therapy. Presence of aortic complications such as aneurysm or dissection can be ruled out within the same investigation. Large thoracic vessel FDG-uptake is seen in the majority of patients with GCA, especially at the subclavian arteries and the aorta. However, FDG-PET can not predict which patients are bound to relapse, and once steroids are started, interpretation is hazardous, which makes its role in follow-up uncertain. Increased thoracic aortic FDG-uptake at diagnosis of GCA may be a bad prognostic factor for later aortic dilatation. In patients with isolated polymyalgia rheumatica – who have less intense vascular FDG uptake – symptoms are caused by inflammation around the shoulders, hips, and spine. The authors concluded that ultrasonography, MRI, and PET remain promising techniques in the scientific and clinical approach of large-vessel vasculitis.

Neuroblastoma

In a phase I trial, Taggart et al (2009) compared 2 functional imaging modalities for neuroblastoma: (i) metaiodobenzylguanidine (MIBG) scan for uptake by the norepinephrine transporter and (ii) (18)F(FDG-PET) uptake for glucose metabolic activity. Patients were eligible for inclusion if they had concomitant FDG-PET and MIBG scans. (131)I-MIBG therapy was administered on days 0 and 14. For each patient, these researchers compared all lesions identified on concomitant FDG-PET and MIBG scans and gave scans a semi-quantitative score. The overall concordance of positive lesions on

concomitant MIBG and FDG-PET scans was 39.6 % when examining the 139 unique anatomical lesions. MIBG imaging was significantly more sensitive than FDG-PET overall and for the detection of bone lesions ($p < 0.001$). There was a trend for increased sensitivity of FDG-PET for detection of soft tissue lesions. Both modalities showed similar improvement in number of lesions identified from day 0 to day 56 scan and in semi-quantitative scores that correlated with overall response. FDG-PET scans became completely negative more often than MIBG scans after treatment. The authors concluded that MIBG scan is significantly more sensitive for individual lesion detection in relapsed neuroblastoma than FDG-PET, though FDG-PET can sometimes play a complementary role, particularly in soft tissue lesions. Complete response by FDG-PET metabolic evaluation did not always correlate with complete response by MIBG uptake.

Cardiac Sarcoidosis and Cardiomyopathies

Sharma (2009) reviewed the role of various imaging modalities in the evaluation of cardiac sarcoidosis and other cardiomyopathies. No study prospectively established the accuracy of each of the various techniques for diagnosing myocardial involvement in patients with suspected cardiac sarcoidosis. Cardiac magnetic resonance imaging (CMR) is demonstrated to have a sensitivity of 100 % and specificity of approximately 80 %, and positive predictive value of approximately 55 % in diagnosing cardiac sarcoidosis. Recent studies have shown that FDG-PET has 100 % sensitivity of detecting earlier stages of sarcoidosis. Both FDG-PET and CMR may provide complementary information for the diagnosis and assessment of efficacy of therapy in patients with cardiac involvement from sarcoidosis. The author concluded that clinical and sub-clinical cardiac involvement is common among patients with sarcoidosis. A structured clinical assessment incorporating advanced cardiac imaging with CMR and FDG-PET scanning is more sensitive than the established clinical criteria. Cardiac MRI is an established imaging modality in the diagnosis of various other cardiomyopathies. The author stated that well designed prospective clinical trials are awaited to define the exact role of these imaging studies in the diagnosis and guidance of therapy.

Kawano et al (2012) stated that sarcoidosis is a multi-systemic granulomatous disease of unknown etiology. These researchers reported an unusual case of sarcoidosis in a woman presenting with cardiac sarcoidosis and massive splenomegaly with a familial history of

cardiac sarcoidosis. Cardiac sarcoidosis was diagnosed based on electrocardiogram, echocardiogram, 18F-fluoro-2-deoxyglucose positron emission tomography (18F-FDG-PET) and skin histological findings. They performed splenectomy to rule out malignant lymphoma, and histological findings confirmed sarcoidosis. After splenectomy, these investigators initiated prednisolone therapy. At 20 months following diagnosis, she was symptom free. The authors concluded that echocardiography and 18F-FDG-PET may be a key diagnostic tool and prednisolone therapy may be safe, effective, and feasible for cardiac sarcoidosis.

A review in UpToDate on "Clinical manifestations and diagnosis of cardiac sarcoidosis" (Blankstein and Stewart, 2018) states that FDG-PET can detect active myocardial inflammation, which can be used to determine the likelihood of cardiac sarcoidosis (CS) when used in the appropriate clinical context. The authors further state that FDG-PET is more sensitive than gallium-67 scintigraphy, thallium-201, or technicium-99m single-photon emission computed tomography (SPECT). The authors state that in patients who have known extracardiac sarcoidosis, characteristic cardiac magnetic resonance imaging (CMR) or 18F-fluorodeoxyglucose-positron emission tomography (FDG-PET) findings can be used to confirm the diagnosis of CS. Endomyocardial biopsy is often not required; and in the absence of biopsy-proven extracardiac sarcoidosis, or when data are inconclusive, integrating data from CMR imaging and FDG-PET may be useful for establishing the likelihood of CS.

An UpToDate review on "Management and prognosis of cardiac sarcoidosis" (Blankstein and Cooper, 2018) states that several studies have demonstrated that advanced cardiac imaging with cardiac magnetic resonance (CMR) or 18F-fluorodeoxyglucose-positron emission tomography (FDG-PET) has predictive value for adverse cardiovascular events including death. Furthermore, retrospective studies in patients with known or suspected CS have found that abnormal cardiac PET findings are associated with adverse cardiac events including sustained VT and death. "A retrospective study of 118 patients referred for cardiac PET with known or suspected CS with a mean follow-up of 1.5 years found that 31 patients (26 percent) developed death or VT requiring therapy. Patients who had both abnormal uptake of FDG by the myocardium (eg, focal inflammation) and a resting perfusion defect (eg, scar or compression of the

microvasculature) had a fourfold increase in the annual rate of VT or death compared with patients with normal imaging. These findings remained significant even after accounting for the Japanese Ministry of Health and Welfare (JMHW) criteria and LVEF. Individuals who had evidence of focal FDG uptake involving the right ventricle had the highest rate of death or VT. The presence of evidence of extracardiac inflammation was not associated with adverse events, suggesting that such features should not be used in deciding on the role of ICD therapy."

Experpt consensus is that immunosuppressive therapy is recommended for selected patients with CS, thus Blankstein and Stewart (2018) recommend FDG-PET in patients who are candidates for immunosuppressive therapy for CS. In this setting, FDG-PET would be useful for the following reasons: the presence of myocardial as well as extracardiac inflammation would indicate a greater role for systemic immunosuppressive therapy, and the FDG-PET scan could serve as a baseline upon which to compare future FDG-PET studies to determine response to therapy.

Bone Marrow Involvement

In a meta-analysis, Chen et al (2011) evaluated the ability of FDG PET or PET/CT scan to ascertain the presence of bone marrow (BM) involvement in aggressive and indolent non-Hodgkin's lymphoma (NHL). These researchers conducted a systematic Medline search of articles published (last update, May 2010). Two reviewers independently assessed the methodological quality of each study. A meta-analysis of the reported sensitivity and specificity of each study was performed. A total of 8 studies met the inclusion criteria. The studies had several design deficiencies. Pooled sensitivity and specificity for the detection of non-Hodgkin aggressive lymphoma were 0.74 (95 % CI: 0.65 to 0.83) and 0.84 (95 % CI: 0.80 to 0.89), respectively. Pooled sensitivity and specificity for the detection of non-Hodgkin indolent lymphoma were 0.46 (95 % CI: 0.33 to 0.59) and 0.93 (95 % CI: 0.88 to 0.98), respectively. The authors concluded that diagnostic accuracy of FDG PET or PET/CT scans was slightly higher but without significant statistical difference (p = 0.1507) in patients with non-Hodgkin aggressive lymphoma as compared with those with non-Hodgkin indolent lymphoma. The sensitivity to detect indolent lymphoma BM infiltration was low for FDG PET or PET/CT.

Sentinel Lymph Node Biopsy for Melanoma

Dengel et al (2011) stated that the false-negative rate for sentinel lymph node biopsy (SLNB) for melanoma is approximately 17 %, for which failure to identify the sentinel lymph node (SLN) is a major cause. Intra-operative imaging may aid in detection of SLN near the primary site, in ambiguous locations, and after excision of each SLN. In a pilot study, these researchers (2011) evaluated the sensitivity and clinical utility of intra-operative mobile gamma camera (MGC) imaging in SLNB in melanoma. From April to September 2008, 20 patients underwent Tc99 sulfur colloid lymphoscintigraphy, and SLNB was performed with use of a conventional fixed gamma camera (FGC), and gamma probe followed by intra-operative MGC imaging. Sensitivity was calculated for each detection method. Intra-operative logistical challenges were scored. Cases in which MGC provided clinical benefit were recorded. Sensitivity for detecting SLN basins was 97 % for the FGC and 90 % for the MGC. A total of 46 SLN were identified: 32 (70 %) were identified as distinct hot spots by pre-operative FGC imaging, 31 (67 %) by pre-operative MGC imaging, and 43 (93 %) by MGC imaging pre- or intra-operatively. The gamma probe identified 44 (96 %) independent of MGC imaging. The MGC provided defined clinical benefit as an addition to standard practice in 5 (25 %) of 20 patients. Mean score for MGC logistic feasibility was 2 on a scale of 1 to 9 (1 = best). The authors concluded that intra-operative MGC imaging provides additional information when standard techniques fail or are ambiguous. Sensitivity is 90 % and can be increased. This pilot study has identified ways to improve the usefulness of an MGC for intra-operative imaging, which holds promise for reducing false negatives of SLNB for melanoma.

Schwannoma (Acoustic Neuroma)

Schwannoma (also known as an acoustic neuromas) are benign nerve sheath tumors composed of Schwann cells, which normally produce the insulating myelin sheath covering peripheral nerves. They are mostly benign and less than 1 % become malignant, degenerating into a form of cancer known as neuro-fibrosarcoma. Schwannomas can arise from a genetic disorder called neurofibromatosis. Schwannomas can be removed surgically, but can then recur. The imaging procedure of choice for schwannomas is magnetic resonance imaging, with or without gadolinium contrast, which can detect tumors as small as 1 to 2

mm in diameter. There are studies reporting FDG uptake in schwannomas, but no studies demonstrating better accuracy or improvements in clinical outcomes with PET over MRI.

Focal Eosinophilic Liver Disease (FELD)

Kim et al (2010) reviewed the FDG PET findings of focal eosinophilic liver disease (FELD) and correlated them with radiologic and pathologic findings. A total of 14 patients, who were clinically or pathologically diagnosed as FELD and underwent CT and/or MR and PET, were enrolled. Two radiologists analyzed CT and MRI regarding size, shape, margin, attenuation, signal intensity (SI), and enhancement patterns of the lesion, both qualitatively and quantitatively. One pathologist determined whether the lesion is eosinophilic abscess (EA) or infiltration. One nuclear medicine physician reviewed the PET images and calculated the peak SUV of the lesion. PET findings were then correlated with CT or MRI, and pathologic findings. Eighty-five lesions were detected on CT (n = 85) and MRI (n = 10). Only 4 of the lesions showed FDG uptake and their mean SUV was 4.0. The size of the lesions with FDG uptake (26.5 mm) was significantly larger than those without uptake (11.8 mm). Mean attenuation and SI differences between the lesion and adjacent liver on CT and T2-weighted MRI tended to be larger in the uptake group (64.3 and 124.5) than the group without uptake (28.5 and 43.5). Among the 4 histologically confirmed lesions, 2 EAs and 1 of the 2 EIs showed FDG uptake. The authors concluded that most FELD do not show FDG uptake on PET. However, larger nodules with greater attenuation or SI differences from the background liver on CT or T2-weighted MRI or those with EA on pathology tend to show FDG uptake on PET.

Also, an UpToDate review on "Clinical manifestations, pathophysiology, and diagnosis of the hypereosinophilic syndromes" (Roufosse et al, 2012) does not mention the use of PET/positron emission tomography.

Erdheim-Chester Disease

An UpToDate review on "Erdheim-Chester disease" (Jacobsen, 2012) states that "imaging studies include magnetic resonance imaging (MRI) of the brain, a computed tomography (CT) scan or an MRI of the entire aorta, a cardiac MRI, a CT scan of the chest, abdomen, and pelvis (which can also be used to image the entire aorta), and a transthoracic

echocardiography. An MRI of the spinal cord is only necessary if the patient has signs or symptoms of spinal cord involvement. The utility of positron emission tomography (FDG-PET) scanning is unclear and not routinely recommended at present".

Myoclonus

UpToDate reviews on "Opsoclonus myoclonus ataxia" (Dalmau and Rosenfeld, 2012) and "Symptomatic (secondary) myoclonus" (Caviness, 2012) do not mention the use of PET and/or CT.

Penile Cancer

Available evidence on the use of PET for penile cancer is limited to use of PET in evaluation of inguinal lymph nodes, with most of the evidence limited to case reports. A recent review article in UpToDate (Lynch, 2012) on "Carcinoma of the penis: Diagnosis, treatment, and prognosis" states describes PET scans as a "evolving imaging technique" that is "promising". Furthermore, the NCCN's clinical practice guideline on penile cancer (NCCN, version 2.2018) states that (for evaluation and risk stratification) "while studies have looked at the use of nanoparticle-enhanced MRI, positron emission tomography-CT (PET/CT), and 18F-fluorodeoxyglucose (FDG) PET/CT, their small sample requires validation in larger prospective studies".

Endometrial Cancer and Uterine Sarcoma

The Society for Gynecologic Oncology guidelines on serous papillary endometrial cancer made no recommendation for PET (Boruta et al, 2009).

Kakhki et al (2013) systematically searched the available literature on the accuracy of 18F-FDG PET imaging for staging of endometrial cancer. PubMed, SCOPUS, ISI Web of Knowledge, Science Direct, and Springer were searched using "endometr* and PET" as the search terms. All studies evaluating the accuracy of 18F-FDG PET in the staging of endometrial carcinoma were included. Statistical pooling of diagnostic accuracy indices was done using random-effects model. Cochrane Q test and I index were used for heterogeneity evaluation. A total of 16 studies (807 patients in total) were included in the meta-analysis. Sensitivity and specificity for detection of the primary lesions were 81.8 % (77.9 % to 85.3 %) and 89.8 % (79.2 % to 96.2 %); for

lymph node staging were 72.3 % (63.8 % to 79.8 %) and 92.9 % (90.6 % to 94.8 %); and for distant metastasis detection were 95.7 % (85.5 % to 99.5 %) and 95.4 % (92.7 % to 97.3 %). The authors concluded that because of low sensitivity, diagnostic utility of 18F-FDG PET imaging is limited in primary tumor detection and lymph node staging of endometrial cancer patients. However, high specificities ensure high positive-predictive values in these 2 indications. Diagnostic performance of 18F-FDG PET imaging is much better in detection of distant metastases. Moreover, they stated that larger studies with better design are needed to draw any more definite conclusion.

Sadeghi et al (2013) reviewed the medical literature on the application of 18F-FDG PET imaging in the management of uterine sarcomas and presented the results in systematic review and meta-analysis format. Medline, SCOPUS, and ISI Web of Knowledge were searched electronically with "PET and (uterine or uterus)" as key words. All studies evaluating the accuracy of 18F-FDG imaging in the staging or restaging of uterine sarcomas were included if enough data could be extracted for calculation of sensitivity and/or specificity. A total of 8 studies were included in the systematic review. Only 2 studies reported the accuracy of 18F-FDG PET imaging in the primary staging of uterine sarcoma with low sensitivity for lymph node staging. For re-staging (detection of recurrence), all 8 included studies had quantitative data, and the patient-based pooled sensitivity and specificity were 92.1 % (95 % CI: 82.4 to 97.4) and 96.2 % (95 % CI: 87 to 99.5), respectively. On a lesion-based analysis, sensitivity was 86.3 % (95 % CI: 76.7 to 92.9), and specificity was 94.4 % (95 % CI: 72.7 to 99.9). Device used (PET versus PET/CT), spectrum of studied patients, and histology of the sarcoma seem to be factors influencing the overall accuracy of 18F-FDG PET imaging. The authors concluded that 18F-FDG PET and PET/CT seem to be accurate methods for detection and localization of recurrence in patients with uterine sarcoma. Moreover, they stated that further large multi-center studies are needed to validate these findings and to correlate both sarcoma type and spectrum of patients to the diagnostic performance of 18F-FDG PET imaging in recurrence detection. The studies evaluating the accuracy of 18F-FDG PET imaging for the primary staging of uterine sarcoma are very limited, and no definite conclusion can be made in this regard.

The National Comprehensive Cancer Network (NCCN) on "Uterine neoplasms" (Version, 1.2019) states to consider whole body PET/CT for initial workup of endometrial carcinoma if metastasis is suspected in select patients, and for select patients who may be candidates for surgery/locoregional therapy. For uterine sarcoma, NCCN recommends consideration of whole body PET/CT for initial workup or post-treatment surveillance of uterine sarcoma to clarify ambiguous findings, or if metastasis is suspected in select patients.

Spindle Cell Sarcoma and Soft Tissue Sarcoma

Spindle cell sarcoma is a type of connective tissue cancer in which the cells are spindle-shaped when examined under a microscope. It is considered a type of soft tissue sarcoma. An UpToDate review on "Clinical presentation, histopathology, diagnostic evaluation, and staging of soft tissue sarcoma" (Ryan and Meyer, 2012) states that "A number of studies report that PET and integrated PET/CT using fluorodeoxyglucose (FDG) can distinguish benign soft tissue tumors from sarcomas, with the greatest sensitivity for high grade sarcomas. However, the ability to differentiate benign soft tissue tumors from low or intermediate grade sarcomas is limited, and PET and PET/CT are not routinely recommended for the initial work-up of a soft tissue mass. One exception may be in the characterization of a suspected peripheral nerve sheath tumor in a patient with neurofibromatosis; in this scenario, PET imaging can be helpful in distinguishing an MPNST from a neurofibroma. Consensus guidelines for workup of a soft tissue sarcoma of the extremity and trunk issued by the National Comprehensive Care Network (NCCN) suggest that PET scan may be useful in the prognostication, grading, and determining response to neoadjuvant chemotherapy in patients with soft tissue sarcoma. However, this recommendation is based upon a single study from the University of Washington that found that FDG-PET was useful to predict the outcomes of patients with high-grade extremity soft tissue sarcomas who were treated initially with chemotherapy. Patients with a baseline tumor SUV max ≥ 6 who had a < 40 percent decrease in FDG uptake after neoadjuvant chemotherapy were found to be at high risk of systemic disease recurrence. The clinical utility of having this information prior to surgical treatment is unclear. At present, the use of PET for prognostication or assessment of treatment response is not considered routine at most institutions .... PET scanning can achieve whole body imaging, and it is widely considered to be more sensitive than CT for the detection of occult distant metastases in a variety of

solid tumors. However, the utility of PET alone or with integrated CT for staging of distant disease extent in STS (soft tissue sarcomas) is unclear as evidenced by the following reports".

Wilms Tumor

The National Wilms Tumor Study Group and the International Society for Paediatric Oncology protocols recommend chest x-ray and CT imaging for lung metastases (Bhatnager et al, 2009). There is no recommendation for use of PET in these protocols.

A review on renal neoplasms in childhood in Radiology Clinics of North America (Geller and Kochan, 2011) states that current Central Oncology Group (COG) protocols call for the use of chest CT for documentation and follow-up of pulmonary metastases. The review makes no recommendation for use of PET in Wilm's tumor.

A GeneReviews review of Wilms tumor (Dome & Huff, 2011) states that: "Positron emission tomography (PET) is not a routine component of the initial evaluation of Wilms tumor, though most Wilms tumors take up the radiotracer fluoro-deoxyglucose. PET may play a role in the detection of occult metastatic sites at recurrence." This GeneReviews article provides one reference in support of the use of PET for detection of occult metastases (Moinul Hossain,et al, 2010) of 27 patients with Wilm's tumor, reporting that there were 34 positive scans, of which 8 were in lungs. The Moinus Hossain article noted, however, that lung lesions less than 10 mm were not consistently visualized on PET scans. The study was done in persons with known Wilms tumor, and did not report whether the PET scans were more accurate than other imaging modalities.

Current NCCN guidelines on "Kidney cancer" (Version 3.2025) make no recommendation for use of PET. The guidelines state that the value of PET in kidney cancer "remains to be determined" and that "PET alone is not a tool that is standardly used to diagnose kidney cancer or follow for evidence of relapse after nephrectomy." Although NCCN guidelines address kidney cancer, they do not have specific recommendations on Wilms tumors.

PET-Probe Guided Surgery

PET-probe guided (assisted) surgery is used for intraoperative localization of PET-positive recurrent/metastatic lesions. The surgery utilizes a hand-held PET probe, essentially is a high energy gamma probe designed to process the 511 keV photons of PET tracers, to localize areas of uptake and guide excision. There is no clinical evidence to support the use of PET-probe guided surgical resection for recurrent ovarian cancer.

Pilar Tumor

Siddha et al (2007) stated that pilar tumor is a rare neoplasm arising from the external root sheath of the hair follicle and is most commonly observed on the scalp. These tumors are largely benign, often cystic, and are characterized by trichilemmal keratinization. Wide local excision has been the standard treatment. Recent reports have described a rare malignant variant with an aggressive clinical course and a propensity for nodal and distant metastases which, therefore, merits aggressive treatment.

Khachemoune et al (2011) stated that a proliferating pilar tumor is a rare neoplasm arising from the isthmus region of the outer root sheath of the hair follicle. It is also commonly called a proliferating trichilemmal cyst. It was first described by Wilson-Jones as a proliferating epidermoid cyst in 1966. Proliferating pilar tumor was then distinguished from proliferating epidermoid cysts in 1995. It occurs most commonly on the scalp in women older than 50 years. Most tumors arise within a pre-existing pilar cyst. Even though they usually are benign in nature, malignant transformation with local invasion and metastasis has been described. A tentative stratification of proliferating pilar tumors into groups as benign, low-grade malignancy, and high-grade malignancy has been introduced. They may be inherited in an autosomal-dominant mode, linked to chromosome 3. Imaging studies are not usually indicated, but they may show a lobulated cystic mass, coarse calcification, or ring-like mineralization. Because some subcutaneous tumors located in the midline of the body may have connections to the central nervous system (e.g., scalp cavernous angioma, which may be part of the symptom complex known as sinus pericranii), imaging tumors in this location with CT or MRI prior to removal should be considered. The best modality to determine bony invasion or erosion is CT scanning, and proliferating pilar tumors are

frequently found as incidental subcutaneous nodules on brain CT scans. They most frequently display iso-intensity on T1-weighted images and heterogeneous signal on T2-weighted images. However, for deeper tissue invasion, MRI is best.

Pancreatic Cancer

Currently, there is insufficient evidence to support PET scan for restaging of pancreatic cancer.

Topkan et al (2013) examined the impact of [(18)F]fluorodeoxyglucose-PET/computed tomography (CT) restaging on management decisions and outcomes in patients with locally advanced pancreatic carcinoma (LAPC) scheduled for concurrent chemoradiotherapy (CRT). A total of 71 consecutive patients with conventionally staged LAPC were restaged with PET/CT before CRT, and were categorized into non-metastatic (M0) and metastatic (M1) groups. M0 patients received 50.4 Gy CRT with 5-fluorouracil followed by maintenance gemcitabine, whereas M1 patients received chemotherapy immediately or after palliative radiotherapy. In 19 patients (26.8 %), PET/CT restaging showed distant metastases not detected by conventional staging. PET/CT restaging of M0 patients showed additional regional lymph nodes in 3 patients and tumors larger than CT-defined borders in 4. PET/CT therefore altered or revised initial management decisions in 26 (36.6 %) patients. At median follow-up times of 11.3, 14.5, and 6.2 months for the entire cohort and the M0 and M1 cohorts, respectively, median overall survival was 16.1, 11.4, and 6.2 months, respectively; median loco-regional progression-free survival was 9.9, 7.8, and 3.4 months, respectively; and median progression-free survival was 7.4, 5.1, and 2.5 months, respectively ($p < 0.05$ each). The authors concluded that these findings suggested that PET/CT-based restaging may help select patients suitable for CRT, sparing those with metastases from futile radical protocols, and increasing the accuracy of estimated survival. (This was a small study examining the use of PET/CT for restaging in loco-regional pancreatic cancer; and the findings were preliminary)

Javery et al (2013) evaluated the impact of FDG-PET or PET/CT (PI) on pancreatic cancer management when added to CT or MRI (CDI). A total of 49 patients underwent 79 PI examinations. Discordant findings on PI and CDI were assessed for clinical impact. Overall, 15 of 79 PI-CDI pairs were discordant; 10 of 79 PI favorably; and 5 of 79

unfavorably altered management. PI favorably altered management more often when ordered for therapy monitoring compared to staging [risk ratio 13.00 (95 % confidence interval [CI]: 1.77 to 95.30)] or restaging [risk ratio 18.5 (95 % CI: 2.50 to 137.22)]. The authors concluded that PI favorably alters management more often when used for therapy monitoring compared to staging or restaging.

Furthermore, a UpToDate review on "Clinical manifestations, diagnosis, and staging of exocrine pancreatic cancer" (Fernandez-del Castillo, 2013) states that "The utility of PET scans in the diagnostic and staging evaluation of suspected pancreatic cancer, particularly whether PET provides information beyond that obtained by contrast-enhanced MDCT, remains uncertain …. Taken together, the data are insufficient to conclude that PET or integrated PET/CT provides useful information above that provided by contrast-enhanced CT. Consensus-based guidelines for staging of pancreatic cancer from the NCCN state that the role of PET/CT remains unclear. Definitive assessment of the role of PET as a component of the diagnostic and/or staging evaluation awaits a large prospective study designed to assess the benefit of PET (preferably integrated PET with a contrast-enhanced CT) in patients with a negative or indeterminate CT scan, with a prospectively designed cost effectiveness analysis".

The American Society of Clinical Oncology (ASCO)'s clinical practice guideline on "Locally advanced, unresectable pancreatic cancer" (Balaban et al, 2016) stated that "The routine use of positron emission tomography imaging for the management of locally advanced, unresectable pancreatic cancer is not recommended".

The National Comprehensive Cancer Network (NCCN) on "Pancreatic adenocarcinoma" (Version 2.2018) states that "PET/CT may be considered after formal pancreatic CT protocol in high-risk patients to detect extra pancreatic metastasis. It is not a substitute for high-quality, contrast-enhanced CT". Furthermore, the "role of PET/CT (without iodinated intravenous contrast) remains unclear".

Adult Onset Still's Disease

Funauchi et al (2008) reported the case of a 35-year old woman was admitted to the authors' hospital because of high fever and skin rash, and subsequently diagnosed as having adult onset Still's disease (AOSD). Because of resistance to the steroid hormones, high levels of

the serum-soluble form of the interleukin-2 receptor and splenomegaly, these researchers suspected a possible diagnosis of malignant lymphoma and performed PET, which disclosed an intense accumulation of 2-deoxy-2 [F18] FDG in the liver and spleen. However, bone marrow aspiration and liver biopsy did not reveal any malignant cells. After the treatment of high-dose adreno-corticosteroids and plasma exchange, her symptoms and laboratory data, including PET findings, gradually improved. The authors concluded that this was a rare case of severe AOSD in which an intense accumulation of FDG was detected by PET, and a differential diagnosis from malignant lymphoma may be difficult by FDG-PET alone, so that careful evaluation by techniques including histopathological examination may be necessary.

Assessment of Splenic Lesions

An UpToDate review on "Approach to the adult patient with splenomegaly and other splenic disorders" (Landaw and Schrier, 2014) states that "A variety of imaging techniques are available for assessment of splenic lesions (e.g., splenic cysts, other space-occupying lesions), including CT scanning, magnetic resonance imaging, ultrasound, Tc-99m sulfur colloid scintigraphy, and 18F-FDG PET. Although the age of the patient, clinical symptomatology, and imaging characteristics might help the radiologist arrive at the correct diagnosis, one study has concluded that PET scanning offered no additional information over that obtained using CT scanning alone, and that a history of prior malignancy was the only independent predictor for a splenic lesion being malignant (odds ratio 6.3; 95 % CI 2.3-17)".

X-Linked Adrenoleukodystrophy (X-ALD)

Salsano et al (2014) investigated the cerebral glucose metabolism in subjects with X-linked adrenoleukodystrophy (X-ALD) by using brain FDG-PET. This was a cross-sectional study in which 12 adults with various forms of X-ALD underwent clinical evaluation and brain MRI, followed by brain FDG-PET, neuropsychological assessment, and personality and psychopathology evaluation using the Symptom Checkist-90-Revised (SCL-90-R) and the Millon Clinical Multiaxial Inventory-III (MCMI-III). When compared to healthy control subjects (n = 27) by using Statistical Parametric Mapping 8 software, the patients with X-ALD-with or without brain MRI changes-showed a pattern of increased glucose metabolism in frontal lobes and reduced glucose

metabolism in cerebellum and temporal lobe areas. On single case analysis by Scenium software, these researchers found a similar pattern, with significant ($p < 0.02$) correlation between the degree of hyper-metabolism in the frontal lobes of each patient and the corresponding X-ALD clinical scores. With respect to personality, these investigators found that patients with X-ALD usually present with an obsessive-compulsive personality disorder on the MCMI-III, with significant ($p < 0.05$) correlation between glucose uptake in ventral striatum and severity of score on the obsessive-compulsive subscale. The authors concluded that they examined cerebral glucose metabolism using FDG-PET in a cohort of patients with X-ALD and provided definite evidence that in X-ALD the analysis of brain glucose metabolism reveals abnormalities independent from morphologic and signal changes detected by MRI and related to clinical severity. They stated that brain FDG-PET may be a useful neuroimaging technique for the characterization of X-ALD and possibly other leukodystrophies.

The drawbacks of this study were: (i) the number of patients was limited because of the rarity of X-ALD, (ii) patients were not randomized, and (iii) the use of some drugs (e.g., corticosteroids, baclofen and valproic acid) by some symptomatic patients; these drugs might influence FDG-PET data. Furthermore, these investigators stated that the findings of this study lay the foundations of larger studies that might assess whether the abnormal brain glucose metabolism detected in X-ALD can be used as a surrogate marker.

Melanoma

The National Comprehensive Cancer Network (NCCN) on "Cutaneous melanoma" (Version 1.2019) states that "routine cross-sectional imaging (CT, PET/CT, or MRI) is not recommended for patients with stage 0, I, and II disease." However, "in patients with stage III disease, PET/CT scan may be more useful. In particular, PET/CT scans can help to further characterize lesions found to be indeterminate on CT scan, and can image areas of the body not studied by the routine body CT scan." Thus, the NCCN panel provides a category 2B recommendation for cross-sectional imaging at baseline for staging of stage III sentinel node positive disease, or a category 2A recommendation to assess specific signs or symptoms. The NCCN panel "encourage baseline chest/abdominal/pelvic CT with or without PET/CT in patients with stage IV melanoma".

Castleman Disease

A review of PET in HIV-associated multi-centric Castleman disease (Rossotti et al, 2012) concluded "So far, FDG-PET/CT use for diagnosing Castleman disease has been reported in only a small number of patients. Data defining sensitivity and specificity of FDG-PET/CT for Castleman disease diagnosis are lacking". Guidelines from the European Association of Nuclear Medicine and the Society for Nuclear Medicine and Molecular Imaging on PET in inflammation and infection (Jamar et al, 2013) stated that Castleman disease is one of several "well-described applications, but without sufficient evidence-based indication" for PET. Furthermore, current British guidelines for HIV-associated malignancies (Bower et al, 2014) states that regarding Castleman disease, "The role of functional imaging such as FDG-PET scans is uncertain; although a small study indicated that in individuals with active MCD, FDG-PET scans more frequently detected abnormal uptake than CT". Guidelines from the National Comprehensive Cancer Network "B-cell lymphomas" (NCCN, 2018) recommend FDG-PET in the workup of patients with Castleman disease.

Gallium Ga 68 Dotatate PET (NetSpot) for Neuroendocrine Tumors

The FDA labeling that states that NETSPOT "is a radioactive diagnostic agent indicated for use with PET for localization of somatostatin receptor positive neuroendocrine tumors (NETs) in adult and pediatric patients." The labeling describes the three open label single center studies of NETSPOT that were submitted to the FDA. One study of subjects with neuroendocrine tumors (n = 97) reported on the correlations between the NETSPOT to CT, MRI, and SPECT images obtained within the previous three years. A second study of subjects with suspected neuroendocrine tumors (n = 104) compared NETSPOT to histopathology or clinical follow-up. A third study evaluated subjects (n = 63) for neuroendocrine tumor recurrence using histopathology or clinical follow-up as a reference standard.

Consensus guidelines from the National Comprehensive Cancer Network (NCCN, 2018) states: "Several studies have also shown diagnostic utility, as well as high sensitivity, of PET/CT imaging using the radiolabeled somatostatin analog gallium-68 (68Ga) dotatate.

Unless otherwise indicated, somatostatin receptor-based imaging in this discussion includes imaging with either somatostatin receptor scintigraphy or 68-Ga dotatate PET/CT."

A systematic evidence review of somatostatin imaging for neuroendocrine tumors (Chou, et al., 2017) found somatostatin-receptor PET was associated with greater sensitivity than OctreoScan and FDG-PET, and also higher sensitivity for identification of primary neuroendocrine tumors than CT/MRI. The review found, however, that, for metastatic lesions, three studies reported inconsistent findings, with some studies finding no clear differences between somatostatin-receptor PET and MRI, or MRI associated with slightly higher sensitivity. The review also concluded that most studies reported no clear differences in specificity between somatostatin-receptor PET and alternative imaging modalities.

The review found that "[e]vidence on how comparative diagnostic accuracy varies according to tumor characteristics is limited" (Chou et al, 2017). Most studies appeared to evaluate accuracy for diagnosis of more well-differentiated/indolent neuroendocrine tumors, though details about tumor grade and type were relatively limited.

The review noted that, although studies found that somatostatin receptor PET was associated with changes in management in a substantial proportion of patients, the studies had important methodological limitations. In particular, the studies did not pre-define "changes in management" or report use of standardized protocols to guide management decisions in response to somatostatin-receptor PET imaging findings, and no study included a comparison group of patients who underwent somatostatin-receptor PET without alternative imaging. The review also found that no study was designed to assess clinical outcomes associated with use of somatostatin-receptor PET.

The investigators noted that limitations of this review include the relatively small number of studies available for specific imaging comparisons and types of neuroendocrine tumors, the lack of evidence on how patient and tumor characteristics impact diagnostic accuracy, and methodological limitations in the studies, including suboptimal and heterogeneous reference standards and use of a case-control design in a number of studies (Chou, et al., 2017). Most studies reported results based on per-lesion analyses, which may not be as clinically

relevant as per-patient analyses. Per-lesion analyses may also result in higher precision of estimates than warranted, since one patient may have many lesions. Some studies were not designed to or failed to report specificity, providing incomplete information regarding diagnostic accuracy.

Copper Cu-64 Dotatate PET (DetectNet) for Neuroendocrine Tumors

The use of a PET-emitter with a longer half-life such as copper-64 (T1/2 = 12.7 hours), such as Cu-64 dotatate (DetectNet) is being investigated as an alternative to Ga-68 DOTATE for neuroendocrine tumors, as the latter requires a generator because of its very short half life. A review (Eychenne et al, 2020) concluded that despite a higher dosimetric impact for copper-64 (only 17.6% of radioactive decay lead to positron emission), copper-64 somatostatin analogs appear to be an advantageous alternative to gallium-68 radiopharmaceuticals. Compared to gallium-68, in addition to economic advantages, copper-64 has a lower positron range which leads to a better PET intrinsic resolution and a higher half-life which allows for a more flexible scanning window. The authors concluded, however, that "[t]he better patient care management and outcomes remain to be proven and the work is in progress to establish these points" [citing Carlsen et al, 2020; Delpassand et al, 2020].

Giant Cell Tumor of the Bone

Muheremu and Niu (2015) examined PET/CT and its applications for the diagnosis and treatment of bone tumors. The advantages and disadvantages of PET/CT were also evaluated and compared with other imaging methods and the prospects of PET/CT were discussed. The PubMed, Medline, Elsevier, Wanfang and China International Knowledge Infrastructure databases were searched for studies published between 1995 and 2013, using the terms "PET/CT", "positron emission tomography", "bone tumor", "osteosarcoma", "giant cell bone tumor" and "Ewing sarcoma". All the relevant information was extracted and analyzed. A total of 73 studies were selected for the final analysis. The extracted information indicated that at present, PET/CT is the imaging method that exhibits the highest sensitivity, specificity and accuracy. The authors concluded that although

difficulties and problems remain to be solved, PET/CT is a promising non-invasive method for the diagnostic evaluation of and clinical guidance for bone tumors.

An UpToDate review on "Giant cell tumor of bone" (Thomas and Desai, 2015) states that "There are limited data regarding the utility of 18F-FDG-PET for newly diagnosed GCTB. Unlike many benign bone tumors, GCTB accumulate the FDG tracer, presumably because the osteoclast-like giant cells are intensely metabolically active. However, whether there are any advantages to evaluation with FDG PET as compared to conventional imaging with CT, MRI, and bone scan is unclear".

Furthermore, NCCN's clinical practice guideline on "Bone cancer" (Version 1.2019) does not mention PET imaging as a management tool for giant cell tumor of the bone.

NaF-18 PET for Bone Metastasis

According to CMS (2010), there is insufficient evidence to determine that the results of NaF-18 PET imaging to identify bone metastases improve health outcomes of beneficiaries with cancer. Thus, the CMS decided that this use is not reasonable and necessary under §1862(a)(1)(A) of the Social Security Act.

Plasmacytoid Dendritic Cell Neoplasm

Kharfan-Dabaja et al (2013) stated that blastic plasmacytoid dendritic cell neoplasm (BPDCN) is an exceedingly rare disorder categorized under acute myeloid leukemia by the World Health Organization (WHO). Phenotypically, malignant cells co-express CD4(+) and CD56(+) without co-expressing common lymphoid or myeloid lineage markers. BPDCN frequently expresses CD123, TCL1, BDCA-2, and CD2AP. Restriction of CD2AP expression to plasmacytoid dendritic cells makes it a useful tool to help confirm diagnosis. Clonal complex chromosome aberrations are described in 2/3 of cases; 80 % of BPDCN cases present with non-specific dermatological manifestations, prompting inclusion in the differential diagnosis of atypical skin rashes refractory to standard treatment. Prognosis is poor, with a median survival of less than 18 months. No prospective randomized data exist to define the most optimal frontline chemotherapy. Current practice considers acute myeloid leukemia-like or acute lymphoblastic

leukemia-like regimens acceptable for induction treatment. Unfortunately, responses are short-lived, with second remissions difficult to achieve, underscoring the need to consider hematopoietic cell transplantation early in the disease course. Allografting, especially if offered in first remission, can result in long-term remissions. The authors concluded that pre-clinical data suggested a potential role for immunomodulatory agents in BPCDN. However, further research efforts are needed to better understand BPDCN biology and to establish evidence-based treatment algorithms that might ultimately improve overall prognosis of this disease.

Also, an UpToDate review on "Blastic plasmacytoid dendritic cell neoplasm" (Gurbuxani, 2015) does not mention PET scan as a management tool.

Squamous Cell Carcinoma (e.g., Cutaneous (SCC) and Vaginal Cancer)

An UpToDate review on "Clinical features and diagnosis of cutaneous squamous cell carcinoma (SCC)" (Lim and Asgari, 2012) does not mention the use of PET.

Bentivegna et al (2011) stated that FDG-PET/CT is commonly used for advanced stage cervical cancer but its efficiency is discussed in early stage. These researchers evaluated false negative rate of FDG-PET/CT in early-stage cervical and vaginal cancer. Patients treated between 2005 and 2008 for stage IB1 cervical cancer and stage I vaginal cancer and who underwent a FDG-PET/CT followed by a pelvic lymphadenectomy were studied. A total of 18 patients were included with bilateral pelvic lymphadenectomy (16 cervical cancers, 2 vaginal cancers). The median age of patients was 41 years. Radical hysterectomy was performed for 16 patients, by a laparoscopic approach in 15 cases and by a laparotomic approach in 1 case. One patient had a simple hysterectomy and 1 had exclusive radiotherapy. No patient had pelvic or para-aortic fixation on FDG-PET/CT; 3 patients have proven pelvic involvement and 1 had para-aortic metastases. The false-negative rate and negative predictive value of FDG-PET/CT were 17 % and 83 %, respectively. The authors concluded that the accuracy of FDG-PET/CT imaging in predicting the pelvic nodal status is very low in patients with early-stage cervical and vaginal cancer and is not able to replace surgical exploration.

An UpToDate review on "Vaginal cancer" (Karam et al, 2015) states that "Imaging studies – The only imaging studies that are part of the International Federation of Gynecology and Obstetrics (FIGO) staging for vaginal cancer are chest and skeletal radiography . However, advanced imaging, such as computed tomography (CT), magnetic resonance imaging (MRI) and 18-fluoro-2-deoxyglucose-positron emission tomography and CT (FDG-PET/CT) can be helpful for treatment planning. MRI can assist in determining the primary vaginal tumor size and local extent. Vaginal tumors generally are best seen on T2 imaging, and instilling gel into the vaginal canal, which distends the vaginal walls, often aids in visualizing and assessing the thickness of the vaginal tumor. FDG-PET can also be helpful for evaluating the primary vaginal tumor and abnormal lymph nodes …. Routine use of imaging studies was not recommended. Computed tomography (CT) and/or positron emission tomography (PET) should be performed ONLY if recurrence is suspected".

The National Comprehensive Cancer Network (NCCN) Imaging Appropriate Use Criteria (2018) for "Squamous cell skin cancer" includes a category 2A recommendation for the option of PET/CT for staging. PET/CT option for palpable regional lymph nodes or abnormal lymph nodes identified by imaging studies; FNA or core biopsy positive indications. For imaging to determine size, number, and location of nodes and to rule out distant disease. PET/CT of the nodal basin can be useful for RT planning.

Alzheimer's Disease

The Centers for Medicare & Medicaid Services (CMS) released a decision memorandum on PET for suspected dementia. Although CMS announced limited coverage of PET to distinguish Alzheimer's disease from fronto-temporal dementia when the distinction is uncertain and other criteria are met, the decision memorandum recognized that there is no available literature that directly evaluated the impact on patient outcomes of adding PET in patients with early dementia who have undergone standard evaluation who do not meet the criteria for Alzheimer disease due to variations in the onset, presentation, or clinical course (suggesting other neurodegenerative causes for the disorder such as fronto-temporal dementia). In addition, CMS found no trials that examined the impact of PET in changing the management as a surrogate for evaluating PET impact on health outcomes in patients with this sort of difficult differential diagnosis. The assessment also

recognized that there are no established cures for either Alzheimer disease or fronto-temporal dementia. A paucity of medications are available for Alzheimer's disease, which have a limited ability to decrease the rate of cognitive decline when administered early in the course of the disease. CMS coverage determination was primarily based on the value of PET in providing information useful in "non-medical decision-making." Aetna, however, does not consider non-medical decision-making a medically necessary indication for testing. Because of a lack of adequate evidence that PET scanning alters clinical management of such persons such that clinical outcomes are improved, Aetna considers PET scanning for differentiating Alzheimer disease from fronto-temporal dementia experimental, investigational, or unproven

An assessment prepared for the California Technology Assessment Forum (CTAF) concluded that PET for Alzheimer's disease does not meet CTAF's criteria (Feldman, 2004). The assessment stated: "The critical question that remains unanswered by this and the other studies of PET in the evaluation of AD/dementia is: To what extent does PET improve diagnostic accuracy beyond what can be obtained with a thorough clinical evaluation? Given that the sensitivity of clinical criteria are reported to be about 80 % to 90 %, it is difficult for any diagnostic test to significantly improve diagnostic accuracy. And given the fact that treatment of the most common non-AD dementias (e.g., Dementia of Lewy Bodies or vascular dementias) with cholinesterase inhibitor drugs is not likely to be harmful and in fact may be beneficial to these patients, it may be that an empirical approach of ruling out reversible causes of dementia and treating all others with cholinesterase inhibitor drugs is appropriate and cost effective."

The assessment noted that the greatest promise of PET in Alzheimer disease is likely to be in improving a clinician's ability to identify at-risk patients and to offer them treatment before they are significantly affected by Alzheimer disease. Few studies, however, have enrolled patients with mild symptoms or mild cognitive impairment so it is unclear what role PET is destined to play in identifying this subgroup of patients most likely to benefit from current and emerging therapies for Alzheimer's disease.

Clark et al (2011) examined if florbetapir F 18 PET imaging performed during life accurately predicts the presence of β-amyloid in the brain at autopsy. Prospective clinical evaluation conducted February 2009 through March 2010 of florbetapir-PET imaging performed on 35 patients from hospice, long-term care, and community health care facilities near the end of their lives (6 patients to establish the protocol and 29 to validate) compared with immunohistochemistry and silver stain measures of brain β-amyloid after their death used as the reference standard. PET images were also obtained in 74 young individuals (18 to 50 years) presumed free of brain amyloid to better understand the frequency of a false-positive interpretation of a florbetapir-PET image. Major outcome measures were correlation of florbetapir-PET image interpretation (based on the median of 3 nuclear medicine physicians' ratings) and semi-automated quantification of cortical retention with post-mortem β-amyloid burden, neuritic amyloid plaque density, and neuropathological diagnosis of Alzheimer disease in the first 35 participants autopsied (out of 152 individuals enrolled in the PET pathological correlation study). Florbetapir-PET imaging was performed a mean of 99 days (range of 1 to 377 days) before death for the 29 individuals in the primary analysis cohort. Fifteen of the 29 individuals (51.7 %) met pathological criteria for Alzheimer disease. Both visual interpretation of the florbetapir-PET images and mean quantitative estimates of cortical uptake were correlated with presence and quantity of β-amyloid pathology at autopsy as measured by immunohistochemistry (Bonferroni ρ, 0.78 [95 % confidence interval, 0.58 to 0.89]; $p < 0.001$]) and silver stain neuritic plaque score (Bonferroni ρ, 0.71 [95 % confidence interval [CI]: 0.47 to 0.86]; $p < 0.001$). Florbetapir-PET images and postmortem results rated as positive or negative for β-amyloid agreed in 96 % of the 29 individuals in the primary analysis cohort. The florbetapir-PET image was rated as amyloid negative in the 74 younger individuals in the non-autopsy cohort. The authors concluded that florbetapir-PET imaging was correlated with the presence and density of β-amyloid. These data provided evidence that a molecular imaging procedure can identify β-amyloid pathology in the brains of individuals during life. Moreover, they stated that additional studies are needed to understand the appropriate use of florbetapir-PET imaging in the clinical diagnosis of Alzheimer disease and for the prediction of progression to dementia.

In an editorial that accompanied the afore-mentioned study, Breteler (2011) stated that "[o]nly through future studies using rigorous study design can the role of either florbetapir-PET imaging or plasma β-amyloid 42/40 in diagnosis or prediction of AD be determined".

Yeo et al (2015) noted that amyloid imaging using fluorine 18-labeled tracers florbetapir, florbetaben, and flutemetamol has recently been reported in Alzheimer's disease (AD). These researchers systematically searched Medline and Embase for relevant studies published from January 1980 to March 2014. Studies comparing imaging findings in AD and normal controls (NCs) were pooled in a meta-analysis, calculating pooled weighted sensitivity, specificity, and diagnostic odds ratio (OR) using the DerSimonian-Laird random-effects model. A total of 19 studies, investigating 682 patients with AD, met inclusion criteria. Meta-analysis demonstrated a sensitivity of 89.6 %, a specificity of 87.2 %, and an OR of 91.7 for florbetapir in differentiating AD patients from NCs, and a sensitivity of 89.3 %, a specificity of 87.6 %, and a diagnostic OR of 69.9 for florbetaben. There were insufficient data to complete analyses for flutemetamol. The authors concluded that these findings suggested favorable sensitivity and specificity of amyloid imaging with fluorine 18-labeled tracers in AD. They stated that prospective studies are needed to determine optimal imaging analysis methods and resolve outstanding clinical uncertainties.

In a Cochrane review, Smailagic and colleagues (2015) determined the diagnostic accuracy of the ^{18}F-FDG PET index test for detecting people with mild cognitive impairment (MCI) at baseline who would clinically convert to AD dementia or other forms of dementia at follow-up. These investigators searched the Cochrane Register of Diagnostic Test Accuracy Studies, Medline, Embase, Science Citation Index, PsycINFO, BIOSIS previews, LILACS, MEDION, (Meta-analyses van Diagnostisch Onderzoek), DARE (Database of Abstracts of Reviews of Effects), HTA (Health Technology Assessment Database), ARIF (Aggressive Research Intelligence Facility) and C-EBLM (International Federation of Clinical Chemistry and Laboratory Medicine Committee for Evidence-based Laboratory Medicine) databases to January 2013. They checked the reference lists of any relevant studies and systematic reviews for additional studies. The authors included studies that evaluated the diagnostic accuracy of ^{18}F-FDG PET to determine the conversion from MCI to AD dementia or to other forms of dementia,

i.e., any or all of vascular dementia, dementia with Lewy bodies, and fronto-temporal dementia. These studies necessarily employ delayed verification of conversion to dementia and are sometimes labelled as 'delayed verification cross-sectional studies'. Two blinded review authors independently extracted data, resolving disagreement by discussion, with the option to involve a third review author as arbiter if necessary. They extracted and summarized graphically the data for 2-by-2 tables. They conducted exploratory analyses by plotting estimates of sensitivity and specificity from each study on forest plots and in receiver operating characteristic (ROC) space. When studies had mixed thresholds, they derived estimates of sensitivity and likelihood ratios at fixed values (lower quartile, median and upper quartile) of specificity from the hierarchical summary ROC (HSROC) models. These researches included 14 studies (421 participants) in the analysis. The sensitivities for conversion from MCI to AD dementia were between 25 % and 100 % while the specificities were between 15 % and 100 %. From the summary ROC curve we fitted we estimated that the sensitivity was 76 % (95 % confidence interval (CI): 53.8 to 89.7) at the included study median specificity of 82 %. This equated to a positive likelihood ratio of 4.03 (95 % CI: 2.97 to 5.47), and a negative likelihood ratio of 0.34 (95 % CI: 0.15 to 0.75). Three studies recruited participants from the same Alzheimer's Disease Neuroimaging Initiative (ADNI) cohort but only the largest ADNI study (Herholz 2011) was included in the meta-analysis. In order to demonstrate whether the choice of ADNI study or discriminating brain region (Chetelat 2003) or reader assessment (Pardo 2010) made a difference to the pooled estimate, the authors performed 5 additional analyses. At the median specificity of 82 %, the estimated sensitivity was between 74 % and 76 %. There was no impact on their findings. In addition to evaluating AD dementia, 5 studies evaluated the accuracy of ^{18}F-FDG PET for all types of dementia. The sensitivities were between 46 % and 95 % while the specificities were between 29 % and 100 %; however, they did not conduct a meta-analysis because of too few studies, and those studies which we had found recruited small numbers of subjects. These findings were based on studies with poor reporting, and the majority of included studies had an unclear risk of bias, mainly for the reference standard and participant selection domains. According to the assessment of Index test domain, more than 50 % of studies were of poor methodological quality. The authors concluded that it was difficult to determine to what extent the findings from the meta-analysis can be applied to clinical practice. Given the

considerable variability of specificity values and lack of defined thresholds for determination of test positivity in the included studies, the current evidence does not support the routine use of ^{18}F-FDG PET scans in clinical practice in people with MCI. The ^{18}F-FDG PET scan is a high-cost investigation, and it is therefore important to clearly demonstrate its accuracy and to standardize the process of ^{18}F-FDG PET diagnostic modality prior to its being widely used. They stated that future studies with more uniform approaches to thresholds, analysis and study conduct may provide a more homogeneous estimate than the one available from the included studies we have identified.

Morris et al (2016) stated that imaging or tissue biomarker evidence has been introduced into the core diagnostic pathway for AD; and PET using (18)F-labelled beta-amyloid PET tracers has shown promise for the early diagnosis of AD. However, most studies included only small numbers of participants and no consensus has been reached as to which radiotracer has the highest diagnostic accuracy. First, these researchers performed a systematic review of the literature published between 1990 and 2014 for studies exploring the diagnostic accuracy of florbetaben, florbetapir and flutemetamol in AD. The included studies were analyzed using the QUADAS assessment of methodological quality. A meta-analysis of the sensitivity and specificity reported within each study was performed. Pooled values were calculated for each radiotracer and for visual or quantitative analysis by population included. The systematic review identified 9 studies eligible for inclusion. There were limited variations in the methods between studies reporting the same radiotracer. The meta-analysis results showed that pooled sensitivity and specificity values were in general high for all tracers. This was confirmed by calculating likelihood ratios. A patient with a positive ratio is much more likely to have AD than a patient with a negative ratio, and vice versa. However, specificity was higher when only patients with AD were compared with healthy controls. This systematic review and meta-analysis found no marked differences in the diagnostic accuracy of the 3 beta-amyloid radiotracers. All tracers performed better when used to discriminate between patients with AD and healthy controls. The sensitivity and specificity for quantitative and visual analysis were comparable to those of other imaging or biomarker techniques used to diagnose AD. The authors concluded that further research is needed to identify the

combination of tests that provides the highest sensitivity and specificity, and to identify the most suitable position for the tracer in the clinical pathway.

Tiepolt et al (2016) investigated the value of early PET images using the novel Aβ tracer [(18)F]FBB in the diagnosis of AD. This retrospective analysis included 22 patients with MCI or dementia who underwent dual time-point PET imaging with either [(11)C]PiB (11 patients) or [(18)F]FBB (11 patients) in routine clinical practice. Images were acquired 1 to 9 mins after administration of both tracers and 40 to 70 mins and 90 to 110 mins after administration of [(11)C]PiB and [(18)F]FBB, respectively. The patients also underwent [(18)F]FDG brain PET imaging; PET data were analyzed visually and semi-quantitatively. Associations between early Aβ tracer uptake and dementia as well as brain atrophy were investigated. Regional visual scores of early Aβ tracer and [(18)F]FDG PET images were significantly correlated (Spearman's $\rho = 0.780$, $p < 0.001$). Global brain visual analysis revealed identical results between early Aβ tracer and [(18)F]FDG PET images. In a VOI-based analysis, the early Aβ tracer data correlated significantly with the [(18)F]FDG data ($r = 0.779$, $p < 0.001$), but there were no differences between [(18)F]FBB and [(11)C]PiB. Cortical SUVRs in regions typically affected in AD on early Aβ tracer and [(18)F]FDG PET images were correlated with MMSE scores ($\rho = 0.458$, $p = 0.032$, and $\rho = 0.456$, $p = 0.033$, respectively). A voxel-wise group-based search for areas with relatively higher tracer uptake on early Aβ tracer PET images compared with [(18)F]FDG PET images revealed a small cluster in the midbrain/pons; no significant clusters were found for the opposite comparison. The authors concluded that early [(18)F]FBB and [(11)C]PiB PET brain images were similar to [(18)F]FDG PET images in AD patients, and these tracers could potentially be used as biomarkers in place of [(18)F]FDG. Thus, Aβ tracer PET imaging has the potential to provide biomarker information on AD pathology and neuronal injury. They stated that the potential of this approach for supporting the diagnosis of AD needs to be confirmed in prospective studies in larger cohorts.

Stefaniak and O'Brien (2016) stated that management strategies for dementia are still very limited, and establishing effective disease modifying therapies based on amyloid or tau remains elusive. Neuroinflammation has been increasingly implicated as a pathological mechanism in dementia and demonstration that it is a key event

accelerating cognitive or functional decline would inform novel therapeutic approaches, and may aid diagnosis. Much research has therefore been done to develop technology capable of imaging neuroinflammation in-vivo. The authors performed a systematic search of the literature and found 28 studies that used in- vivo neuroimaging of 1 or more markers of neuroinflammation on human patients with dementia. The majority of the studies used PET imaging of the 18kDa translocator protein (TSPO) microglial marker and found increased neuroinflammation in at least 1 neuroanatomical region in dementia patients, most usually AD, relative to controls, but the published evidence to date does not indicate whether the regional distribution of neuroinflammation differs between dementia types or even whether it is reproducible within a single dementia type between individuals. It is less clear that neuroinflammation is increased relative to controls in MCI than it is for dementia, and therefore it is unclear whether neuroinflammation is part of the pathogenesis in early stages of dementia. The authors concluded that despite its great potential, this review demonstrated that imaging of neuroinflammation has not thus far clearly established brain inflammation as an early pathological event. They stated that further studies are needed, including those of different dementia subtypes at early stages, and newer, more sensitive, PET imaging probes need to be developed.

Rodriguez-Vieitez et al (2017) stated that for amyloid PET tracers, the simplified reference tissue model derived ratio of influx rate in target relative to reference region (R1) has been shown to serve as a marker of brain perfusion, and, due to the strong coupling between perfusion and metabolism, as a proxy for glucose metabolism. In the present study, 11 prodromal AD and 9 AD dementia patients underwent [18F]THK5317, carbon-11 Pittsburgh Compound-B ([11C]PIB), and 2-deoxy-2-[18F]fluoro-D-glucose ([18F]FDG) PET to assess the possible use of early-phase [18F]THK5317 and R1 as proxies for brain perfusion, and thus, for glucose metabolism. Discriminative performance (prodromal versus AD dementia) of [18F]THK5317 (early-phase SUVr and R1) was compared with that of [11C]PIB (early-phase standard uptake value ratio [SUVr] and R1) and [18F]FDG. Strong positive correlations were found between [18F]THK5317 (early-phase, R1) and [18F]FDG, particularly in frontal and temporo-parietal regions. Differences in correlations between early-phase and R1 ([18F]THK5317 and [11C]PIB) and [18F]FDG, were not statistically significant, nor were differences in area under the curve values in the

discriminative analysis. The authors concluded that these findings suggested that early-phase [18F]THK5317 and R1 provide information on brain perfusion, closely related to glucose metabolism. As such, a single PET study with [18F]THK5317 may provide information about both tau pathology and brain perfusion in AD, with potential clinical applications.

Breast Cancer

An assessment by the Blue Cross and Blue Shield Association Technology Evaluation Center on PET for breast cancer (2003) stated that FDG-PET for evaluating breast cancer does not meet its criteria for initial staging of axillary lymph nodes, for detection of loco-regional recurrence or distant metastasis/recurrence, or for evaluating response to treatment.

Pritchard et al (2012) studied the use of 2-[(18)F] FDG PET in assessing lymph nodes and detecting distant metastases in women with primary breast cancer. Women diagnosed with operable breast cancer within 3 months underwent FDG-PET at 1 of 5 Ontario study centers followed by axillary lymph node assessment (ALNA) consisting of sentinel lymph node biopsy (SLNB) alone if sentinel lymph nodes (SLNs) were negative, SLNB with axillary lymph node dissection (ALND) if SLNB or PET was positive, or ALND alone if SLNs were not identified. Between January 2005 and March 2007, a total of 325 analyzable women entered this study. Sentinel nodes were found for 312 (96 %) of 325 women and were positive for tumor in 90 (29 %) of 312. ALND was positive in 7 additional women. Using ALNA as the gold standard, sensitivity for PET was 23.7 % (95 % CI: 15.9 % to 33.6 %), specificity was 99.6 % (95 % CI: 97.2 % to 99.9 %), positive-predictive value was 95.8 % (95 % CI: 76.9 % to 99.8 %), negative-predictive value was 75.4 % (95 % CI: 70.1 % to 80.1 %), and prevalence was 29.8 % (95 % CI: 25.0 % to 35.2 %). Using logistic regression, tumor size was predictive for prevalence of tumor in the axilla and for PET sensitivity. PET scan was suspicious for distant metastases in 13 patients; 3 (0.9 %) were confirmed as metastatic disease and 10 (3.0 %) were false-positive. The authors concluded that FDG-PET is not sufficiently sensitive to detect positive axillary lymph nodes, nor is it sufficiently specific to appropriately identify distant metastases. However, the very high positive-predictive value (96 %) suggests that PET when positive is indicative of disease in axillary nodes, which may influence surgical care.

Infectious Endocarditis

Yan and colleagues (2016) systematically conducted a meta-analysis of the available evidence for PET/CT using 18F-fluorodeoxyglucose as tracers in the imaging of infectious endocarditis (IE). Databases, including PubMed, Embase, and Web of Science, were searched for studies that examined the diagnostic value of 18F-FDG PET/CT for patients with IE. The reference lists following review articles and those of the included articles were checked to complement the electronic searches. The data extraction and methodological quality assessment were completed by 2 independent reviewers, and the meta-analysis was then conducted using Meta-Disc software, version 1.4. A total of 6 studies involving 246 patients was included. The results of the meta-analysis indicated that the pooled sensitivity was 61 % (95 % CI: 52 to 88 %), and the pooled specificity was 88 % (95 % CI: 80 to 93 %). The positive likelihood ratio was 3.24 (95 % CI: 1.67 to 6.28, p = 0.224), and the negative likelihood ratio was 0.50 (95 % CI: 0.32 to 0.77, p = 0.015). The diagnostic odds ratio (OR) was 6.98 (95 % CI: 2.55 to 19.10, p = 0.145), the overall area under the curve (AUC) was 0.8230 (SE = 0.1085), and the Q^* value was 0.7563 (SE = 0.0979). The authors concluded that 18F-FDG PET/CT is currently not sufficient for the diagnosis of IE because of its low sensitivity.

Diagnosis of Gallbladder Cancer

National Comprehensive Cancer Network's clinical practice guideline on "Hepatobiliary cancers"(Version 4.2018) states that PET/CT has limited sensitivity but high specificity in the detection of regional lymph node metastases in gallbladder cancer. PET/CT may be considered when there is an equivocal finding on CT/MRI. "The routine use of PET/CT in preoperative setting has not been established in prospective trials."

Interim FDG-PET for Prognosis in Follicular Lymphoma

Adams and colleagues (2016) systematically reviewed the prognostic value of interim and end-of-treatment FDG-PET in follicular lymphoma during and after first-line therapy. The PubMed/Medline database was searched for relevant original studies. Included studies were methodologically assessed, and their results were extracted and descriptively analyzed. A total of 3 studies on the prognostic value of interim FDG-PET and 8 studies on the prognostic value of end-of-

treatment FDG-PET were included. Overall, studies were of poor methodological quality. In addition, there was incomplete reporting of PFS and OS data by several studies, and none of the studies incorporated the Follicular Lymphoma International Prognostic Index (FLIPI) in the OS analyses. Two studies reported no significant difference in PFS between interim FDG-PET positive and negative patients, whereas 1 study reported a significant difference in PFS between the 2 groups. Two studies reported no significant difference in OS between interim FDG-PET positive and negative patients; 5 studies reported end-of-treatment FDG-PET positive patients to have a significantly worse PFS than end-of-treatment FDG-PET negative patients, and 1 study reported a non-significant trend towards a worse PFS for end-of-treatment FDG-PET positive patients. Three studies reported end-of-treatment FDG-PET positive patients to have a significantly worse OS than end-of-treatment FDG-PET negative patients. The authors concluded that the available evidence does not support the use of interim FDG-PET in follicular lymphoma. Although published studies suggested end-of-treatment FDG-PET to be predictive of PFS and OS, they suffered from numerous biases and failure to correct OS prediction for the FLIPI.

Mesothelioma

Positron emission tomography has limited sensitivity for mesothelioma. Furthermore, current guidelines have not incorporated PET scanning into the management of persons with mesothelioma. The available literature on the effect of PET on clinical outcomes of malignant mesothelioma are limited. In a small feasibility study, Francis and colleagues (2007) evaluated the role of serial (18)F-FDG PET in the assessment of response to chemotherapy in patients with mesothelioma. Patients were prospectively recruited and underwent both (18)F-FDG PET and conventional radiological response assessment before and after 1 cycle of chemotherapy. Quantitative volume-based (18)F-FDG PET analysis was performed to obtain the total glycolytic volume (TGV) of the tumor. Survival outcomes were measured. A total of 23 patients were suitable for both radiological and (18)F-FDG PET analysis, of whom 20 had CT measurable disease. After 1 cycle of chemotherapy, 7 patients attained a partial response and 13 had stable disease on CT assessment by modified RECIST (Response Evaluation Criteria in Solid Tumors) criteria. In the 7 patients with radiological partial response, the median TGV on quantitative PET analysis fell to 30 % of baseline (range of 11 % to 71

%). After 1 cycle of chemotherapy, Cox regression analysis demonstrated a statistically significant relationship between a fall in TGV and improved patient survival (p = 0.015). Neither a reduction in the maximum standardized uptake value (p = 0.097) nor CT (p = 0.131) demonstrated a statistically significant association with patient survival. The authors concluded that semi-quantitative (18)F-FDG PET using the volume-based parameter of TGV is feasible in mesothelioma and may predict response to chemotherapy and patient survival after 1 cycle of treatment. Therefore, metabolic imaging has the potential to improve the care of patients receiving chemotherapy for mesothelioma by the early identification of responding patients. This technology may also be useful in the assessment of new systemic treatments for mesothelioma.

Ceresoli et al (2007) noted that most patients with malignant pleural mesothelioma (MPM) are candidates for chemotherapy during the course of their disease. Assessment of the response with conventional criteria based on computed tomography (CT) measurements is challenging, due to the circumferential and axial pattern of growth of MPM. Such difficulties hinder an accurate evaluation of clinical study results and make the clinical management of patients critical. Several radiological response systems have been proposed, but neither WHO criteria nor the more recent RECIST uni-dimensional criteria nor hybrid uni- and bi-dimensional criteria seem to apply to tumor measurement in this disease. Recently, modified RECIST criteria for MPM have been published. Although they are already being used in current clinical trials, they have been criticized based on the high grade of inter-observer variability and on theoretical studies of mesothelioma growth according to non-spherical models. Computer-assisted techniques for CT measurement are being developed. The use of FDG-PET for prediction of response and, more importantly, of survival outcomes of MPM patients is promising and warrants validation in large prospective series. New serum markers such as osteopontin and mesothelin-related proteins are under evaluation and in the future might play a role in assessing the response of MPM to treatment.

Sorensen et al (2008) stated that extra-pleural pneumonectomy (EPP) in MPM may be confined with both morbidity and mortality, and careful pre-operative staging identifying resectable patients is important. Staging is difficult and the accuracy of pre-operative CT scan, 18F-FDG PET/CT scan (PET/CT), and mediastinoscopy is unclear. These

investigators compared these staging techniques to each other and to surgical-pathological findings. Subjects were patients with epithelial subtype MPM, aged less than or equal to 70 years, and had lung function test allowing pneumonectomy. Pre-operative staging after 3 to 6 courses of induction chemotherapy included conventional CT scan, PET/CT, and mediastinoscopy. Surgical-pathological findings were compared to pre-operative findings. A total of 42 consecutive patients were without T4 or M on CT scan. PET/CT showed inoperability in 12 patients (29 %) due to T4 (7 patients) and M1 (7 patients). Among 30 patients with subsequent mediastinoscopy, including 10 with N2/N3 on PET/CT, N2 were histologically verified in 6 (20 %). Among 24 resected patients, T4 occurred in 2 patients (8 %), and N2 in 4 (17 %), all being PET/CT negative. The PET/CT accuracy of T4 and N2/N3 compared to combined histological results of mediastinoscopy and EPP showed sensitivity, specificity, positive predictive value, negative predictive value, and positive and negative likelihood ratios of 78 % and 50 %, 100 % and 75 %, 100 % and 50 %, 94 % and 75 %, not applicable and 5.0, and 0.22 and 0.67, respectively. The authors concluded that non-curative surgery is avoided in 29 % out of 42 MPM patients by pre-operative PET/CT and in further 14 % by mediastinoscopy. Even though both procedures are valuable, there are false negative findings with both, urging for even more accurate staging procedures.

A recent review on malignant peritoneal mesothelioma (Turner et al, 2012) stated the role of PET in the diagnosis of malignant peritoneal mesothelioma is "unclear". The British Thoracic Society pleural disease guideline on "Investigation of a unilateral pleural effusion in adults" (Hooper et al, 2010) mentioned the use of CT, but not the use of PET.

Zahid et al (2011) addressed the question – which diagnostic modality [CT, PET, combination PET/CT and magnetic resonance imaging (MRI)] provides the best diagnostic and staging information in patients with malignant pleural mesothelioma (MPM). Overall, 61 papers were found using the reported search, of which 14 represented the best evidence to answer the clinical question. The authors, journal, date and country of publication, patient group studied, study type, relevant outcomes and results are tabulated. These investigators concluded that FDG-PET is superior to MRI and CT but inferior to PET-CT, in terms of diagnostic specificity, sensitivity and staging of MPM. Four

studies reported outcomes using FDG-PET to diagnose MPM. PET diagnosed MPM with high sensitivity (92 %) and specificity (87.9 %). Mean standardized uptake value (SUV) was higher in malignant than benign disease (4.91 versus 1.41, p < 0.0001). Lymph node metastases were detected with higher accuracy (80 % versus 66.7 %) compared to extra-thoracic disease. Three studies assessed the utility of PET-CT to diagnose MPM. Mean SUV was higher in malignant than benign disease (6.5 versus 0.8, p < 0.001). MPM was diagnosed with high sensitivity (88.2 %), specificity (92.9 %) and accuracy (88.9 %). PET-CT had low sensitivity for stage N2 (38 %) and T4 (67 %) disease. CT-guided needle biopsy definitively diagnosed MPM after just 1 biopsy (100 % versus 9 %) much more often than a 'blind' approach. CT had a lower success rate (92 % versus 100 %) than thoracoscopic pleural biopsy but was equivalent to MRI in terms of detection of lymph node metastases (p = 0.85) and visceral pleural tumor (p = 0.64). CT had a lower specificity for stage II (77 % versus 100 %, p < 0.01) and stage III (75 % versus 100 %, p < 0.01) disease compared to PET-CT. Overall, the high specificity and sensitivity rates seen with open pleural biopsy make it a superior diagnostic modality to CT, MRI or PET for diagnosing patients with MPM.

An UpToDate review on "Diagnostic evaluation of pleural effusion in adults: Additional tests for undetermined etiology" (Lee, 2012) states that "Positron emission tomography (PET)/CT has an emerging role: 18-fluorodeoxyglucose (FDG)-avidity confirms, but cannot differentiate between inflammatory and malignant disease. Focal increased uptake of FDG in the pleura and the presence of solid pleural abnormalities on CT are suggestive of malignant pleural disease. A PET-CT pattern composed of pleural uptake and increased effusion activity had an accuracy of 90 percent in predicting malignant pleural effusions in 31 patients with known extrapulmonary malignancy and a pleural effusion [24]. A negative PET/CT would favor a benign etiology [25]. PET/CT may also highlight extra-pleural abnormalities that may be the cause of the effusion …. CT scan of the thorax with contrast should be performed in virtually all patients with an undiagnosed pleural effusion. Additional imaging modalities that may be helpful are CT pulmonary angiogram and positron emission tomography (PET)/CT scans".

Furthermore, the NCCN clinical practice guideline on "Malignant Pleural Mesothelioma" (Version 2.2018) states that pretreatment evaluation for patients diagnosed with MPM is performed to

stage and assess whether patients are candidates for surgery. Evaluation may include FDG-PET/CT but only for patients being considered for surgery. PET/CT "should be obtained before pleurodesis, if practical, because talc produces pleural inflammation, which can affect the FDG avidity."

Non-Small Cell Lung Cancer

Spiro et al (2008) stated that guidelines issued by the National Institute for Clinical Excellence (NICE) in the England and Wales recommend that rapid access to (18)F-deoxyglucose positron emission tomography (FDG-PET) is made available to all appropriate patients with non-small-cell lung cancer (NSCLC). The clinical evidence for the benefits of PET scanning in NSCLC is substantial, showing that PET has high accuracy, sensitivity and specificity for disease staging, as well as pre-therapeutic assessment in candidates for surgery and radical radiotherapy. Moreover, PET scanning can provide important information to assist in radiotherapy treatment planning, and has also been shown to correlate with responses to treatment and overall outcomes. If the government cancer waiting time targets are to be met, rapid referral from primary to secondary healthcare is essential, as is early diagnostic referral within secondary and tertiary care for techniques such as PET. Studies are also required to explore new areas in which PET may be of benefit, such as surveillance studies in high-risk patients to allow early diagnosis and optimal treatment, while PET scanning to identify treatment non-responders may help optimize therapy, with benefits both for patients and healthcare resource use. Further studies are needed into other forms of lung cancer, including small-cell lung cancer and mesothelioma. The authors concluded that PET scanning has the potential to improve the diagnosis and management of lung cancer for many patients. Further studies and refinement of guidelines and procedures will maximize the benefit of this important technique.

The NCCN Imaging Appropriate Use Criteria (2018) provides category 2A recommendations for FDG PET/CT (if not previously done) for IA, IB, IIA, IIB, IIIA, IIIB, IIIC, IV, or IVA (M1b) as diagnostic workup in non-small cell lung cancer (NSCLC). PET/CT if not previously performed, PET/CT performed skull base to knees or whole body. Positive PET/CT findings for distant disease need pathologic or other radiologic confirmation. If PET/CT is positive in the mediastinum, lymph node status needs pathologic confirmation. CT and PET used for staging

should be within 60 days before proceeding with surgical evaluation. For stage IA, consider EBUS or EUS. In addition, NCCN provides a 2A recommendation for FDG PET/CT for treatment response assessment for locoregional recurrence or symptomatic local disease, post therapy.

Moyamoya Disease

An UpToDate review on "Moyamoya disease: Etiology, clinical features, and diagnosis" (Suwanwela, 2016) states that "Although supporting evidence is limited, additional methods that may be useful to determine the extent of inadequate resting brain perfusion and blood flow reserve in patients with Moyamoya disease prior to and after treatment include perfusion CT, Xenon-enhanced CT, perfusion-weighted MRI, PET, and SPECT with acetazolamide challenge".

Osteomyelitis

A proposed decision memo for FDG-PET for infection and inflammation from the CMS (Phurrough et al, 2007) stated that there is insufficient evidence to conclude that FDG- PET for chronic osteomyelitis, infection of hip arthroplasty and fever of unknown origin are reasonable and necessary. Thus, CMS proposed to continue national non-coverage for these indications.

Pre-Transplant PET Scans for Prognosis in Refractory/Relapsed Hodgkin Lymphoma Treated with Autologous Stem Cell Transplantation

Adams and Kwee (2016) systematically reviewed the prognostic value of pre-transplant FDG-PET in refractory/relapsed Hodgkin lymphoma treated with ASCT. Medline was searched for appropriate studies; included studies were methodologically appraised. Results of individual studies were meta-analyzed, if possible. A total of 11 studies, comprising 745 refractory/relapsed Hodgkin lymphoma patients who underwent FDG-PET before autologous SCT, were included. The overall methodological quality of these studies was moderate. The proportion of pre-transplant FDG-PET positive patients ranged between 25 and 65.2 %; PFS ranged between 0 and 52 % in pre-transplant FDG-PET positive patients, and between 55 and 85 % in pre-transplant FDG-PET negative patients; OS ranged between 17 and 77 % in pre-transplant FDG-PET positive patients, and between 78 and 100 % in FDG-PET negative patients. Based on 5 studies that provided sufficient data for meta-analysis, pooled sensitivity and specificity of

pre-transplant FDG-PET in predicting treatment failure (i.e., either progressive, residual, or relapsed disease) were 67.2 % (95 % CI: 58.2 to 75.3 %) and 70.7 % (95 % CI: 64.2 to 76.5 %), respectively. Based on 2 studies that provided sufficient data for meta-analysis, pooled sensitivity and specificity of pre-transplant FDG-PET in predicting death during follow-up were 74.4 % (95 % CI: 58.8 to 86.5 %) and 58.0 % (95 % CI: 49.3 to 66.3 %), respectively. The authors concluded that the moderate quality evidence suggested pre-transplant FDG-PET to have value in predicting outcome in refractory/relapsed Hodgkin lymphoma patients treated with ASCT.

Post-Autologous Stem Cell Transplantation (ASCT)

Dickinson et al (2010) noted that the utility of (18)F-FDG-PET for predicting outcome after autologous stem cell transplantation (ASCT) for diffuse large B cell lymphoma (DLBCL) is uncertain – existing studies include a range of histological subtypes or have a limited duration of follow-up. A total of 39 patients with primary-refractory or relapsed DLBCL with pre-ASCT PET scans were analyzed. The median follow-up was 3 years. The 3-year progression-free survival (PFS) for patients with positive PET scans pre-ASCT was 35 % versus 81 % for those who had negative PET scans (p = 0.003). The overall survival (OS) in these groups was 39 % and 81 % (p = 0.01), respectively. In a multi-variate analysis, PET result, number of salvage cycles and the presence of relapsed or refractory disease were shown to predict a longer PFS; PET negativity (p = 0.04) was predictive of a longer OS. PET is useful for defining those with an excellent prognosis post-ASCT. Although those with positive scans can still be salvaged with current treatments, PET may be useful for selecting patients eligible for novel consolidation strategies after salvage therapies. The findings of this small study need to be validated by well-designed studies.

Rheumatoid Arthritis

Beckers et al (2006) evaluated rheumatoid arthritis (RA) synovitis with FDG-PET in comparison with dynamic MRI and ultrasonography (US). A total of 16 knees in 16 patients with active RA were assessed with FDG-PET, MRI and US at baseline and 4 weeks after initiation of anti-TNF-alpha treatment. All studies were performed within 4 days. Visual and semi-quantitative (standardized uptake value, SUV) analyses of the synovial uptake of FDG were performed. The dynamic

enhancement rate and the static enhancement were measured after intravenous gadolinium injection and the synovial thickness was measured in the medial, lateral patellar and suprapatellar recesses by US. Serum levels of C-reactive protein (CRP) and metalloproteinase-3 (MMP-3) were also measured. FDG-PET was positive in 69 % of knees, while MRI and US were positive in 69 % and 75 %, respectively. Positivity on one imaging technique was strongly associated with positivity on the other two. FDG-PET-positive knees exhibited significantly higher SUVs, higher MRI parameters and greater synovial thickness compared with PET-negative knees, whereas serum CRP and MMP-3 levels were not significantly different. SUVs were significantly correlated with all MRI parameters, with synovial thickness and with serum CRP and MMP-3 levels at baseline. Changes in SUVs after 4 weeks were also correlated with changes in MRI parameters and in serum CRP and MMP-3 levels, but not with changes in synovial thickness. The authors concluded that FDG-PET is a unique imaging technique for assessing the metabolic activity of synovitis. The FDG-PET findings are correlated with MRI and US assessments of the pannus in RA, as well as with the classical serum parameter of inflammation, CRP, and the synovium-derived parameter, serum MMP-3. They stated that further studies are needed to establish the place of metabolic imaging of synovitis in RA.

Solitary Fibrous Tumor

A Medscape review on "Solitary fibrous tumor workup" (Ng, 2015), an UpToDate review on "Solitary fibrous tumor" (Demicco and Meyer, 2016).

Soft Tissue Sarcoma

The National Comprehensive Cancer Network (NCCN) guidelines on "Soft tissue sarcoma" (version 2.2018) state that PET/CT scan may be useful in staging, prognostication, and grading as part of additional imaging studies for soft tissue sarcoma of the extremity/superficial truck and head/neck. PET/CT may be useful in determining response to neoadjuvant chemotherapy for lesions that are larger than 3 cm, firm, and deep (not superficial) in stage II/III disease. The NCCN Imaging Appropriate Use Criteria (2018) provides a category 2A recommendation for use of PET/CT under certain circumstances for diagnostic purpose for the indication of lesion that has a reasonable

chance of being malignant., or for treatment response assessment in IIA, IIB, III disease with lesions larger than 3 cm, firm and deep, post preoperative chemotherapy.

NCCN Imaging Appropriate Use Criteria (2018) provides the following category 2A recommendations for PET/CT in soft tissue sarcoma - gastrointestinal stromal tumors (GIST):

- Diagnostic:
  - GIST is resectable with negative margins but with risk of significant morbidity; Planned treatment with imatinib: Obtain baseline PET/CT if using PET/CT during follow-up. PET is not a substitute for CT.
  - GIST is definitively unresectable, recurrent, or metastatic: Obtain baseline PET/CT if using PET/CT during follow-up. PET is not a substitute for CT.
- Treatment Response Assessment:
  - GIST is resectable with negative margins but with risk of significant morbidity; Post imatinib: Imaging to assess preoperative imatinib response. PET may give indication of imatinib activity after 2-4 weeks of therapy when rapid readout of activity is necessary. Progression may be determined by abdominal/pelvic CT or MRI with clinical interpretation. PET/CT scan may be used to clarify if CT or MRI are ambiguous. Routine long-term PET follow-up is rarely indicated.
  - GIST is definitively unresectable, recurrent, or metastatic; Post imatinib: Imaging to assess preoperative imatinib response. PET may give indication of imatinib activity after 2-4 weeks of therapy when rapid readout of activity is necessary. Progression may be determined by abdominal/pelvic CT or MRI with clinical interpretation. PET/CT scan may be used to clarify if CT or MRI are ambiguous. Routine long-term PET follow-up is rarely indicated. In some patients it may be appropriate to image before 3 months.
  - Post resection; Persistent gross residual disease with or without prior preoperative imatinib; Post continuation of

imatinib: Imaging to assess preoperative imatinib response and evaluate patient compliance. PET may give indication of imatinib activity after 2-4 weeks of therapy when rapid readout of activity is necessary. Progression may be determined by abdominal/pelvic CT or MRI with clinical interpretation. PET/CT scan may be used to clarify if CT or MRI are ambiguous. Routine long-term PET follow-up is rarely indicated.
- Disease progression; Dose escalation of imatinib as tolerated or treatment changed to sunitinib: Reassess therapeutic response with CT or MRI. Consider PET only if CT or MRI results are ambiguous.

- Follow-up:

  Post resection; Discovery of metastatic disease or persistent gross residual disease with or without prior preoperative imatinib; Post continuation of imatinib: PET/CT may be used to clarify if CT or MRI are ambiguous.

NCCN Imaging Appropriate Use Criteria (2018) does not make a recommendation for PET/CT for soft tissue sarcoma - retroperitoneal/intra-abdominal tumors; however, they do provide a category 2A recommendation for PET or PET/CT for staging of rhabdomyosarcoma. PET or PET/CT may be useful for initial staging because of the possibility of nodal metastases and the appearance of unusual sites of initial metastatic disease in adult patients.

PET Scan for Acute Myeloid Leukemia

National Comprehensive Cancer Network's clinical practice guideline on "Acute myeloid leukemia" (Version 3.2017) states that "If extramedullary disease is suspected, a PET/CT is recommended".

Amyloid-PET for the Diagnosis of Sporadic Cerebral Amyloid Angiopathy

Charidimou and colleagues (2017) performed a meta-analysis synthesizing evidence of the value and accuracy of amyloid-PET in diagnosing patients with sporadic cerebral amyloid angiopathy (CAA). In a PubMed systematic literature search, these researchers identified

all case-control studies with extractable data relevant for the sensitivity and specificity of amyloid-PET positivity in symptomatic patients with CAA (cases) versus healthy participants or patients with spontaneous deep intra-cerebral hemorrhage (ICH) (control groups). Using a hierarchical (multi-level) logistic regression model, these investigators calculated pooled diagnostic test accuracy. A total of 7 studies, including 106 patients with CAA (greater than 90% with probable CAA) and 151 controls, were eligible and included in the meta-analysis. The studies were of moderate-to-high quality and varied in several methodological aspects, including definition of PET-positive and PET-negative cases and relevant cutoffs. The sensitivity of amyloid-PET for CAA diagnosis ranged from 60 % to 91 % and the specificity from 56 % to 90 %. The overall pooled sensitivity was 79 % (95 % CI: 62 to 89) and specificity was 78 % (95 % CI: 67 to 86) for CAA diagnosis. A pre-defined subgroup analysis of studies restricted to symptomatic patients presenting with lobar ICH CAA (n = 58 versus 86 controls) resulted in 79 % sensitivity (95 % CI: 61 % to 90 %) and 84 % specificity (95 % CI: 65 % to 93 %). In pre-specified bivariate diagnostic accuracy meta-analysis of 2 studies using 18F-florbetapir-PET, the sensitivity for CAA-ICH diagnosis was 90 % (95 % CI: 76 % to 100 %) and specificity was 88 % (95 % CI: 74 % to 100 %). The authors concluded that amyloid-PET appeared to have moderate-to-good diagnostic accuracy in differentiating patients with probable CAA from cognitively normal healthy controls or patients with deep ICH. They stated that given that amyloid-PET labels both cerebrovascular and parenchymal amyloid, a negative scan might be useful to rule out CAA in the appropriate clinical setting.

These investigators noted that not all studies reviewed in this meta-analysis were prospective; they had a relatively small sample size, with different demographics, disease duration, and severity and were in general restricted to survivors of lobar ICH who were able to undergo research imaging. This is a well-appreciated bias source in diagnostic test accuracy studies. A main drawback pertains to how to classify cases as amyloid-PET positive or negative. The methods and threshold used to define PET positive versus negative cases varied across the 7 studies of this analysis, something incorporated into data synthesis by the use of a hierarchical model. Nevertheless, the pooled estimates might still naturally depend on the exact method and cut-offs used to discriminate PET positive cases. For 3 studies, these researchers used data from visual assessment, which was practical for

wide clinical application but required a good inter-observer reproducibility and validation against quantitative assessment. One study that used both visual assessment and quantitative Pittsburgh compound B (PiB)-PET methods reported slightly worse performance of the former. One 18F-florbetapir-PET study reported perfect agreement (κ = 1) between 2 trained neurologists from a tertiary CAA center. For 4 studies, these investigators used commonly applied quantitative cutoffs toward theoretically more objective categorization. They stated that further systematic studies are needed to directly compare the reliability and diagnostic value of amyloid-PET according to the methods used to define positivity, i.e., visual versus quantitative analysis. The authors stated that despite the drawbacks, their findings were hypothesis-generating and provided preliminary diagnostic accuracy data based on the totality of current evidence about amyloid-PET use in patients with CAA. They stated that further larger prospective studies, including longitudinal PET cohorts, are needed to provide external validation and to examine if amyloid-PET can be useful for detecting CAA pathology in patients in whom diagnostic certainty would have substantial clinical relevance and for patient selection for trials.

In an editorial that accompanied the afore-mentioned study by Charidimou et al (2017), Raposo and Sonnen (2017) stated that "Increasing evidence suggests that amyloid-PET can label vascular amyloid deposits. This meta-analysis promotes the hypothesis that amyloid-PET can accurately discriminate patients without dementia with probable CAA from cognitively normal healthy controls or patients with deep ICH. Further well-powered studies with pathologically proven CAA are warranted to assess the diagnostic value of amyloid-PET in challenging cases (single lobar ICH, mixed hemorrhage, patients with suspected CAA with cognitive impairment) and to determine a standardized method to define PET positivity".

Florbetapir Imaging

Yang and colleagues (2012) noted that important florbetapir scan limitations are (i) positive scan does not establish a diagnosis of AD or other cognitive disorder, and (ii) the scan has not been shown to be useful in predicting the development of dementia or any other neurologic condition, nor has usefulness been shown for monitoring responses to therapies. The authors stated that "the

ultimate clinical value of florbetapir imaging awaits further studies to assess the role, if any, that it plays in providing prognostic and predictive information. For example, the prognostic usefulness of florbetapir imaging in identifying persons with mild cognitive impairment or cognitive symptoms who may be at risk for progression to dementia has not been determined. Nor are data available to determine whether florbetapir imaging could prove useful for predicting responses to medication. These concerns prompted the FDA to require a specific "Limitations of Use" section in the florbetapir label".

Florbetapir-PET Imaging for the Diagnosis of Cerebral Amyloid Angiopathy-Related Hemorrhages

Raposo and colleagues (2017) examined if 18F-florbetapir, a PET amyloid tracer, could bind vascular amyloid in CAA by comparing cortical florbetapir retention during the acute phase between patients with CAA-related lobar ICH and patients with hypertension-related deep ICH. Patients with acute CAA-related lobar ICH were prospectively enrolled and compared with patients with deep ICH. 18F-florbetapir PET, brain MRI, and APOE genotype were obtained for all participants. Cortical florbetapir SUVr was calculated with the whole cerebellum used as a reference. Patients with CAA and those with deep ICH were compared for mean cortical florbetapir SUVr values. A total of 15 patients with acute lobar ICH fulfilling the modified Boston criteria for probable CAA (mean age of 67 ± 12 years) and 18 patients with acute deep ICH (mean age of 63 ± 11 years) were enrolled. Mean global cortical florbetapir SUVr was significantly higher among patients with CAA-related ICH than among patients with deep ICH (1.27 ± 0.12 versus 1.12 ± 0.12, p = 0.001). Cortical florbetapir SUVr differentiated patients with CAA-ICH from those with deep ICH (AUC = 0.811; 95 % CI: 0.642 to 0.980) with a sensitivity of 0.733 (95 % CI: 0.475 to 0.893) and a specificity of 0.833 (95 % CI: 0.598 to 0.948). The authors concluded that these findings suggested that cortical florbetapir binding was increased in patients with CAA with lobar ICH relative to patients with deep ICH. In-vivo detection of vascular Aβ deposits might promote early diagnosis of CAA. However, they noted that 18F-florbetapir PET used as a diagnostic tool for acute CAA-related ICH appeared limited by its sensitivity. They stated that larger studies with pathologically proven CAA and age-matched healthy controls are needed to establish its relevance in clinical practice.

Flortaucipir F18 injection (Tauvid)

On May 28, 2022, the U.S. Food and Drug Administration approved flortaucipir F18 for intravenous injection (Tauvid) to help image tau pathology in the brain, a distintive feature of Alzheimer's disease. Tauvid is a radioactive diagnostic agent for adult patients with cognitive impairment who are undergoing evaluation for Alzheimer's disease (AD). Specifically, Tauvid is indicated for positron emission tomography (PET) imaging of the brain to estimate the density and distribution of aggregated tau neurofibrillary tangles (NFTs), a central marker of Alzheimer's disease (FDA, 2020). Tauvid is not indicated for use in the evaluation of patients for chronic traumatic encephalopathy.

Tian and colleagues (2022) noted in their discussion of ^{18}F-flortaucipir PET imaging depicting tau depositin in the brain, that it is important to interpret a scan positive for tau deposition in association with amyloid PET imaging and current clinical information to diagnose AD. In the event of negative results of ^{18}F-flortaucipir PET imaging accompanied with positive amyloid PET imaging, this would indicate individuals who are possibly developed with AD. Positive results of ^{18}F-flortaucipir PET imaging accompanied with positive amyloid PET imaging would indicate the existence of AD in these individuals. Notably, a positive ^{18}F-flortaucipir PET imaging does not rule out other coexistent neurodegenerative disorders. Negative results of amyloid PET imaging indicate individuals who are unlikely to have AD and negative ^{18}F-flortaucipir PET results among mild cognitive impairment individuals may indicate that they are unlikely to advance to AD.

According to eviCore Head Imaging Guidelines (Version 2.0.2022), "amyloid PET studies may be approved one time for Medicare individuals enrolled in approved clinical trials under Coverage with Evidence Development (CED) program". Additionally, "only one study will be paid per beneficiary and the radiopharmaceutical must be FDA-approved. Examples of radiopharmaceuticals which met this qualification include Amyvid (forbetapir F18), Neuraceq (florbetaben F18), Tauvid (flortaucipir F18) and Vizamyl (flutemetamol F18)."

PET for Pituitary Tumor

Seok et al (2013) stated that although magnetic resonance imaging (MRI) is a good visual modality for the evaluation of pituitary lesions, it has limited value in the diagnosis of mixed nodules and some cystic

lesions. These investigators evaluated the usefulness of FDG-PET for patients with pituitary lesions. (18)F-FDG PET and MRI were performed simultaneously in 32 consecutive patients with pituitary lesions. The relationships between FDG uptake patterns in PET and MRI findings were analyzed. Of 24 patients with pituitary adenomas, 19 (79.2 %) showed increased uptake of (18)F-FDG in the pituitary gland on PET scans. All patients with pituitary macroadenomas showed increased (18)F-FDG uptake on PET scans. Meanwhile, only 5 (50 %) of the 10 patients with pituitary microadenomas showed positive PET scans. Interestingly, of 2 patients with no abnormal MRI findings, 1 showed increased (18)F-FDG uptake on PET. For positive (18)F-FDG uptake, maximum standardized uptake values (SUV(max)) greater than 2.4 had 94.7 % sensitivity and 100 % specificity. In addition, SUV(max) increased in proportion to the size of pituitary adenomas. Most cystic lesions did not show (18)F-FDG uptake on PET scans. The authors concluded that about 80 % of pituitary adenomas showed positivity on PET scans, and SUV(max) was related to the size of the adenomas. They stated that PET may be used as an ancillary tool for detection and differentiation of pituitary lesions. They stated that further PET studies determining the right threshold of SUVmax or conjugating various tracer molecules will be helpful.

Chittiboina et al (2015) noted that high-resolution PET (hrPET) performed using a high-resolution research tomograph is reported as having a resolution of 2 mm and could be used to detect corticotroph adenomas through uptake of (18)F-FDG. To determine the sensitivity of this imaging modality, the authors compared(18)F-FDG hrPET and MRI detection of pituitary adenomas in Cushing disease (CD). Consecutive patients with CD who underwent preoperative(18)F-FDG hrPET and MRI (spin echo [SE] and spoiled gradient recalled [SPGR] sequences) were prospectively analyzed; SUVs were calculated from hrPET and were compared with MRI findings. Imaging findings were correlated to operative and histological findings. A total of 10 patients (7 females and 3 males) were included (mean age of 30.8 ± 19.3 years; range of 11 to 59 years). MRI revealed a pituitary adenoma in 4 patients (40 % of patients) on SE and 7 patients (70 %) on SPGR sequences. (18)F-FDG hrPET demonstrated increased(18)F-FDG uptake consistent with an adenoma in 4 patients (40 %; adenoma size range of 3 to 14 mm). Maximum SUV was significantly higher for(18)F-FDG hrPET-positive tumors (difference = 5.1, 95 % confidence interval [CI]: 2.1 to 8.1; p = 0.004) than for(18)F-FDG hrPET-negative tumors.

(18)F-FDG hrPET positivity was not associated with tumor volume (p = 0.2) or dural invasion (p = 0.5). Midnight and morning ACTH levels were associated with(18)F-FDG hrPET positivity (p = 0.01 and 0.04, respectively) and correlated with the maximum SUV (R = 0.9; p = 0.001) and average SUV (R = 0.8; p = 0.01). All(18)F-FDG hrPET-positive adenomas had a less than a 180 % ACTH increase and(18)F-FDG hrPET-negative adenomas had a greater than 180 % ACTH increase after CRH stimulation (p = 0.03). Three adenomas were detected on SPGR MRI sequences that were not detected by(18)F-FDG hrPET imaging. Two adenomas not detected on SE (but no adenomas not detected on SPGR) were detected on(18)F-FDG hrPET. The authors concluded that while(18)F-FDG hrPET imaging can detect small functioning corticotroph adenomas and is more sensitive than SE MRI, SPGR MRI is more sensitive than(18)F-FDG hrPET and SE MRI in the detection of CD-associated pituitary adenomas. Response to CRH stimulation can predict(18)F-FDG hrPET-positive adenomas in CD. This was a small study (n = 10).

Yao et al (2017) noted that pituitary adenomas (PAs) are the most common intra-sellar mass. Functional PAs constitute most of pituitary tumors and can produce symptoms related to hormonal over-production. Timely and accurate detection is therefore of vital importance to prevent potentially irreversible sequelae. Magnetic resonance imaging is the gold standard for detecting PAs, but is limited by poor sensitivity for microadenomas and an inability to differentiate scar tissue from tumor residual or predict treatment response. Several new modalities that detect PAs have been proposed. These researchers performed a systematic review of the PubMed database for imaging studies of PAs since its inception. Data concerning study characteristics, clinical symptoms, imaging modalities, and diagnostic accuracy were collected. After applying exclusion criteria, 25 studies of imaging PAs using PET, magnetic resonance spectroscopy (MRS), and single photon emission computed tomography (SPECT) were reviewed. PET reliably detects PAs, particularly where MRI is equivocal, although its efficacy is limited by high cost and low availability; SPECT possesses good sensitivity for neuroendocrine tumors but its use with PAs is poorly documented. MRS consistently detects cellular proliferation and hormonal activity, but warrants further study at higher magnetic field strength. The authors concluded that PET and MRS appeared to have the strongest predictive value in detecting PAs. MRS has the advantage of low cost, but the literature is

lacking in specific studies of the pituitary. Due to high recurrence rates of functional PAs and low sensitivity of existing diagnostic work-ups, further investigation of metabolic imaging is needed.

An UpToDate review on "Clinical manifestations and diagnosis of gonadotroph and other clinically nonfunctioning pituitary adenomas" (Synder, 2017) states that "Pituitary imaging – We suggest MRI for the initial imaging study for suspected adenomas, because of its superior resolution and its ability to demonstrate the optic chiasm. MRI with gadolinium is preferred to computed tomography (CT) because it provides superior resolution of the mass and its relation to surrounding structures. MRI is also able to detect blood, thereby permitting recognition of hemorrhage into the pituitary and distinction of an aneurysm from other intrasellar lesions". This review does not mention PET imaging.

The National Comprehensive Cancer Network (NCCN) on "Neuroendocrine and adrenal tumors" (Version 3.2018) does not mention PET scan for pituitary tumors.

PET for Prostate Cancer

Although PET scans using the radioactive glucose analog FDG have proven to be a highly accurate imaging test for diagnosing and staging a variety of non-urologic cancer types, its role in the management of prostate malignancies is still being defined. The use of PET scanning in the diagnosis and staging of prostate cancer is hampered by the generally low metabolic activity of most prostate tumors and their metastases. It has shown promise for staging and re-staging persons with advanced-stage disease and aggressive tumors suspected by a high tumor grade and high prostate-specific antigen velocity. Further investigations are needed to ascertain the eventual place of PET scans in prostate cancer.

Vees et al (2007) evaluated the value of PET/computed tomography (CT) with either (18)F-choline and/or (11)C-acetate, of residual or recurrent tumor after radical prostatectomy (RP) in patients with a prostate-specific antigen (PSA) level of less than 1 ng/ml and referred for adjuvant or salvage radiotherapy. In all, 22 PET/CT studies were performed, 11 with (18)F-choline (group A) and 11 with (11)C-acetate (group B), in 20 consecutive patients (2 undergoing PET/CT scans with both tracers). The median (range) PSA level before PET/CT was 0.33

(0.08 to 0.76) ng/ml. Endorectal-coil magnetic resonance imaging (MRI) was used in 18 patients. A total of 19 patients were eligible for evaluation of biochemical response after salvage radiotherapy. There was abnormal local tracer uptake in 5 and 6 patients in group A and B, respectively. Except for a single positive obturator lymph node, there was no other site of metastasis. In the 2 patients evaluated with both tracers there was no pathological uptake. Endorectal MRI was locally positive in 15 of 18 patients; 12 of 19 responded with a marked decrease in PSA level (half or more from baseline) 6 months after salvage radiotherapy. The authors concluded that although (18)F-choline and (11)C-acetate PET/CT studies succeeded in detecting local residual or recurrent disease in about 50 % of the patients with PSA levels of less than 1 ng/ml after RP, these studies can not yet be recommended as a standard diagnostic tool for early relapse or suspicion of subclinical minimally persistent disease after surgery. Endorectal MRI might be more helpful, especially in patients with a low likelihood of distant metastases. Nevertheless, further research with (18)F-choline and/or (11)C-acetate PET with optimal spatial resolution might be needed for patients with a high risk of distant relapse after RP even at low PSA values.

Takahashi et al (2007) noted that 2-(18)F-fluoro-2-deoxy-D-glucose (FDG)-PET imaging in prostate cancer is challenging because glucose utilization in well-differentiated prostate cancer is often lower than in other tumor types. Nonetheless, FDG-PET has a high positive predictive value for untreated metastases in viscera, but not lymph nodes. A positive FDG-PET can provide useful information to aid the clinician's decision on future management in selected patients who have low PSA levels and visceral changes as a result of metastases. On the other hand, FDG-PET is limited in the identification of prostate tumors, as normal urinary excretion of radioisotope can mask pathological uptake. Moreover, there is an overlap in the degree of uptake between prostate cancer, benign prostatic hyperplasia and inflammation. The tracer choice is also important. (11)C-choline has the advantage of reduced urinary excretion, and thus (11)C-choline PET may provide more accurate information on the localization of main primary prostate cancer lesions than MRI or MR spectroscopy. (11)C-choline PET is sensitive and accurate in the pre-operative staging of pelvic lymph nodes in prostate cancer. A few studies are available but

there were no PET or PET/CT studies with a large number of patients for tissue confirmation of prostate cancer; further investigations are required.

Greco et al (2008) stated that the patient population with a rising PSA post-therapy with no evidence of disease on standard imaging studies currently represents the second largest group of prostate cancer patients. Little information is still available regarding the specificity and sensitivity of PET tracers in the assessment of early biochemical recurrence. Ideally, PET imaging would allow one to accurately discriminate between local versus nodal versus distant relapse, thus enabling appropriate selection of patients for salvage local therapy. The vast majority of studies show a relatively poor yield of positive scans with PSA values less than 4 ng/ml. So far, no tracer has been shown to be able to detect local recurrence within the clinically useful 1 ng/ml PSA threshold, clearly limiting the use of PET imaging in the post-surgical setting. Preliminary evidence, however, suggested that 11C-choline PET may be useful in selecting out patients with early biochemical relapse (PSA less than 2 ng/ml) who have pelvic nodal oligo-metastasis potentially amenable to local treatment. The authors concluded that the role of PET imaging in prostate cancer is gradually evolving but still remains within the experimental realm. Well-conducted studies comparing the merits of different tracers are needed.

In a systematic review and meta-analysis, Moshen and colleagues (2013) evaluated 23 studies of 11C-acetate PET in prostate cancer. For evaluation of primary tumor, pooled sensitivity was 75.1 (69.8 to 79.8) % and specificity was 75.8 (72.4 to 78.9) %. For detection of recurrence, sensitivity was 64 (59 to 69) % and specificity was 93 (83 to 98) %. Sensitivity for recurrence detection was higher in post-surgical versus post-radiotherapy patients and in patients with PSA at relapse of greater than 1 ng/ml. Studies using PET/computed tomography versus PET also showed higher sensitivity for detection of recurrence. Imaging with 11C-acetate PET can be useful in patients with prostate cancer. This is especially true for evaluation of patients at PSA relapse, although the sensitivity is overall low. For primary tumor evaluation (localization of tumor in the prostate and differentiation of malignant from benign lesions), 11C-acetate is of limited value due to low sensitivity and specificity. The authors concluded that due to the poor quality of the included studies, the results should be interpreted with caution and further high-quality studies are needed. They stated

that the quality of the conducted studies on the application of 11C-acetate in prostate cancer was usually poor. This is especially true for studies on re-staging at PSA relapse. The major drawback in the quality of the included studies in the present meta-analysis was poor and/or inconsistent gold standard. The researchers rarely provided histology results as the main gold standard method and usually relied on follow-up and conventional imaging (i.e., CT, bone scan, etc.). This is a major issue of the present study and these findings should be interpreted with the knowledge of this limitation. They noted that high quality prospective studies with consecutive patient recruitment and applying the best gold standard (histology confirmation preferably) are definitely needed.

Ceci et al (2014) investigated the clinical impact of (11)C-choline PET/CT on treatment management decisions in patients with recurrent prostate cancer (rPCa) after radical therapy. Enrolled in this retrospective study were 150 patients (95 from Bologna, 55 from Wurzburg) with rPCa and biochemical relapse (PSA mean ± SD 4.3 ± 5.5 ng/ml, range of 0.2 to 39.4 ng/ml) after radical therapy. The intended treatment before PET/CT was salvage radiotherapy of the prostatic bed in 95 patients and palliative androgen deprivation therapy (ADT) in 55 patients. The effective clinical impact of (11)C-choline PET/CT was rated as major (change in therapeutic approach), minor (same treatment, but modified therapeutic strategy) or none. Multi-variate binary logistic regression analysis included PSA level, PSA kinetics, ongoing ADT, Gleason score, tumor, node, metastasis (TNM), age and time to relapse. Changes in therapy after (11)C-choline PET/CT were implemented in 70 of the 150 patients (46.7 %). A major clinical impact was observed in 27 patients (18 %) and a minor clinical impact in 43 (28.7 %). (11)C-choline PET/CT was positive in 109 patients (72.7 %) detecting local relapse (prostate bed and/or iliac lymph nodes and/or para-rectal lymph nodes) in 64 patients (42.7 %). Distant relapse (para-aortic and/or retroperitoneal lymph nodes and/or bone lesions) was seen in 31 patients (20.7 %), and both local and distant relapse in 14 (9.3 %). A significant difference was observed in PSA level and PSA kinetics between PET-positive and PET-negative patients ($p < 0.05$). In multi-variate analysis, PSA level, PSA doubling time and ongoing ADT were significant predictors of a positive scan ($p < 0.05$). In statistical analysis no significant differences were observed between the Bologna and Wurzburg patients ($p > 0.05$). In both centers the same criteria to validate PET-positive findings were used:

in 17.3 % of patients by histology and in 82.7 % of patients by correlative imaging and/or clinical follow-up (mean follow-up of 20.5 months, median of 18.3 months, range of 6.2 to 60 months). The authors concluded that (11)C-Choline PET/CT had a significant impact on therapeutic management in rPCa patients. It led to an overall change in 46.7 % of patients, with a major clinical change implemented in 18 % of patients. Moreover, they stated that further prospective studies are needed to evaluate the effect of such treatment changes on patient survival.

Giovacchini et al (2015) examined if [11C]choline PET/CT can predict survival in hormone-naive PCa patients with biochemical failure. This retrospective study included 302 hormone-naive PCa patients treated with radical prostatectomy who underwent [11C]choline PET/CT from December 1, 2004 to July 31, 2007 because of biochemical failure (prostate-specific antigen, PSA, greater than 0.2 ng/ml). Median PSA was 1.02 ng/ml. PCa-specific survival was estimated using Kaplan-Meier curves. Cox regression analysis was used to evaluate the association between clinicopathological variables and PCa-specific survival. The coefficients of the covariates included in the Cox regression analysis were used to develop a novel nomogram. Median follow-up was 7.2 years (1.4 to 18.9 years). [11C]Choline PET/CT was positive in 101 of 302 patients (33 %). Median PCa-specific survival after prostatectomy was 14.9 years (95 % CI: 9.7 to 20.1 years) in patients with positive [11C]choline PET/CT. Median survival was not achieved in patients with negative [11C]choline PET/CT. The 15-year PCa-specific survival probability was 42.4 % (95 % CI: 31.7 to 53.1 %) in patients with positive [11C]choline PET/CT and 95.5 % (95 % CI: 93.5 to 97.5 %) in patients with negative [11C]choline PET/CT. In multi-variate analysis, [11C]choline PET/CT (hazard ratio [HR] 6.36, 95 % CI: 2.14 to 18.94, $p < 0.001$) and Gleason score greater than 7 (HR 3.11, 95 % CI: 1.11 to 8.66, $p = 0.030$) predicted PCa-specific survival. An internally validated nomogram predicted 15-year PCa-specific survival probability with an accuracy of 80 %. The authors concluded that positive [11C]choline PET/CT after biochemical failure predicts PCa-specific survival in hormone-naive PCa patients. Moreover, they stated that prospective studies are needed to confirm our results before more extensive use of [11C]choline PET/CT for prognostic stratification of PCa patients.

An UpToDate review on "Rising serum PSA following local therapy for prostate cancer: Diagnostic evaluation" (Moul and Lee, 2015) states that "Newer imaging techniques using 18F-NaF positron emission tomography (PET)/computed tomography (CT) and 11-choline PET/CT are under development and appear to offer improved sensitivity and specificity compared with technetium 99 radionuclide bone scans. However, false-positive scans remain a significant concern and there is a steep learning curve associated with interpretation of this approach. The Prostate Cancer Radiographic Assessments for Detection of Advanced Recurrence (RADAR) Working Group recommends using 18F-NaF PET/CT for skeletal assessment in biochemical recurrence as an initial scan with PSA >5 ng/mL or for doubling of PSA after a prior negative scan …. Early studies suggest that 18-fluorine-labeled choline, 18-F sodium fluoride, and 11-carbon-labeled acetate may be better tracers for use in recurrent prostate cancer. PET and PET/CT with these tracers is still considered investigational in men with a PSA-only recurrence".

The National Comprehensive Cancer Network (NCCN) Imaging Appropriate Use Criteria (2018) for prostate cancer states that 18F sodium fluoride PET/CT can be considered for equivocal results on initial bone scan. Bone imaging should be performed for any patient with symptoms consistent with bone metastases. NCCN provides the following category 2A recommendations:

- Treatment Response Assessment:
    - Metastatic disease; post ADT or localized on observation: 11C choline or 18F fluciclovine PET/CT for further soft tissue imaging may be considered. 18F sodium fluoride PET/CT can be considered for equivocal results on initial bone scan, or can be considered after bone scan for further evaluation when clinical suspicion of bone metastases is high.
    - Post radical prostatectomy; PSA persistence or PSA recurrence; Post treatment; 11C choline or 18F fluciclovine PET/CT for further soft tissue imaging may be considered. 18F sodium fluoride PET/CT can be considered for equivocal results on initial bone scan or can be considered after bone scan for further evaluation when clinical suspicion of bone metastases is high.

- Radiation therapy recurrence; PSA persistence/recurrence or positive DRE; Candidate for local therapy: original clinical stage T1-T2, NX or N0, life expectancy > 10 yr, PSA now < 10ng/mL; Studies negative for distant metastases; Post treatment; May consider: 11C choline PET/CT or PET/MRI, or 18F fluciclovine PET/CT or PET/MRI, or 18F sodium fluoride PET/CT.
- Advanced disease; Post systemic therapy for castration-naïve disease; May consider: 11C choline PET/CT or PET/MRI, or 18F fluciclovine PET/CT or PET/MRI, or 18F sodium fluoride PET/CT.

- Staging:

  - Post radical prostatectomy; PSA persistence (Failure of PSA to fall to undetectable levels) or PSA recurrence (Undetectable PSA after RP with a subsequent detectable PSA that increases on 2 or more determinations). PET/CT using 11C choline tracers may identify sites of metastatic disease in men with biochemical recurrence after primary treatment failure. If PET imaging is used, histologic confirmation is recommended whenever feasible due to significant rates of false positivity. 18F sodium fluoride PET/CT can be considered for equivocal results on initial bone scan, or can be considered after bone scan for further evaluation when clinical suspicion of bone metastases is high.
  - Radiation therapy recurrence; PSA persistence/recurrence or positive DRE; Candidate for local therapy: original clinical stage T1-T2, NX or N0, life expectancy > 10 yr, PSA now < 10ng/mL; May consider: 11C choline PET/CT or PET/MRI, or18F fluciclovine PET/CT or PET/MRI.
  - Advanced disease; Post systemic therapy for castration-naïve disease; Studies negative for distant metastases; Post systemic therapy for M0 CRPC; PSA increasing; May consider: 11C choline PET/CT or PET/MRI, or 18F fluciclovine PET/CT or PET/MRI, or 18F sodium fluoride PET/CT.
  - Advanced disease; CRPC; Studies positive for metastases; No visceral metastases or visceral metastases with adenocarcinoma histology; Post systemic therapy for M1 CRPC; May consider: 11C choline PET/CT or PET/MRI, or18F

fluciclovine PET/CT or PET/MRI, or 18F sodium fluoride PET/CT.

PET for Ureteral Cancer

An UpToDate review on "Malignancies of the renal pelvis and ureter" (Richie and Kantoff, 2017) states that "The diagnosis of a tumor of the renal pelvis or ureter is usually suspected based upon an abnormal computed tomography (CT) or retrograde pyelography". It does not mention PET as a diagnostic tool.

Cerianna (Fluoroestradiol F-18) PET Imaging for Breast Cancer

Evangelista et al (2016) determined the correlation between 16α-18F-fluoro- 17β-estradiol (18F-FES) uptake and the expression and functionality of estrogen receptors (ERs), and examined the ability of 18F-FES PET to predict the response to hormonal therapy (HT) in patients with locally advanced or metastatic breast cancer (BC). These researchers carried out literature searches in the major literature data-bases to select English-language articles dealing with 18F-FES PET and BC. Studies that included patients with BC undergoing 18F-FES PET alone or in combination with other imaging modalities and included the absolute numbers of true-positive, true-negative, false-positive and false-negative test results were selected. They found 23 studies, published between 1988 and December 2014, that critically examined the role of 18F-FES PET in BC patients. Two separate meta-analyses were performed: the 1st one evaluated the correlation between 18F-FES uptake and ER expression; and the 2nd one determined the predictive value of 18F-FES in response to HT. For the 1st meta-analysis, these investigators considered 9 selected studies with a total of 238 patients. A pooled sensitivity of 82 % (95 % CI: 74 to 88 %) and a pooled specificity of 95 % (95 % CI: 86 to 99 %) for the evaluation of ER functional status by 18F-FES PET were found; 7 studies, with a total of 226 patients, were considered eligible for the analysis of predict ion for response. The pooled sensitivities and specificities were 63.9 % (95 % CI: 46.2 to 79.2 %) versus 66.7 % (95 % CI: 52.1 to 79.2 %), and 28.6 % (95 % CI: 17.3 to 42.2 %) versus 62.1 % (95 % CI: 48.4 to 74.5%), for a SUV cut-off of 1.5 and 2.0, respectively. The authors concluded that a good correlation between 18F-FES uptake and ER expression by immunohistochemistry emerges, while the role of 18F-FES in predicting the response to endocrine therapy in advanced BC remains undetermined.

Jones et al (2019) noted that dedicated breast PET (dbPET) is an emerging technology with high sensitivity and spatial resolution that enables detection of sub-centimeter lesions and depiction of intra-tumoral heterogeneity. These investigators reported their initial experience with dbPET using [F-18]fluoroestradiol (FES) in assessing ER+ primary BCs. A total of 6 patients with greater than 90 % ER+ and HER2- BCs were imaged with dbPET and breast MRI. Two patients had invasive lobular carcinoma (ILC), 3 had invasive ductal carcinoma (IDC), and 1 had an unknown primary tumor. One ILC patient was treated with letrozole, and another patient with IDC was treated with neoadjuvant chemotherapy without endocrine treatment. In this small cohort, these researchers observed FES uptake in ER+ primary breast tumors with specificity to ER demonstrated in a case with tamoxifen blockade. FES uptake in ILC had a diffused pattern compared to the distinct circumscribed pattern in IDC. In evaluating treatment response, the reduction of SUVmax was observed with residual disease in an ILC patient treated with letrozole, and an IDC patient treated with chemotherapy. The authors concluded that future study is critical to understand the change in FES SUVmax after endocrine therapy and to consider other tracer uptake metrics with SUVmax to describe ER-rich BC. These researchers stated that limitations of this study included variations of FES uptake in different ER+ BC diseases and exclusion of posterior tissues and axillary regions. However, FES-dbPET has a high potential for clinical utility, especially in measuring response to neoadjuvant endocrine treatment. These investigators stated that further studies involving larger numbers of ER+ BC patients are needed to validate these preliminary findings.

Katzenellenbogen (2020) stated that many BC and prostate cancer are driven by the action of steroid hormones on their cognate receptors in primary tumors and in metastases, and endocrine therapies that inhibit hormone production or block the action of these receptors provide clinical benefit to many but not all of these cancer patients. Because it is difficult to predict which individuals will be helped by endocrine therapies and which will not, PET imaging of estrogen receptor (ER) and progesterone receptor (PgR) in BC, and androgen receptor (AR) in prostate cancer can provide useful, often functional, information on the likelihood of endocrine therapy response in individual patients. The author discussed the development of 3 PET imaging agents, 16α-[18F]fluoroestradiol (FES) for ER, 21-[18F]fluoro-furanyl-nor-progesterone (FFNP) for PgR, and 16β-[18F]fluoro-5α-

dihydrotestosterone (FDHT) for AR, and the evolution of their clinical use. A number of active clinical trials (some multi-center) continue to examine the predictive value of FES-PET, often together with FDG-PET or other targeted imaging agents, or to examine its ability to identify metastases and intra-patient heterogeneity. Suggestions for best practices in the clinical use of FES-PET have been published, as has a simulation of the cost-effectiveness of FES-PET versus conventional work-up of patients with metastatic BC. The author concluded that the use of PET imaging of steroid receptors enables observation of the extent of cancer metastasis and predicts responsiveness to endocrine therapies. Its use in evaluating new aspects of tumor clinical status non-invasively suggests a robust future for PET imaging in providing meaningful benefit for patients with BC and prostate cancer.

Ulaner et al (2021) noted that invasive lobular carcinoma (ILC) demonstrates lower conspicuity on 2-deoxy-2-[18F]fluoro-D-glucose ([18F]FDG) PET than the more common invasive ductal carcinoma (IDC). Other molecular imaging methods may be needed for evaluation of this malignancy. As ILC is nearly always (95 %) ER-positive, ER-targeting PET tracers such as 16α-[18F]-fluoroestradiol ([18F]FES) may have value. These investigators reviewed prospective trials at Memorial Sloan Kettering Cancer Center (MSK) utilizing [18F]FES PET/CT to evaluate metastatic ILC patients with synchronous [18F]FDG and [18F]FES PET/CT imaging, which allowed a head-to-head comparison of these 2 PET tracers. A total of 6 prospective clinical trials utilizing [18F]FES PET/CT in patients with metastatic BC were carried out at MSK from 2008 to 2019. These trials included 92 patients, of which 14 (15 %) were of ILC histology; 7 of 14 patients with ILC had an [18F]FDG PET/CT performed within 5 weeks of the research [18F]FES PET/CT and no intervening change in management. For these 7 patients, the [18F]FES and [18F]FDG PET/CT studies were analyzed to determine the total number of tracer avid lesions, organ systems of involvement, and SUVmax of each organ system for both tracers. In the 7 comparable pairs of scans, there were a total of 254 [18F]FES-avid lesions (SUVmax 2.6 to 17.9) and 111 [18F]FDG-avid lesions (SUVmax 3.3 to 9.9) suspicious for malignancy. For 5 of 7 (71 %) of ILC patients, [18F]FES PET/CT detected more metastatic lesions than [18F]FDG PET/CT. In the same 5 of 7 patients, the SUVmax of [18F]FES-avid lesions was greater than the SUVmax of [18F]FDG-avid lesions; 1 patient had [18F]FES avid metastases with no corresponding [18F]FDG avid metastases. There

were no patients with [18F]FDG avid distant metastases without [18F]FES avid distant metastases, although in 1 patient liver metastases were evident on [18F]FDG but not [18F]FES PET. The authors concluded that [18F]FES PET/CT compared favorably with [18F]FDG PET/CT for detection of metastases in patients with metastatic ILC. Moreover, these researchers stated that larger, prospective trials of [18F]FES PET/CT in ILC should be considered to examine ER-targeted imaging for clinical value in patients with this histology of BC.

Boers et al (2020) noted that molecular imaging with PET is a powerful tool to visualize BC characteristics. Nonetheless, implementation of PET imaging into cancer care is challenging, and essential steps have been outlined in the international "imaging biomarker roadmap". These investigators identified hurdles and provide recommendations for implementation of PET biomarkers in BC care, focusing on the PET tracers 2-[18F]-fluoro-2-deoxyglucose ([18F]-FDG), sodium [18F]-fluoride ([18F]-NaF), 16α-[18F]-fluoroestradiol ([18F]-FES), and [89Zr]-trastuzumab. Technical validity of [18F]-FDG, [18F]-NaF, and [18F]-FES is established and supported by international guidelines. However, support for clinical validity and utility is still pending for these PET tracers in BC, due to variable end-points and procedures in clinical studies. The authors concluded that evaluation of clinical validity and utility is essential towards implementation; however, these steps are still lacking for PET biomarkers in BC. This could be solved by adding PET biomarkers to randomized trials, development of imaging data warehouses, and harmonization of end-points and procedures.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Breast cancer" (Version 6.2020) does not mention 18F-fluoroestradiol PET as a management tool.

DOPA-PET/CT Scan for Prognosis of Brain Cancer

Isal et al (2018) stated that in diffuse grade II to III gliomas, a high 3,4-dihydroxy-6-(18F)-fluoro-L-phenylalanine (18F-FDOPA) PET uptake, with a standardized uptake value (SUVmax) / contralateral brain tissue ratio greater than 1.8, was previously found to be consistently associated with the presence of an isocitrate dehydrogenase (IDH) mutation, whereas this mutation is typically associated with a better prognosis. In a pilot study, these researchers determined the prognostic value of this high 18F-FDOPA uptake in diffuse grade II to III

gliomas with regard to the velocity of diameter expansion (VDE), which represents an established landmark of better prognosis when below 4 mm/year. A total of 20 patients (42 ± 10 years, 10 women) with newly-diagnosed diffuse grade II to III gliomas (17 with IDH mutation) were retrospectively included. All had a 18F-FDOPA PET, quantified with SUVmax ratio, along with a serial MRI enabling VDE determination. SUVmax ratio was above 1.8 in 5 patients (25 %) all of whom had a VDE less than 4 mm/year (100 %) and IDH mutation (100 %). Moreover, a SUVmax ratio above 1.8 was associated with higher rates of VDE less than 4 mm/year in the overall population (45 % versus 0 %, p = 0.04) and also in the subgroup of patients with IDH mutation (45 % versus 0 %, p = 0.10). The authors concluded that the findings of this pilot study showed that in diffuse grade II to III gliomas, a high 18F-FDOPA uptake would be predictive of low tumor growth, with a different prognostic significance than IDH mutation. These investigators noted that 18F-FDOPA PET in a single session imaging could have prognostic value in initial diagnosis of diffuse grade II to III gliomas.

The authors stated that the most important limitations of the present study were its limited sample size and its retrospective and single-center nature. They stated that further larger-scale studies are needed, especially for the subgroups of patients with IDH-mutant 1p/19q co-deleted oligodendrogliomas or wild-type IDH astrocytoma, in order to confirm the low growth rate as well as the good prognosis of patients showing a SUVmax ratio of greater than 1.8. Secondly, the average time between MRI 1 and MRI 2 (82 days ± 29) could be sometimes short to define precisely tumor growth rate. However, a minimum delay of 40 days was observed between the 2 investigations. Finally, the growth rate of tumor was chosen as a surrogate for clinical course in this study since no OS could be obtained in this population owing to the long clinical course duration of diffuse low-grade gliomas.

Patel et al (2018) reported the potential value of pre-operative 18F-FDOPA PET and anatomic MRI in diagnosis and prognosis of glioma patients. A total of 45 patients with a pathological diagnosis of glioma with pre-operative 18F-FDOPA PET and anatomic MRI were retrospectively examined. The volume of contrast enhancement and T2 hyperintensity on MRI images along with the ratio of maximum 18F-FDOPA SUV in tumor to normal tissue (T/N SUVmax) were measured and used to predict tumor grade, molecular status, and OS. A significant correlation was observed between WHO grade and: the

volume of contrast enhancement (r = 0.67), volume of T2 hyperintensity (r = 0.42), and 18F-FDOPA uptake (r = 0.60) (p < 0.01 for each correlation). The volume of contrast enhancement and 18F-FDOPA T/N SUVmax were significantly higher in glioblastoma (WHO IV) compared with lower grade gliomas (WHO I to III), as well as for high-grade gliomas (WHO III to IV) compared with low-grade gliomas (WHO I to II). Receiver-operator characteristic (ROC) analyses confirmed the volume of contrast enhancement and 18F-FDOPA T/N SUVmax could each differentiate patient groups. No significant differences in 18F-FDOPA uptake were observed by IDH or MGMT status. Multi-variable Cox regression suggested age (hazard ratio [HR] 1.16, p = 0.0001) and continuous measures of 18F-FDOPA PET T/N SUVmax (HR 4.43, p = 0.016) were significant prognostic factors for OS in WHO I to IV gliomas. The authors concluded that the findings of this study suggested a potential role for the use of pre-operative 18F-FDOPA PET in suspected glioma. Increased 18F-FDOPA uptake may not only predict higher glioma grade, but also worse OS. These researchers stated that prospective studies are needed to validate these observations and to determine the impact of routine pre-operative implementation of 18F-FDOPA PET scanning combined with MRI in clinical decision-making, cost of care, and patient outcomes.

The authors stated that the findings of this study should be interpreted in the context of its limitations. The retrospective nature of the study did not allow for the control of potential confounding variables in the analysis (e.g., physician biases regarding when to order the 18F-FDOPA scan). Additionally, the use of SUVmax within the tumor region instead of strictly examining the distribution in MR-defined regions (e.g., CE or T2/FLAIR hyperintense regions) was a potential limitation; but was chosen based on similar approaches and the ease of rapid evaluation in clinical contexts. The highest 18F-FDOPA extra-striatal uptake was noticed outside of the MRI-based tumor region in 11 cases (24.4 %) during the blinded analysis, and 9 of these instances occurred in the posterior fossa in normal brain tissue. This may represent a pattern of normal brain bio-distribution of 18F-FDOPA uptake. The reasons for this finding of increased 18F-FDOPA uptake in the posterior fossa despite absence of glioma remain unclear. Previous studies in canine and rodent brain have identified expression of the L-type amino acid transporter 1 (LAT1, the primary transporter responsible for 18F-FDOPA uptake in the brain) in the cerebellum, which was found to correlate with 18F-FDOPA uptake. The fact that no recurrences

occurred in the non-co-localizing regions (predominantly posterior fossa) suggested that high 18F-FDOPA uptake in the posterior fossa should not pose a clinical concern if the suspected glioma was outside of the posterior fossa based on MRI.

Humbert et al (2019) evaluated the therapeutic impact and diagnostic accuracy of 18F-DOPA PET/CT in patients with glioblastoma or brain metastases. Patients with histologically proven glioblastoma or brain metastases were prospectively included in this mono-centric clinical trial (IMOTEP). Patients were included either due to a clinical suspicion of relapse or to assess residual tumor infiltration after treatment. Multi-modality brain MRI and 18F-DOPA PET were performed. Patient data were discussed during a Multidisciplinary Neuro-oncology Tumor Board (MNTB) meeting. The discussion was first based on clinical and MRI data, and an initial diagnosis and treatment plan were proposed. Secondly, a new discussion was conducted based on the overall imaging results, including 18F-DOPA PET. A second diagnosis and therapeutic plan were proposed. A retrospective and definitive diagnosis was obtained after a 3-month follow-up and considered as the reference standard. A total of 106 cases were prospectively examined by the MNTB. All patients with brain metastases (n = 41) had a clinical suspicion of recurrence. The addition of 18F-DOPA PET data changed the diagnosis and treatment plan in 39.0 % and 17.1 % of patients' cases, respectively. Concerning patients with a suspicion of recurrent glioblastoma (n = 12), the implementation of 18F-DOPA PET changed the diagnosis and treatment plan in 33.3 % of cases. In patients evaluated to assess residual glioblastoma infiltration after treatment (n = 53), 18F-DOPA PET data had a lower impact with only 5.7 % (3/53) of diagnostic changes and 3.8 % (2/53) of therapeutic plan changes. The definitive reference diagnosis was available in 98/106 patients. For patients with tumor recurrence suspicion, the adjunction of 18F-DOPA PET increased the Younden's index from 0.44 to 0.53 in brain metastases and from 0.2 to 1.0 in glioblastoma, reflecting an increase in diagnostic accuracy. The authors concluded that 18F-DOPA PET had a significant impact on the management of patients with a suspicion of brain tumor recurrence, either glioblastoma or brain metastases, but a low impact when used to evaluate the residual glioblastoma infiltration after a 1st-line radio-chemotherapy or 2nd-line bevacizumab.

Chiaravalloti et al (2019) examined the PFS and the OS in a population affected by primary brain tumors (PBT) evaluated by 18F-DOPA PET/CT. A total of 133 subjects with PBT (65 women and 68 men, mean age of 45 ± 10 years) underwent 18F-FDOPA PET/CT after treatment. Of them, 68 (51.2 %) were grade II, 34 (25.5 %) were grade III and 31 (23.3%) were grade IV. PET/CT was scored as positive or negative and standardized uptake value ratio (SUVr) was calculated as the ratio between SUVmax of the lesion versus that of the background. Patients have been observed for a mean of 24 months. The outcome of 18F-FDOPA PET/CT scan was significantly related to the OS and PFS in grade II gliomas. In grade II PBT, the OS proportions at 24 months were 100 % in subjects with a negative PET/CT scan and 82 % in those with a positive scan. Gehan-Breslow-Wilcoxon test showed a significant difference in the OS curves (p = 0.03) and the HR was equal to 5.1 (95 % CI: 1.1 to 23.88). As for PFS, the proportion at 24 months was 90 % in subjects with a negative PET/CT scan and 58 % in those with a positive scan. Gehan-Breslow-Wilcoxon test showed a significant difference in the OS curves (p = 0.007) and the HR was equal to 4.1 (95 % CI: 1.3 to 8). These researchers did not find any significant relationship between PET outcome and OS and PFS in grade III and IV PBT. The authors concluded that a positive 18F-FDOPA PET/CT scan was related to a poor OS and PFS in subjects with low-grade PBT. This imaging modality could be considered as a prognostic factor in these subjects.

Cicone et al (2019) noted that the role of amino acid PET in glioma grading and outcome prognostication has not yet been well established. This is particularly true in the context of the new WHO 2016 classification, which introduced a definition of glioma subtypes primarily based on molecular fingerprints. These investigators correlated F-DOPA uptake parameters with IDH mutation, 1p/19q status, and survival outcomes in patients with glioma. The study population consisted of 33 patients (17 men/16 women, mean age of 46 ± 13 years) who underwent F-DOPA PET/CT for the evaluation of tumor extent before the start of chemo- or radiotherapy. The presence of IDH mutation and 1p/19q status was assessed in all the cases. Tumor volume and semi-quantitative uptake parameters, namely SUVmax, tumor-to-normal brain ratio and tumor-to-normal striatum ratio, were calculated for each tumor. Imaging-derived parameters were compared between patients stratified according to molecular fingerprints, using parametric or non-parametric tests, where

appropriate. The Kaplan-Meier method was used to assess differences of OS and PFS between groups. PET parameters were also tested as prognostic factors in univariate Cox survival regression models. There were 12 IDH-wild-type and 21 IDH-mutant patients. Stratification according to 1p/19q co-deletion resulted in 20 non-co-deleted and 13 co-deleted patients. Median follow-up time from PET/CT examination was 30.5 months (range of 3.5 to 74 months). Semi-quantitative uptake parameters did correlate neither with IDH mutation nor with 1p/19q status. Uptake was similar in low-grade and high-grade tumors, respectively. In addition, F-DOPA uptake parameters, macroscopic tumor volume, or tumor grade did not stratify OS, while a correlation between SUVmax and PFS was shown in the subgroup of astrocytoma. On the other hand, IDH mutation status and presence of 1p/19q co-deletion had a significant impact on survival outcomes. The prognostic value of IDH mutation status was also confirmed in the subgroup of patients with astrocytic tumors. The authors concluded that F-DOPA uptake parameters did not correlate with tumor molecular and histological characteristics; the predictive value of PET-derived parameters on outcomes of survival was limited.

Furthermore, an UpToDate review on “Overview of the clinical features and diagnosis of brain tumors in adults” (Wong and Wu, 2020) does not mention DOPA PET/CT as a management option.

FDG-PET/CT Scan for Initial Staging of Kaposi's Sarcoma

Bertagna et al (2010) reported on the case of a 38-year old man who was affected by the human immunodeficiency virus (HIV) with a diagnosis of Castleman's disease, plasmablastic type human herpes virus 8 (HPV-8) infection, and Kaposi sarcoma (KS) based on a histological examination of 1 cervical lymph node biopsy. The patient underwent (18)F-FDG-PET/CT. The authors concluded that (18)F-FDG-PET/ CT appeared to be a valuable tool in patients with HIV-associated Castleman's disease and KS; it allowed accurate staging and identified more sites of disease than conventional CT.

Ozdemir et al (2014) stated that although mucocutaneous sites are the most frequently encountered sites of involvement, KS may also occasionally involve the breast and the skeletal, endocrine, urinary and nervous systems. Various imaging modalities may be used to delineate the extent of the disease by detecting unexpected sites of involvement. These investigators reported a case of classical type KS,

in whom staging with (18)F-FDG PET/CT imaging disclosed widespread disease and unexpected findings of bone and salivary gland involvement.

Furthermore, an UpToDate review on “AIDS-related Kaposi sarcoma: Staging and treatment” (Groopman, 2020) does not mention PET/CT as a staging tool.

Gallium-68 PSMA PET/CT and Piflufolastat F-18 (Pylarify) Scan for Prostate Cancer

National Comprehensive Cancer Network’s clinical practice guideline on “Prostate cancer” (Version 2.2020) states that “Newer tracers under development, but they are neither FDA cleared nor readily available and are considered investigational at this time. For instance, gallium-68 prostate-specific membrane antigen (PSMA) may provide better detection of recurrences at lower PSA levels than reported for FDA-approved imaging agents; and has comparable sensitivity (76 % - 88 %) and specificity (86 % - 100 %)“.

Calais et al (2019) stated that NCCN guidelines consider 18F-fluciclovine PET-CT for PCa biochemical recurrence localization after radical prostatectomy, whereas European Association of Urology (EAU) guidelines recommend PSMA PET-CT. To the best of the authors’ knowledge, no prospective head-to-head comparison between these tests has been carried out so far. In a prospective, single-center, open-label, single-arm study, these researchers compared paired 18F-fluciclovine and PSMA PET-CT scans for localizing biochemical recurrence of PCa after radical prostatectomy in patients with low PSA concentrations (less than 2.0 ng/ml). Patients older than 18 years of age with PCa biochemical recurrence after radical prostatectomy and PSA levels ranging from 0.2 to 2.0 ng/ml without any prior salvage therapy and with a Karnofsky performance status (KPS) of at least 50 were eligible. Patients underwent 18F-fluciclovine (reference test) and PSMA (index test) PET-CT scans within 15 days. Detection rate of biochemical recurrence at the patient level and by anatomical region was the primary endpoint. A statistical power analysis demonstrated that a sample size of 50 patients was needed to show a 22 % difference in detection rates in favor of PSMA (test for superiority). Each PET scan was interpreted by 3 independent masked readers and a consensus majority interpretation was generated (2 versus 1) to determine positive findings. Between February 26, 2018 and

September 20, 2018, a total of 143 patients were screened for eligibility, of whom 50 patients were enrolled into the study. Median follow-up was 8 months (inter-quartile range [IQR] 7 to 9). The primary endpoint was met; detection rates were significantly lower with 18F-fluciclovine PET-CT (13 [26 %; 95 % CI: 15 to 40] of 50) than with PSMA PET-CT (28 [56 %; 41 to 70] of 50), with an OR of 4.8 (95 % CI: 1.6 to 19.2; p = 0.·0026) at the patient level; in the sub-analysis of the pelvic nodes region (4 [8 %; 2 to 19] with 18F-fluciclovine versus 15 [30 %; 18 to 45] with PSMA PET-CT; OR 12.0 [1.8 to 513.0], p = 0.0034); and in the sub-analysis of any extra-pelvic lesions (none [0 %; 0 to 6] versus 8 [16 %; 7 to 29]; OR non-estimable [95 % CI: non-estimable], p = 0.0078). The authors concluded that with higher detection rates, PSMA should be the PET tracer of choice when PET-CT imaging is considered for subsequent treatment management decisions in patients with PCa and biochemical recurrence after radical prostatectomy and low PSA concentrations (less than 2.0 ng/ml).

Perera and co-workers (2020) stated that accurate staging of high-risk localized, advanced, and metastatic PCa is becoming increasingly more important in guiding local and systemic treatment. Gallium-68 prostate-specific membrane antigen (PSMA) PET has increasingly been used globally to evaluate the local and metastatic burden of PCa, typically in biochemically recurrent or advanced disease. Following the authors' previous meta-analysis, a high-volume series has been reported highlighting the use of 68Ga-PSMA PET in this setting. These researchers carried out a systematic review and meta-analysis to update reported predictors of positive 68Ga-PSMA PET according to prior therapy and proportion of positivity in various anatomical locations with sensitivity and specificity profiles. They performed critical reviews of Medline, Embase, ScienceDirect, Cochrane Libraries, and Web of Science databases in July 2018 according to the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) statement. Quality assessment was conducted using Quality Assessment if Diagnostic Accuracy Studies-2 tool. Meta-analyses of proportions were carried out using a random-effect model. Summary sensitivity and specificity values were obtained by fitting bi-variate hierarchical regression models. A total of 37 articles including 4,790 patients were analyzed. For patients with biochemical recurrence, positive 68Ga-PSMA PET scans increased with higher pre-PET PSA levels. For PSA categories 0 to 0.19, 0.2 to 0.49, 0.5 to 0.99, 1 to 1.99, and greater than or equal to 2 ng/ml, the percentages of positive scans were 33 %,

45 %, 59 %, 75 %, and 95 %, respectively. No significant differences in positivity were noted between Gleason sums less than or equal to 7 and greater than or equal to 8. Significant differences in positivity after biochemical recurrence in the prostate bed were noted between radical prostatectomy (22 %) and radiotherapy (52 %) patients. On per-node analysis, high sensitivity (75 %) and specificity (99 %) were observed. The authors concluded that Ga-68-PSMA PET improved detection of metastases with biochemical recurrence, especially at low pre-PET PSA levels of greater than 0.2 ng/ml (33 %) and 0.2 to 0.5 ng/ml (45 %). Ga-68-PSMA-PET produced favorable sensitivity and specificity profiles on meta-analysis of pooled data. This analysis highlighted different anatomic patterns of metastatic spread according to PSMA PET in the primary and biochemically recurrent settings.

Kimura et al (2020) noted that salvage lymph node dissection (sLND) for nodal recurrence in PCa patients with biochemical recurrence (BCR) is still not recommended in current guidelines because of the diagnostic inaccuracy of current conventional imaging. In a systematic review and meta-analysis, these researchers examined the performance of Ga-68 PSMA-PET in detecting PCa lymph node metastasis using pathologic confirmation via sLND. They carried out literature search using the Medline, SCOPUS, Web of Science, and Cochrane Library on November 11, 2018 to identify the eligible studies. Studies were eligible if they examined the diagnostic performance of PSMA-PET before sLND in PCa patients with BCR and reported the number of true positive, false positive, false negative, and true negative on a lesion-based and/or field-based analyses to compare with histopathologic findings in sLND specimens. A total of 14 studies published between 2015 and 2018 comprising 462 patients were selected. The positive predictive value of PSMA-PET before sLND on a patient-based analysis ranged between 0.70 and 0.93. The pooled sensitivity using lesion-based and field-based analyses were 0.84 (95 % CI: 0.61 to 0.95) and 0.82 (95 % CI: 0.72 to 0.89), respectively. The pooled specificity using lesion-based and field-based analyses were 0.97 (95 % CI: 0.95 to 0.99) and 0.95 (95 % CI: 0.70 to 0.99), respectively. The diagnostic odds ratio (DOR) using lesion-based and field-based analyses were 189 (95 % CI: 39 to 920) and 82 (95 % CI: 8 to 832), respectively. The authors concluded that PSMA-PET before sLND provided highly accurate performance with clinically relevant high positive and negative predictive values for detecting lymph node

disease in patients with BCR after local treatment with curative intent for PCa. PSMA-PET can identify the patients who are likely to benefit from sLND and possibly direct to lesion or region-based dissection.

Hofman et al (2020) noted that conventional imaging using CT and bone scan has insufficient sensitivity when staging men with high-risk localized PCa. In a randomized, 2-arm, multi-center study, these investigators examined if novel imaging using PSMA PET-CT might improve accuracy and affect management. They recruited men with biopsy-proven PCa and high-risk features at 10 hospitals in Australia. Patients were randomly assigned to conventional imaging with CT and bone scanning or Ga-68 PSMA-11 PET-CT. First-line imaging was carried out within 21 days following randomization. Patients crossed over unless 3 or more distant metastases were identified. The primary outcome was accuracy of 1st-line imaging for identifying either pelvic nodal or distant-metastatic disease defined by the receiver-operating curve using a pre-defined reference-standard including histopathology, imaging, and biochemistry at 6-month follow-up. From March 22, 2017 to November 2, 2018, a total of 339 men were examined for eligibility and 302 men were randomly assigned -- 152 (50 %) men were randomly assigned to conventional imaging and 150 (50 %) to PSMA PET-CT. Of 295 (98 %) men with follow-up, 87 (30 %) had pelvic nodal or distant metastatic disease. PSMA PET-CT had a 27 % (95 % CI: 23 to 31) greater accuracy than that of conventional imaging (92 % [88 to 95] versus 65 % [60 to 69]; $p < 0.0001$). These researchers found a lower sensitivity (38 % [24 to 52] versus 85 % [74 to 96]) and specificity (91 % [85 to 97] versus 98 % [95 to 100]) for conventional imaging compared with PSMA PET-CT. Subgroup analyses also showed the superiority of PSMA PET-CT (area under the curve of the receiver operating characteristic curve 91 % versus 59 % [32 % absolute difference; 28 to 35] for patients with pelvic nodal metastases, and 95 % versus 74 % [22 % absolute difference; 18 to 26] for patients with distant metastases). First-line conventional imaging conferred management change less frequently (23 [15 %] men [10 to 22] versus 41 [28 %] men [21 to 36]; $p = 0.008$) and had more equivocal findings (23 % [17 to 31] versus 7 % [4 to 13]) than PSMA PET-CT did. Radiation exposure was 10.9 mSv (95 % CI: 9.8 to 12.0) higher for conventional imaging than for PSMA PET-CT (19.2 mSv versus 8.4 mSv; $p < 0.001$). These investigators found high reporter agreement for PSMA PET-CT ($\kappa = 0.87$ for nodal and $\kappa = 0.88$ for distant metastases). In patients who underwent 2nd-line image, management

change occurred in 7 (5 %) of 136 patients following conventional imaging, and in 39 (27 %) of 146 following PSMA PET-CT. The authors concluded that PSMA PET-CT is a suitable replacement for conventional imaging, providing superior accuracy, to the combined findings of CT and bone scanning.

Ling and associates (2021) noted that in December 2020, the FDA 68Ga-PSMA-11 for PET in patients with suspected PCa metastasis who are candidates for initial definitive therapy. 68Ga-PSMA PET is increasingly carried out for these patients and is usually combined with CT. In recent years, 68Ga-PSMA PET has been combined with high-resolution MRI, which is beneficial for T staging and may further enhance the staging of primary PCa. In a systematic review and meta-analysis, these researchers compared the diagnostic accuracy of 68Ga-PSMA PET/MRI with 68Ga-PSMA PET/CT for staging of primary PCa. They carried out a comprehensive literature search using Embase, PubMed/Medline, Web of Science, Cochrane Library, and Google Scholar up to June 24, 2021 in accordance with the PRISMA guidelines; risk of bias was evaluated using the QUADAS-2 tool. The search identified 2,632 articles, of which 27 were included. The diagnostic accuracy of 68Ga-PSMA PET/MRI, measured as the pooled natural logarithm of diagnostic odds ratio (lnDOR), was 2.27 (95 % CI: 1.21 to 3.32) for detection of extra-capsular extension (ECE), 3.50 (95 % CI: 2.14 to 4.86) for seminal vesicle invasion (SVI), and 4.73 (95 % CI: 2.93 to 6.52) for lymph node metastasis (LNM). For 68Ga-PSMA PET/CT, the analysis showed lnDOR of 2.45 (95 % CI: 0.75 to 4.14), 2.94 (95 % CI: 2.26 to 3.63), and 2.42 (95 % CI: 2.07 to 2.78) for detection of ECE, SVI, and LNM, respectively. The overall risk of bias and applicability concerns were assessed as moderate and low, respectively. The authors concluded that 68Ga-PSMA PET/MRI showed high diagnostic accuracy equivalent to that of 68Ga-PSMA PET/CT for detection of ECE, SVI, and LNM in staging of PCa.

In a systematic review and meta-analysis, Satapathy and colleagues (2021) examined the diagnostic performance of 68Ga-PSMA PET/CT in the initial detection of PCa in patients with clinical or biochemical findings suspicious for PCa. This systematic review followed PRISMA guidelines. Searches in PubMed, Scopus, and Embase were carried out using relevant keywords, and articles published through April 30, 2020, were included. Using histopathology results as the reference standard, the numbers of true- and false-positives and true- and false-

negatives were extracted. Pooled estimates of diagnostic test accuracy, including sensitivity, specificity, positive likelihood ratio, negative likelihood ratio, and summary ROC (SROC) curve were generated using bi-variate random-effects meta-analysis. A total of 7 studies comprising 389 patients were included in the systematic review and meta-analysis. The pooled sensitivity, specificity, positive likelihood ratio, and negative likelihood ratio for the initial diagnosis of PCa using 68Ga-PSMA PET/CT were 0.97 (95 % CI: 0.90 to 0.99), 0.66 (95 % CI: 0.52 to 0.78), 2.86 (95 % CI: 1.95 to 4.20), and 0.05 (95 % CI: 0.01 to 0.15), respectively. The test had high accuracy; the area under the SROC curve was 0.91 (95 % CI: 0.88 to 0.93). The authors concluded that Ga-68-labeled PSMA PET/CT had excellent sensitivity and negative likelihood ratio in the initial diagnosis of PCa in patients with clinical or biochemical findings suspicious for PCa.

An UpToDate review on “Initial staging and evaluation of men with newly diagnosed prostate cancer” (Taplin and Smith, 2021) states that “Given the greater degree of accuracy, we prefer the use of a PSMA-based radiotracer (i.e., Ga-68 PSMA-11 or piflufolastat F-18) for integrated PET/CT, where available, for the initial staging of men with high and very-high risk disease”.

Furthermore, an UpToDate review on “Rising or persistently elevated serum PSA following radical prostatectomy for prostate cancer: Management” (Lee, 2021) states that “Other prostate-specific radiotracers are available and approved for PET scanning in men with a rising PSA after initial definitive local therapy, including GA-68 PSMA-11 and F-18 DCFPyL (piflufolastat F-18), which also have greater sensitivity than conventional imaging for both nodal and distant metastases, even in men with a very low PSA level. Several studies have shown that the inclusion of one of these PET scans in the diagnostic imaging evaluation of a man with a rising PSA after prostatectomy alters management in 50 % or more, with a greater sensitivity than F-18 fluciclovine at PSA levels < 2 ng/ml”.

Piflufolastat F-18 (Pylarify) is a radioactive diagnostic agent indicated for PET of prostate-specific membrane antigen (PSMA) positive lesions in men with prostate cancer with suspected metastasis who are candidates for initial definitive therapy; or with suspected recurrence based on elevated serum PSA level.

PET Scan for Evaluation of Basaloid Squamous Cell Carcinoma

Soriano et al (2005) stated that the course and prognosis of basaloid squamous cell carcinoma (BSCC) are not well-known. These investigators examined the course and prognosis in a population of BSCC patients. They analyzed a retrospective cohort of 49 patients with BSCC in comparison with a cross-matched population of 49 patients treated for well-to-moderately differentiated SCC. The statistical analysis showed that survival in BSCC group was lower than in the SCC group. Local recurrence in the BSCC group was not higher than in the SCC group, but mortality by distant metastasis was 6 times higher than in the SCC population. The authors considered BSCC patients as a high-risk population and these investigators completed diagnosis explorations including a FDG-PET before curative treatment. They also recommended post-operative or exclusive radiotherapy (RT), which may be associated with concomitant chemotherapy.

Soriano et al (2008) described the natural history and evaluated the prognosis of BSCC of the upper aero-digestive tract as compared to the usual SCC. A total of 62 patients with BSCC and 62 patients with SCC were matched with regards to TNM classification, localization and therapeutic modalities. Histological criteria, follow-up and 5-year survival were compared between the 2 groups. Survival rates were significantly higher in patients with SCC as compared to patients with BSCC. The rate of distant metastasis was 6 times higher in cases of BSCC, which was the major cause of mortality. The authors concluded that the findings of this study showed that BSCC has distinct histopathologic features and an aggressive clinical course, justifying its consideration as a separate entity with poor prognosis. The authors proposed to systematically perform a chest CT-scan and FDG-PET to rule out early distant metastasis and to include adjuvant chemotherapy in treatment protocols.

Silverton et al (2016) noted that treatment with tumor necrosis factor (TNF) antagonists may lead to enhanced susceptibility to certain malignancies. In particular, an association is observed emerging between TNF antagonists and development of SCCs of the skin (in association with psoriasis), the oral cavity, and in the anogenital areas (possibly related to prior human papilloma virus [HPV] infection). These researchers presented the case of a 53-year old woman with a history of severe rheumatoid arthritis (RA), most recently treated with the TNF antagonist etanercept plus methotrexate, presented to the

authors' institution after several months of increasing left pelvis and buttock pain. Evaluation with a CT-directed biopsy of a pelvic side wall mass revealed a metastatic SCC. On a FDG-PET, an additional area of uptake was identified in the left posterior rectum corresponding to a 1-cm nodule palpable on digital examination. Colonoscopic biopsy revealed a BSCC of the rectum as the likely primary site. Immunosuppression following TNF antagonist therapy may have resulted in this unrestrained neoplastic growth; thus, it underscored the need for an initial baseline study of risk factors and identification of patients who are at higher risk for development of a malignancy, in order to achieve a diagnosis at an early stage.

National Comprehensive Cancer Network's clinical practice guideline on "Squamous cell skin cancer" (Version 1.2020) does not mention FDG-PET as a management tool.

PET Scan for Evaluation of Intraductal Papillary Mucinous Neoplasm (IPMN) of the Pancreas

Yamashita et al (2019) stated that identifying malignancy in intraductal papillary mucinous neoplasm (IPMN) of the pancreas remains challenging. These researchers examined the usefulness of 18-FDG PET/CT in distinguishing malignant from benign IPMN of the pancreas. The cases of 79 patients with IPMN of the pancreas who underwent surgical resections between 1996 and 2016 at the authors' institution were enrolled in the present retrospective analysis of predictors for malignancy in IPMN of the pancreas. 18-FDG PET/CT evaluations were performed in 38 patients, and the usefulness of 18-FDG PET/CT for detecting malignancy in IPMN of the pancreas was evaluated. Three factors were significantly related to malignancy in IPMN; the diameter of the main pancreatic duct, the serum value of carcinoembryonic antigen (CEA), and the neutrophil-to-lymphocyte ratio (NLR). 18-FDG PET accumulation was significantly related to malignancy in IPMN; sensitivity of 0.82 and specificity of 0.71. Independent predictors for malignancy in IPMN were as follows: 18-FDG PET accumulation, CEA greater than 1.0 ng/ml, and NLR greater than 2.63. The authors concluded that 18-FDG PET accumulation is a potent new marker for distinguishing malignant from benign IPMNs of the pancreas.

Regenet et al (2020) noted that malignant or high-risk IPMN require surgical resection but surgery should be avoided in patients with IPMN carrying a low risk of malignancy; and 18F-FDG PET has been studied mostly in small, single-center, retrospective series. In a prospective, non-comparative, multi-center study, these researchers examined the value of 18F-FDG PET/CT in differentiating between benign and malignant IPMN of the pancreas. The primary end-point was the specificity of PET/CT for identifying malignant IPMN (in-situ or invasive carcinoma). Final diagnosis was obtained from pathological examination of the resected specimen. Among 120 patients analyzed, 99 had confirmed IPMN, including 24 with malignant lesions, namely 9 with carcinoma in-situ and 15 with invasive carcinoma. The 18-F-FDG PET/CT was positive in 44 and 31 patients in the overall and IPMN populations, respectively. In the 99 IPMN patients, PET/CT showed 13 true-positive, 18 false-positive, 57 true-negative and 11 false-negative results. The sensitivity, specificity, negative predictive value (NPV) and positive predictive value (PPV) for the diagnosis of malignancy were 54.2 %, 76.0 %, 83.8 % and 41.9 %, respectively, versus 64.9 %, 75.9 %, 82.9 % and 54.5 % in the overall population. These investigators could not identify a cut-off value for SUVmax to distinguish benign from malignant lesions. Conventional imaging included CT, magnetic resonance cholangiopancreatography (MRCP) and endoscopic US. In IPMN patients who underwent the 3 techniques, sensitivity, specificity, NPV and PPV were 66.7 %, 84.4 %, 84.4 % and 66.7 %, respectively. The authors concluded that in this study, 18-F-FDG PET/CT did not perform better than conventional imaging to differentiate malignant from benign IPMN.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Pancreatic adenocarcinoma" (Version 1.2020) does not mention FDG-PET as a management option.

PET Scan for Re-Staging of Malignant Peripheral Nerve Sheath Tumor

Kim et al (2017) stated that malignant peripheral nerve sheath tumor (MPNST) is a highly malignant tumor and rarely occurs in the head and neck. These researchers described the imaging features of MPNST of the head and neck. They retrospectively analyzed computed tomography (CT; n = 14), magnetic resonance imaging (MRI; n = 16), and 18-fludeoxyglucose (FDG) positron emission tomography/computed tomography (18F-FDG PET/CT; n = 5) imaging

features of 18 MPNSTs of the head and neck in 17 patients. Special attention was paid to determine the nerve of origin from which the tumor might have arisen. All lesions were well-defined (n = 3) or ill-defined (n = 15) masses (mean of 6.1 cm). Lesions were at various locations but most commonly the neck (n = 8), followed by the intra-cranial cavity (n = 3), para-nasal sinus (n = 2), and orbit (n = 2). The nerve of origin was inferred for 11 lesions: 7 in the neck, 2 in the orbit, 1 in the cerebellopontine angle (CPA), and 1 on the parietal scalp. Attenuation, signal intensity, and enhancement pattern of the lesions on CT and MRI were non-specific. Necrosis/hemorrhage/cystic change within the lesion was considered to be present on images in 13 and bone change in 9. On 18F-FDG PET/CT images, all 5 lesions demonstrated various hyper-metabolic foci with maximum standard uptake value (SUVmax) from 3.2 to 14.6 (mean of 7.16 ± 4.57). The authors concluded that MPNSTs could arise from various locations in the head and neck. Moreover, these investigators noted that although non-specific, a mass with an ill-defined margin along the presumed course of the cranial nerves may aid the diagnosis of MPSNT in the head and neck.

Tripathy et al (2019) stated that MPNSTs are a highly malignant group of soft tissue sarcomas and carry a very poor prognosis. Metastasis to bilateral adrenal glands is very rare in such group of neoplasms. These researchers discussed the case of a 85-year old man who was diagnosed with MPNST from pre-vertebral mass with metastases to bilateral adrenal glands and bone marrow from the beginning, and the role of serial 18F-FDG PET/CT scans in response assessment first to sunitinib (tyrosine kinase inhibitor) and then to liposomal doxorubicin.

Schwabe et al (2019) stated that distinguishing between benign peripheral nerve sheath tumors (BPNSTs) and MPNSTs in neurofibromatosis type 1 (NF1) patients before excision could be challenging. How could MPNST be most accurately diagnosed using clinical symptoms, MRI findings (tumor size, depth, and necrosis), PET measures (SUVpeak, SUVmax, SUVmax tumor/SUVmean liver, and qualitative scale), and combinations of the above? All NF1 patients who underwent PET imaging at the authors' institution (January 1, 2007 to December 31, 2016) were included. Medical records were reviewed for clinical findings; MR images and PET images were interpreted by 2 fellowship-trained musculoskeletal and nuclear medicine radiologists, respectively. Receiver operating characteristic

(ROC) curves were created for each PET measurement; the area under the curve (AUC) and thresholds for diagnosing malignancy were calculated. Logistic regression determined significant predictors of malignancy. The study population of 41 patients contained 34 benign and 36 malignant tumors. Clinical findings did not reliably predict MPNST. Tumor depth below fascia was highly sensitive; larger tumors were more likely to be malignant but without a useful cut-off for diagnosis. Necrosis on MRI was highly accurate and was the only significant variable in the regression model. PET measures were highly accurate, with AUCs comparable and cut-off points consistent with prior studies. A diagnostic algorithm was created using MRI and PET findings. The authors concluded that MRI and PET were more effective at diagnosing MPNST than clinical features. These researchers created an algorithm for pre-operative evaluation of PNSTs in NF1 patients, for which additional validation will be indicated.

There is a lack of evidence regarding the use of PET for re-staging of malignant peripheral nerve sheath tumor.

PET Scan for Re-Staging of Waldenstrom Macroglobulinemia

Muz and colleagues (2020) noted that Waldenstrom macroglobulinemia (WM) is a rare B-cell malignancy characterized by secretion of immunoglobulin M (IgM) and cancer infiltration in the bone marrow. Chemokine receptor such as CXCR4 and hypoxic condition in the bone marrow play crucial roles in cancer cell trafficking, homing, adhesion, proliferation, survival, and drug resistance. These investigators used CXCR4 as a potential biomarker to detect hypoxic-metastatic WM cells in the bone marrow and in the circulation by using CXCR4-detecting radio-pharmaceutical. They radio-labeled a CXCR4-inhibitor (AMD3100) with 64Cu and tested its binding to WM cells with different levels of CXCR4 expression using gamma counter in-vitro. The accumulation of this radio-pharmaceutical tracer was tested in-vivo in subcutaneous and intra-tibial models using PET/CT scan. In addition, peripheral blood mononuclear cells (PBMCs) spiked with different amounts of WM cells ex-vivo were detected using gamma counting. In-vitro, 64Cu-AMD3100 binding to WM cell lines demonstrated a direct correlation with the level of CXCR4 expression, which was increased in cells cultured in hypoxia with elevated levels of CXCR4; and decreased in cells with CXCR4 and HIF-1α knockout. Moreover, 64Cu-AMD3100 detected localized and circulating CXCR4high WM cells with high metastatic potential. The authors concluded that they developed a

molecularly targeted system, 64Cu-AMD3100, which bound to CXCR4 and specifically detected WM cells with hypoxic phenotype and metastatic potential in the subcutaneous and intra-tibial models. The researchers stated that these preliminary findings using CXCR4-detecting PET radio-pharmaceutical tracer indicated a potential technology to predict high-risk patients for the progression to WM due to metastatic potential.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Waldenstrom macroglobulinemia/lymphoplasmacytic lymphoma" (Version 2.2020) does not mention PET as a management tool.

Whole Body PET for Unspecified Periodic Fever Syndrome

An UpToDate review on "Periodic fever syndromes and other autoinflammatory diseases: An overview" (Nigrovic, 2020) does not mention whole body PET as a management tool.

Amyloid PET Scan for Persons with Mild Cognitive Impairment due to Alzheimer Disease or Who are Being Considered for Treatment with Aducanumab (Aduhelm)

Kuller and Lopez (2021) noted that the potential benefit of the anti-amyloid drug aducanumab based on results of recent EMERGE and ENGAGE clinical trials has generated great controversy and has very important implications for the future of anti-amyloid drug therapies. The 2 clinical trials of 18-month duration were carried out in patients with mild cognitive impairment (MCI) and early dementia. The ENGAGE trial showed no benefit while the high-dose EMERGE trial initially also showed no benefit but with longer follow-up there was a significant positive benefit. A recent review form the FDA Advisory Committee was negative while the FDA Office of Neurological Drugs was positive and the statisticians negative. This has generated debate regarding whether the drug should be approved, disapproved, require a new clinical trial, or approved for a subsample only.

On June 7, 2021, the FDA approved aducanumab (Aduhelm) as the 1st treatment to attack a likely cause of Alzheimer's disease (AD). This indication is approved under accelerated approval, and continued approval for this indication may be contingent upon verification of clinical benefit in confirmatory trial(s). Aducanumab aims to remove

amyloid beta from the brains of patients in earlier stages of AD in order to stave off its ravages, which include memory loss and the inability to care for oneself. One of the key inclusion criteria of the phase-III clinical trial of aducanumab (BIIB037) in early AD (ENGAGE) entails that subjects must have a positive amyloid PET scan.

CU 64-Dotatate PET (DetectNet) for Neuroendocrine Tumors

Pfeifer et al (2015) stated that neuroendocrine tumors (NETs) can be visualized using radio-labeled somatostatin analogs. These researchers had previously shown the clinical potential of (64)Cu-DOTATATE in a small first-in-human feasibility study. The aim of the present study was, in a larger prospective design, to compare on a head-to-head basis the performance of (64)Cu-DOTATATE and (111)In-diethylenetriaminepentaacetic acid (DTPA)-octreotide ((111)In-DTPA-OC) as a basis for implementing (64)Cu-DOTATATE as a routine. These investigators prospectively enrolled 112 patients with pathologically confirmed NETs of gastroenteropancreatic or pulmonary origin. All patients underwent both PET/CT with (64)Cu-DOTATATE and SPECT/CT with (111)In-DTPA-OC within 60 days. PET scans were acquired 1 hour after injection of 202 MBq (range of 183 to 232 MBq) of (64)Cu-DOTATATE after a diagnostic contrast-enhanced CT scan. Patients were followed for 42 to 60 months for evaluation of discrepant imaging findings. The McNemar test was used to compare the diagnostic performance. A total of 87 patients were congruently PET- and SPECT-positive. No SPECT-positive cases were PET-negative, whereas 10 false-negative SPECT cases were identified using PET. The diagnostic sensitivity and accuracy of (64)Cu-DOTATATE (97 % for both) were significantly better than those of (111)In-DTPA-OC (87 % and 88 %, respectively, p = 0.017). In 84 patients (75 %), (64)Cu-DOTATATE identified more lesions than (111)In-DTPA-OC and always at least as many. In total, twice as many lesions were detected with (64)Cu-DOTATATE than with (111)In-DTPA-OC. Moreover, in 40 of 112 cases (36 %) lesions were detected by (64)Cu-DOTATATE in organs not identified as disease-involved by (111)In-DTPA-OC. The authors concluded that (64)Cu-DOTATATE was far superior to (111)In-DTPA-OC in diagnostic performance in NET patients; thus, they did not hesitate to recommend implementation of (64)Cu-DOTATATE as a replacement for (111)In-DTPA-OC.

Hope et al (2018) noted that somatostatin receptor (SSTR) PET has demonstrated a significant improvement over conventional imaging (CI) in patients with NETs. SSTR PET should replace 111In-pentetreotide scintigraphy (OctreoScan; Mallinckrodt) in all indications in which the latter is currently being used. These appropriate use criteria (AUC) are intended to aid referring medical practitioners in the appropriate use of SSTR PET for imaging of patients with NETs. The indications were evaluated in well-differentiated NETs. Of the 12 clinical scenarios evaluated, 9 were graded as appropriate: initial staging after the histologic diagnosis of NET, evaluation of an unknown primary, evaluation of a mass suggestive of NET not amenable to endoscopic or percutaneous biopsy, staging of NET before planned surgery, monitoring of NET observed predominantly on SSTR PET, evaluation of patients with biochemical evidence and symptoms of a NET, evaluation of patients with biochemical evidence of a NET without evidence on CI or a prior histologic diagnosis, re-staging at time of clinical or laboratory progression without progression on CI, and new indeterminate lesion on CI with unclear progression. Representatives from the Society of Nuclear Medicine and Molecular Imaging (SNMMI), the American College of Radiology (ACR), the American Society of Clinical Oncology (ASCO), the North American Neuroendocrine Tumor Society (NANETS), the European Association of Nuclear Medicine (EANM), the Endocrine Society, the Society of Surgical Oncology, the National Comprehensive Cancer Network (NCCN), the American College of Physicians (ACP), the American Gastroenterological Association (AGA), and the World Conference on Interventional Oncology (WCIO) assembled under the auspices of an autonomous workgroup to develop the following AUC.

Delpassand et al (2020) stated that studies demonstrated that the investigational 64Cu-DOTATATE radiopharmaceutical may provide diagnostic and logistical benefits over available imaging agents for patients with SSTR-positive NETs. In a prospective phase-III clinical trial, these researchers examined the lowest dose of 64Cu-DOTATATE that facilitates diagnostic-quality scans and evaluated the diagnostic performance and safety in patients with SSTR-expressing NETs. A dose-ranging study was carried out on 12 patients divided into 3 dose groups (111 MBq [3.0 mCi], 148 MBq [4.0 mCi], and 185 MBq [5.0 mCi] ± 10 %) to determine the lowest dose of 64Cu-DOTATATE that produced diagnostic-quality PET/CT images. Using the 64Cu-DOTATATE dose identified in the dose-ranging study, 3 independent

nuclear medicine physicians who were masked to all clinical information read PET/CT scans from 21 healthy volunteers and 42 NET-positive patients to determine those with disease or no disease, as well as those with localized versus metastatic status. Masked-reader evaluations were compared with a patient-specific standard of truth, which was established by an independent oncologist who used all previously available pathology, clinical, and conventional imaging data. Diagnostic performance calculated for 64Cu-DOTATATE included sensitivity, specificity, negative predictive value (NPV), positive predictive value (PPV), and accuracy. Inter- and intra-reader reliability, as well as ability to differentiate between localized and metastatic disease, was also determined. Adverse events (AEs) were recorded from 64Cu-DOTATATE injection through 48 hours after injection. The dose-ranging study identified 148 MBq (4.0 mCi) as the optimal dose to obtain diagnostic-quality PET/CT images. After database lock, diagnostic performance from an initial majority read of the 3 independent readers showed a significant 90.9 % sensitivity ($p = 0.0042$) and 96.6 % specificity ($p < 0.0001$) for detecting NETs, which translated to a 100.0 % sensitivity and 96.8 % specificity after correcting for an initial standard-of-truth misread. Excellent inter- and intra-reader reliability, as well as ability to distinguish between localized and metastatic disease, was also noted. No AEs were related to 64Cu-DOTATATE, and no serious AEs were observed. The authors concluded that 64Cu-DOTATATE PET/CT was a safe imaging technique that provided high-quality and accurate images at a dose of 148 MBq (4.0 mCi) for the detection of somatostatin-expressing NETs.

These researchers stated that 64Cu-DOTATATE offers several potentially practical advantages over 68Ga-DOTATATE. First, 64Cu-DOTATATE is a cyclotron-produced positron emitter that can be manufactured in large-scale with a well-controlled process at a centralized location. The production of 68Ga-DOTATATE, by contrast, is largely limited to major tertiary radio-pharmacies with varying levels of quality control. The centralized and large-scale production of 64Cu-DOTATATE may ensure greater quality control and eliminate the need for a 68Ge/68Ga generator locally. Second, the longer half-life of 64Cu-DOTATATE than of 68Ga-DOTATATE (12.7 versus 1.1 hours) and centralized production may allow for wider geographic distribution, more flexible patient scheduling, and less strain for nuclear medicine technologists who must coordinate radioisotope delivery with patient and scanner availability. Third, the shorter positron range of 64Cu-

DOTATATE and associated improvements in resolution may permit the detection of more or smaller lesions than those observed with 68Ga-DOTATATE. Fourth, the longer half-life of 64Cu-DOTATATE also may provide therapeutic benefits. For example, 64Cu-DOTATATE may permit delayed serial imaging with important implications for personalized dosimetry planning in peptide receptor radionuclide therapy, as well as aid in clarifying suspect findings observed on initial scans. Furthermore, the 12.7-hour half-life of 64Cu-DOTATATE may improve radio-guided surgery using a dedicated positron hand-held probe.

National Comprehensive Cancer Network's clinical practice guideline on "Neuroendocrine and adrenal tumors" (Version 1.2021) notes that "SSRT PET tracers include: 68Ga-DOTATATE, 64CU-DOTATATE, 68Ga-DOTATOC".

Full-Body Ga-68 DOTATATE PET/CT for Evaluation of Atypical Meningioma

Unterrainer et al (2019) reported on the case of a 43-year old woman who presented with suspected recurrence of atypical meningioma World Health Organization (WHO) histological Grade-II with extensive intra-cranial lesions with high Ga-DOTATATE uptake. Moreover, numerous gallium(Ga)-DOTATATE-positive bone, lung, and liver lesions were observed. For final diagnosis, biopsies taken from a lung lesion revealed distant metastases of the atypical meningioma. This case underlined the high diagnostic power of Ga-DOTATATE positron emission tomography/computed tomography (PET/CT) for the staging of meningioma even beyond cerebral or spinal lesions; in case of distant lesions in patients with known meningioma, differential diagnosis should also contain metastases despite their rare occurrence. Moreover, this case emphasized radioligand therapy especially in metastatic meningioma.

Pelak and d'Amico (2019) examined the potential value of Ga-68-DOTATATE PET/CT in predicting the risk of progression in non-benign meningioma after definite irradiation. These researchers retrospectively reviewed their patients with meningiomas who had the highest risk of progression: WHO histological Grade-II and III tumors and with macroscopic disease as identified in Ga-68-DOTATATE PET/CT. A total of 13 patients were included in this study. For each tumor, the following quantifiers were measured: maximum and mean

standardized uptake volume (SUV), standard deviation, metabolic tumor volume (MTV), total lesion activity, and coefficient of variation. Each of the quantifiers except for maximum SUV was obtained with 3 different SUV thresholds: muscle-based (ms), liver-based (liv), and gradient-based (gb). The quantifiers were analyzed in univariate Cox model for their prognostic value for progression-free survival (PFS) and overall survival (OS). Mean follow-up of the patients was 28.2 months. The 2-year PFS and OS was 28.1 % and 76.9 %, respectively. The MTVgb was a significant predictor for PFS (risk of progression of disease above versus below the 34 cm3 threshold: 100 % versus 28.3 %, p = 0.0003). Clinically, the male sex also influenced PFS (hazard ratio [HR] = 13.06; 95 % confidence interval [CI]: 1.56 to 109.25; p = 0.018). The mean SUVms (p = 0.041) and SUVgb (p = 0.048) had a prognostic value for predicting the risk of death. The authors concluded that Ga-68-DOTATATE PET/CT has potential to predict disease progression in non-benign meningioma patients. Moreover, these researchers stated that further prospective studies with larger patient cohorts are needed for validating and standardizing these findings.

The authors stated that the principal limitation of this trial was its limited cohort size (n = 13). This could be attributed to the epidemiology and management of meningiomas: currently, there are no guidelines recommending pre-surgical DOTA PET/CT for all patients, and due to the fact that about 90 % of meningiomas are histologically WHO Grade-I, and have excellent outcomes after radiotherapy, these investigators would not expect that it is widely introduced. On the contrary, this trial included highly negatively selected patients featuring the biggest risk of progression and death. Patients with sub-totally resected non-benign meningiomas have been reported to have significantly worse outcomes than ones in whom an oncologically radical resection (Simpson Grade I and II) was performed for both PFS (31 % to 36 % versus 63 % to 74 %) and OS (particularly notable for WHO Grade-III tumors: as low as 10 % to 17 % versus 45 % to 75 %). Furthermore, while the metabolic tumor volume (MTV) parameter appeared to be less dependent on the examination protocol and potentially well reproducible, the same could not be said about the mean SUV. A change in the resolution or acquisition time would significantly alter the mean and maximum SUV values; these researchers were therefore aware of the limited practical usability of the mean SUVs that this trial found prognostic for OS. The most

notable application of well-standardized Ga-68-DOTA PET/CT quantifiers would be a trial escalating the radiation dose in meningiomas most prone to local treatment failure. Before this is possible, bigger patient cohorts must be prospectively studied. Such analysis should also be multi-centric to increase the chance of finding enough individuals and evaluate the reproducibility of PET/CT quantifiers across different protocols, scanners, or software.

Furthermore, UpToDate reviews on “Management of atypical and malignant (WHO grade II and III) meningioma” (Shih and Park, 2021), and “Epidemiology, pathology, clinical features, and diagnosis of meningioma” (Park, 2021) do not mention Ga-68 DOTATATE PET/CT as a management tool.

PET Scan Dotatate for Evaluation of Recurrent Insulinoma

Novruzov e al (2019) reported on the case of a 54-year old man with multiple endocrine neoplasia type 1 (MEN-1) who had previous history of parathyroid surgery and left thyroid lobectomy 5 years earlier; and was referred for recurrent hypoglycemic episodes. Ga-DOTATATE positron emission tomography/computerized tomography (PET/CT) had showed multiple lesions in the right lung, liver, and pancreas. Biopsy from pancreas revealed low-grade neuroendocrine neoplasia. After 2 fractions of Lu-DOTATATE therapy, the size of lesions and its activity reduced on the Ga-DOTATATE scan and the hypoglycemic episodes manifested every day have scaled down to 1 time over 1-year follow-up.

Garg et al (2020) described the case of a 38-year old woman who presented with recurrent episodes of hypoglycemia for 5 years. On 72-hour fast test, critical sample biochemistry was suggestive of endogenous hyper-insulinemic hypoglycemia. Both contrast-enhanced computed tomography (CE-CT) and 68Ga-DOTATATE PET/CT revealed no pancreatic lesion but showed a jejunal lesion suggestive of neuroendocrine tumor (NET) but not confirmatory of insulinoma. 68Ga-Exendin-4 PET/CT showed intense uptake in the proximal jejunum, and this being a specific scan for insulinoma, confirmed it as an ectopic insulinoma. The patient attained normoglycemia following excision of this NET confirming it to be a case of ectopic insulinoma located in the jejunum. Although most insulinomas are located in the pancreas, rarely ectopic cases have been described in the spleen, peri-

splenic tissue, duodeno-hepatic ligament, adjacent to the ligament of Treitz, duodenum, and the jejunum. Functional scanning with 68Ga-Exendin-4 PET/CT scan aided the localization of ectopic insulinoma.

PET Scan for Evaluation of Erdheim-Chester Disease

An UpToDate review on “Erdheim-Chester disease” (Jacobsen, 2021) states that “Bone lesions that are not apparent on plain radiographs can be detected by bone scintigraphy, computed tomography (CT), magnetic resonance imaging (MRI), or positron emission tomography (PET) … We suggest the following studies at the time of diagnosis … Positron emission tomography (PET)/CT; although bone scintigraphy can demonstrate the iconic radiologic features, unlike PET, bone scintigraphy does not assess extra-osseous involvement”.

PET Scan for Evaluation of Hemangiopericytoma

Ito et al (2012) noted that choline is a new PET tracer that is useful for the detection of malignant tumor. Choline is a precursor of the biosynthesis of phosphatidylcholine, a major phospholipid in the cell membrane of eukaryotic cells. Malignant tumors have an elevated level of phosphatidylcholine in cell membrane; thus, choline is a marker of tumor malignancy. In this case-report, the patient was a 51-year old man with repeated recurrent hemangiopericytoma in the skull base. These researchers performed Choline-PET in this patient after various treatments and compared findings with those of FDG-PET. Choline accumulated in this tumor, but FDG did not accumulate. They diagnosed this tumor as residual hemangiopericytoma and performed the resection of the residual tumor. FDG-PET is not appropriate for skull base tumor detection because uptake in the brain is very strong. The authors emphasized the usefulness of choline-PET for the detection of residual hemangiopericytoma in the skull base after various treatments, compared with FDG-PET.

Jong et al (2014) stated that intracranial hemangiopericytoma (IHPC) is a rare tumor representing less than 1 % of all central nervous system (CNS) tumors and is often indistinguishable from meningioma on structural imaging alone. Unlike meningioma, IHPC is an aggressive tumor with the propensity for early locoregional recurrence and distant metastases. Hence, its management strategies differ greatly from that of meningioma. Some investigators have reported the potential role of multi-tracers (F-FDG, C-methionine, and O-H2O) PET imaging in

distinguishing IHPC from meningioma. These researchers described the findings of dual-tracer (C-acetate and F-FDG) PET/CT imaging in a histopathologically proven case of IHPC with extensive extracranial osseous metastases that showed significantly greater C-acetate avidity.

Hammes et al (2017) reported on the case of a 76-year old man with biochemical relapse of prostate cancer who underwent Ga-prostate-specific membrane antigen (PSMA) PET/CT. Besides a local lymph node metastasis, a nodular structure inside the left orbit caudal to the optic nerve showed increased uptake. A metastasis in this location is unlikely. The subsequently performed MRI showed the structure being T1 hypo-intense, T2 indifferent, and strongly gadolinium contrast agent enhancing. Histopathologic examination after surgical removal identified the tumor as hemangiopericytoma, which rarely occurs in the orbit. Regarding the intense uptake observed, PSMA-targeting PET tracers could bear potential for staging purposes of this tumor entity.

Patro et al (2018) presented the case of a 53-year old woman who initially diagnosed as hemangiopericytoma in the posterior fossa on the right side post-excision with immunohistochemistry staining for CD34 being positive. Presently, the patient had difficulty in walking due to back pain and pain in left arm. Imaging with F-FDG PET showed low glucose avidity in disease sites but Ga-PSMA PET unequivocally demonstrated multiple skeletal and liver metastases with intense PSMA avidity. Patient received palliative radiotherapy to bone metastasis and was planned for chemotherapy. This report added to the list of applications of Ga PSMA PET and a possible theragnostic target.

Jehanno et al (2019) reported the case of a 50-year old man, with previous history of grade-3 intracranial hemangiopericytoma with initial complete surgical resection, addressed for local recurrence. Surgical revision performed 18 months after initial surgery allowed only partial resection, leaving residual disease along the optic nerve. Complementary radiotherapy with proton was decided. F-FDG PET/CT and F-choline PET/CT were both performed for treatment planning. F-FDG PET showed no uptake of the residual tumor, whereas F-choline depicted highly metabolic residual disease uptake with excellent delineation of local recurrence. The authors concluded that F-choline PET/CT appeared as a useful PET tracer for hemangiopericytoma imaging.

Lavacchi et al (2020) noted that solitary fibrous tumor / hemangiopericytoma with primary tumor location in the CNS accounts for less than 1 % of all CNS tumors. Despite the relatively indolent clinical course, extracranial metastases are reported in 28 % of cases. In recent years, NAB2-STAT6 gene fusion has been recognized as the pathognomonic molecular feature of solitary fibrous tumor / hemangiopericytoma and STAT6 immunohistochemistry has been shown to be a sensitive and specific surrogate for the identification of the gene fusion in these patients. These investigators reported 2 cases of patients who experienced occurrence of diffuse extracranial metastases several years after successful surgery for an intracranial solitary fibrous tumor / hemangiopericytoma. In the 1st patient, the metastases had maintained similar histological features to the primary tumor; in contrast, in the 2nd case, a dedifferentiation occurred with loss of expression of CD34 and Bcl-2. These different histological features were associated with radically different behaviors. Whereas the 1st case experienced an indolent course of the disease, the 2nd patient had a rapid disease progression and deterioration of clinical conditions. The molecular imaging findings in these 2 cases and the role of functional imaging for tumor detection, disease staging and monitoring in this rare cancer were also discussed. Recurrences and metastases maintained high expression of somatostatin receptors confirmed by somatostatin receptor imaging in the 1st case. In contrast, in the 2nd patient, the abrupt transition into a highly aggressive form was associated with the absence of somatostatin receptors at 111In Pentetreotide scan and intense hyper-metabolism at 18F-FDG PET.

PET Scan for Evaluation of Light Chain Deposition Disease

UpToDate reviews on “Clinical presentation, laboratory manifestations, and diagnosis of immunoglobulin light chain (AL) amyloidosis” (Dispenzieri, 2021), and “Pathogenesis of immunoglobulin light chain (AL) amyloidosis and light and heavy chain deposition diseases” (Rajkumar, 2021) do not mention PET scan as a management tool.

PET Scan for Evaluation of Uterine Leiomyosarcoma

In a retrospective study, Huang et al (2020) analyzed the features of CT, MRI and PET/CT and their diagnostic value for spinal osteoblastomas (OBs). The radiological and clinical data of 21 patients with histopathologically-confirmed spinal OBs were analyzed; 16 of the

21 cases were benign and 5 were aggressive OBs. Tumors were located in the lumbar (n = 11), cervical (n = 4), thoracic (n = 5), and sacral (n = 1) spinal regions; 19 cases were centered in the posterior elements of the spine, 13 of which extended into the vertebral body. Punctate or nodular calcifications were found in all cases on CT with a complete sclerotic rim (n = 12) or incomplete sclerotic rim (n = 8). The flare phenomenon (indicative of surrounding tissue inflammation) was found in 17/21 cases on CT, thin in 11 cases and thick in 6 cases, and in 19/19 cases on MRI, thin in 1 case and thick in 18 cases. On 18F-FDG PET/CT, all cases (8/8) were metabolically active with the SUVmax of 12.3 to 16.0; the flare sign was observed in 8 cases, including 7 cases of hypo-metabolism and 1 case of co-existence of hyper-metabolism and hypo-metabolism. Based on CT, 3, 12, and 6 cases were classified as Enneking stage 1, 2 and 3, respectively. Of 19 cases with MRI, 1 and 18 cases were classified as Enneking stage 2 and 3, respectively. The authors concluded that spinal OB has multiple unique characteristic radiological features. Although a larger sample size is needed, combining CT, MRI and PET may be beneficial to optimize pre-operative diagnosis and care of patients with OBs.

An UpToDate review on “Evaluation and management of complete and impending pathologic fractures in patients with metastatic bone disease, multiple myeloma, and lymphoma” (Damron et al, 2021) states that “The main differential diagnosis in an adult aged 40 and over for a pathologic fracture caused by metastatic cancer or multiple myeloma includes benign insufficiency or stress fracture, and fracture associated with a primary benign tumor (enchondroma, fibrous dysplasia, giant cell tumor of bone, Langerhans cell histiocytosis, osteoblastoma), primary bone sarcoma (Pagetoid or post radiation osteosarcoma, chondrosarcoma, undifferentiated high-grade pleomorphic sarcoma of bone [previously termed malignant fibrous histiocytoma of bone]), plasmacytoma, and primary lymphoma of bone. When the underlying lesion associated with the fracture is aggressive in appearance, the differential diagnosis is limited to metastatic carcinoma, plasma cell dyscrasia (myeloma, solitary plasmacytoma), lymphoma, and bone sarcoma. Radiographic imaging may assist in the differential diagnosis. In one series, compared with pathologic fractures caused by metastatic bone tumors (including multiple myeloma), fractures caused by a primary bone sarcoma had a higher incidence of lytic destruction of the bone cortex, mineralization and a soft tissue mass on plain radiographs and CT scans, and a

periosteal abnormality on magnetic resonance imaging (MRI). Taken together, all of these features had a high negative predictive value (with their absence highly suggestive of a metastatic cancer rather than primary bone sarcoma). However, the absence of these features might also point towards a benign bone lesion as well".

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Bone cancer" (Version 1.2021) does not mention osteoblastoma.

PET Scan for Evaluation of Uterine Leiomyosarcoma

An UpToDate review on "Uterine fibroids (leiomyomas): Differentiating fibroids from uterine sarcomas" (Stewart, 2021) states that "Neither computed tomography (CT) nor positron emission tomography/CT with fluorodeoxyglucose (FDG) can reliably differentiate between leiomyomas and uterine sarcomas. While the FDG uptake is generally high in sarcomas and low in leiomyomas, the uptake varies across individual tumors".

Furthermore, an UpToDate review on "Treatment and prognosis of uterine leiomyosarcoma" (Hensley and Leitao, 2021) states that "There are no data on the ideal imaging evaluation for women with LMS. Chest radiograph, computed tomography (CT), magnetic resonance imaging (MRI), and positron emission tomography (PET) scans are all used to evaluate for metastatic disease for soft tissue sarcomas. We perform a CT perioperatively in all women diagnosed with LMS. There are no data on the utility of PET in perioperative staging of LMS, and the data on PET for the detection of recurrence are limited to retrospective case series. There is no evidence that it provides more useful information than either CT or MRI".

FDG-PET for the Diagnosis and Treatment Rosai-Dorfman Disease

Huang et al (2011) noted that Rosai-Dorfman disease (RDD) or "sinus histiocytosis with massive lymphadenopathy" is a benign systemic proliferative disorder of histiocytes that was first described in 1969. The disease usually presented with bilateral painless lymphadenopathy, fever, leukocytosis, high erythrocyte sedimentation rate (ESR), and polyclonal hyper-gammaglobulinemia. Approximately 50 % of the patients may present with extra-nodal involvement. Although skin involvement is the most common form of extra-nodal

diseases, purely cutaneous RDD with no nodal or other extra-nodal lesions accounting is rare. The authors presented the case of a 41-year-old man with purely cutaneous RDD, who underwent FDG-PET/CT for detection of disease extension. They stated that PET/CT showed accurate localization of increased glucose metabolism.

Vuong et al (2013) stated that RDD is a benign form of non-Langerhans-cell histiocytosis (LCH). It is identified by a particular histological profile first observed in febrile lymph nodes. Extra-nodal sites are frequent. The most common site is the skin, which can reveal the disease despite a difficult and delayed diagnosis. In a retrospective study, 7 cases of cutaneous revelation of RDD were studied to delineate the clinical characteristics and facilitate diagnosis and treatment of this extremely rare disease. A total of 6 cases of RDD from 1990 to 2011 were identified in the photographic and histopathological records of the Saint-Louis Hospital and 1 case came from a Bichat Hospital consultation. The diagnosis in all cases was based on histopathology results. Patients consisted of 4 men and 3 women aged between 31 and 69 years. Cutaneous lesions (3 to 20) revealed the disease in all of them and the time from disease onset to diagnosis ranged from 6 months to 5 years. The clinical presentation was erythematous or orange popular nodules or plaques, usually on the face. Microscopically, a dense dermal infiltration was observed, in some cases extending into the subcutaneous tissue, with pale histiocytic cells characterized by emperipolesis, plasma cells, lymphocytes, some neutrophils and variable fibrosis. The diagnosis, initially erroneous in 4 cases, was rectified by a 2nd reading of histopathology slides, and immuno-histochemical studies showed expression of S-100 protein in histiocytes but not CD1a; 3 patients had pure cutaneous RDD; 2 neurological sites and 1 nasal site were also found, with 1 ENT site and sequelae of previous uveitis in 1 patient. All extra-cutaneous sites were identified by clinical examination. Different treatments were proposed according to the sites and impact of the disease. In 1 case, the lesions regressed spontaneously after 18 months. The authors concluded that few RDD series have been published and they mainly concern Asian patients. The ethnic origin of these patients varied. The main findings included the following: First, common clinical findings (orange or erythematous papules or nodules, mostly on the upper body), which should alert the dermatologist and histopathologist to the possible diagnosis of RRD. Second, the possibility, already mentioned in the literature, of spontaneous

regression and a good prognosis. Third, the need for thorough evaluation by thoracic, abdominal and cerebral CT or more a PET scan to screen for potentially dangerous visceral sites, and also clinical follow-up.

Albano et al (2015) stated that RDD is a rare histiocytic disorder characterized by massive lymphadenopathy and with extra-nodal involvement in 25 % to 43 % of cases. The clinical course of RDD is unpredictable with episodes of exacerbation and remissions that can last many years, and treatment strategies can be different according to organ involvement. These investigators reported a case of a 42-year-old woman with extra-nodal disease followed for almost 10 years from the diagnosis who underwent 7 18F-FDG PET/CT. PET/CT has proven to be a useful method for the management of this patient, mainly for the staging, follow-up, and evaluation of treatment results.

Pucar et al (2018) noted that RDD is a rare systemic histiocytic disorder of unknown etiology characterized by the accumulation of enlarged non-Langerhans histiocytes within lymph nodes and extra-nodal sites. The histiocytes display characteristic emperipolesis (non-destructive engulfment of inflammatory cells) and are CD68- and S100-positive and CD1a-negative. Although extra-nodal disease frequently occurs with nodal involvement, isolated extra-nodal disease is uncommon. These investigators reported a case of isolated localized subcutaneous multi-nodular disease on FDG PET/CT. They also included a companion classic RDD case with extensive nodal involvement and a characteristic benign clinical course with spontaneous improvement.

Hadchiti et al (2018) RDD commonly involves the lymph nodes but may secondarily involve the skin. Purely cutaneous disease without lymphatics or internal organ involvement occurs rarely. The present report detailed a rare case of 18FDG PET-CT performed in a 33-year-old male soldier with a purely cutaneous form of RDD. Staging with 18FDG PET-CT was ordered prior to excisional biopsies of the afore-mentioned masses and pathology reported RDD. The authors stated that the present case demonstrated the importance of 18FDG PET-CT imaging in the screening of extra-nodal RDD, even with cutaneous involvement. In the latter case, 18FDG-avidity was usually attributed to

the infiltrative and inflammatory changes caused by the disease process. Ultimately, this imaging modality may aid with making therapeutic recommendations.

FDG-PET for the Management of Light Chain Amyloidosis

An UpToDate review on “Clinical presentation, laboratory manifestations, and diagnosis of immunoglobulin light chain (AL) amyloidosis” (Dispenzieri, 2021) does not mention PET scanning as a management option. Furthermore, National Comprehensive Cancer Network’s clinical practice guideline on “Systemic light chain amyloidosis” (Version 1.2022) does not mention PET scanning / imaging as a management tool.

Kit for the Preparation of Gallium Ga 68 Gozetotide (Illuccix, Locametz) PET

On December 20, 2021, the FDA approved Illuccix, a kit for the preparation of gallium-68 (68Ga) gozetotide (also known as PSMA-11) injection, which is a radioactive diagnostic agent indicated for positron emission tomography of prostate-specific membrane antigen positive lesions in patients with prostate cancer with:

- Suspected metastasis who are candidates for initial definitive therapy
- Suspected recurrence based on elevated serum prostate-specific antigen level.

On March 23, 2022, the U.S. Food and Drug Administration (FDA) approved Locametz (gallium Ga 68 gozetotide), a radioactive diagnostic agent for positron emission tomography (PET) of prostate-specific membrane antigen (PSMA)-positive lesions, including selection of patients with metastatic prostate cancer for whom Pluvicto (lutetium Lu 177 vipivotide tetraxetan) PSMA-directed therapy is indicated.

Hope and colleagues (2021) evaluated the diagnostic efficacy of gallium Ga 68 gozetotide (Locametz) positron emission tomography (PET) in an investigator-initiated prospective multicenter single-arm open-label phase 3 imaging trial for patients with intermediate- to high-risk prostate cancer considered for prostatectomy. A total of 764 men (median [interquartile range] age, 69 [63-73] years) underwent 1 Locametz imaging scan for primary staging, and 277 of 764 (36%)

subsequently underwent prostatectomy with lymph node dissection (efficacy analysis cohort). Pathology reports noted that 75 of 277 patients (27%) had pelvic nodal metastasis. Positive results were noted in in 40 of 277 (14%), 2 of 277 (1%), and 7 of 277 (3%) of patients for pelvic nodal, extra-pelvic nodal, and bone metastatic disease. Sensitivity, specificity, positive predictive value, and negative predictive value for pelvic nodal metastases were 0.40 (95% Confidence Interval [CI], 0.34-0.46), 0.95 (95% CI, 0.92-0.97), 0.75 (95% CI, 0.70-0.80), and 0.81 (95% CI, 0.76-0.85), respectively. In addition, 487 of 764 patients (64%) did not undergo prostatectomy, of which 108 patients were lost to follow-up. Of the patients who had follow-up, they underwent radiotherapy (262 of 379 [69%]), systemic therapy (82 of 379 [22%]), surveillance (16 of 379 [4%]), or other treatments (19 of 379 [5%]). The investigators concluded that in men with intermediate- to high-risk prostate cancer who underwent radical prostatectomy and lymph node dissection, the sensitivity and specificity of Locametz PET were 0.40 and 0.95, respectively.

Fluorodopa F-18 PET for Evaluation of Adults with Suspected Parkinsonian Syndromes/Parkinson Disease

Fluorodopa F-18 Injection is a radioactive diagnostic agent indicated for use in positron emission tomography (PET) to visualize dopaminergic nerve terminals in the striatum for the evaluation of adult patients with suspected Parkinsonian syndromes (PS). Guidance from the National Institute for Health and Care Excellence (2017) states: "Do not use positron emission tomography (PET) in the differential diagnosis of parkinsonian syndromes, except in the context of clinical trials." The guidance explained: "PET has better spatial resolution than SPECT, so it might be anticipated that PET should be of value in differential diagnosis. However, the evidence for PET's role in differentiating Parkinson's disease from other parkinsonian conditions using FDG requires further confirmation. No work was found on PET's ability to differentiate Parkinson's disease from essential tremor. This lack of evidence stems from the high cost and poor availability of PET. Further research is required in this area." A guideline on procedure standards for dopaminergic imaging in Parkinsonian syndromes from the European Association of Nuclear Medicine (EANM) and the Society of Nuclear Medicine and Molecular Imaging (SNMMI) (2020) report that fluorodopa F-18 PET is now used in some centers for the differential

diagnosis of parkinsonism, although procedural guidelines aiming to define standard procedures for fluorodopa F-18 PET imaging in this setting are still lacking.

Large-Vessel Vasculitis

On behalf of the American College of Rheumatology/Vasculitis Foundation, Maz et al (2021) provided evidence-based recommendations and expert guidance for the management of giant cell arteritis (GCA) and Takayasu arteritis (TAK) as exemplars of large vessel vasculitis. Clinical questions regarding diagnostic testing, treatment, and management were developed in the population, intervention, comparator, and outcome (PICO) format for GCA and TAK (27 for GCA, 27 for TAK). Systematic literature reviews were conducted for each PICO question. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology was used to rate the quality of the evidence. Recommendations were developed by the Voting Panel, comprising adult and pediatric rheumatologists and patients. Each recommendation required 70 % or higher consensus among the Voting Panel. These investigators presented 22 recommendations and 2 ungraded position statements for GCA, and 20 recommendations and 1 ungraded position statement for TAK. These recommendations and statements addressed clinical questions relating to the use of diagnostic testing, including imaging, treatments, and surgical interventions in GCA and TAK. Recommendations for GCA included support for the use of glucocorticoid-sparing immunosuppressive agents and the use of imaging to identify large vessel involvement. Recommendations for TAK include the use of non-glucocorticoid immunosuppressive agents with glucocorticoids as initial therapy. There were only 2 strong recommendations; the remaining recommendations were conditional due to the low quality of evidence available for most PICO questions. The authors concluded that these recommendations provided guidance regarding the evaluation and management of patients with GCA and TAK, including diagnostic strategies, use of pharmacologic agents, and surgical interventions.

Recommendation:

For patients with suspected GCA and a negative temporal artery biopsy result (or results), the authors conditionally recommend non-invasive vascular imaging of the large vessels with clinical assessment to aid in

diagnosis over clinical assessment alone. Imaging the large vessels may provide additional evidence of disease (e.g., extra-cranial GCA) when the diagnosis is uncertain following negative temporal artery biopsy results. Potential diagnostic imaging modalities include MR or computed tomography (CT) angiography of the neck/chest/abdomen/pelvis, ultrasonography (US), and 18F-fluorodeoxyglucose positron emission tomography (FDGPET).

Furthermore, an UpToDate review on “Diagnosis of giant cell arteritis” (Salvarani and Muratore, 2022) states that “Positron emission tomography (PET), computed tomography (CT), CT with angiography (CTA), and conventional MRA lack sufficient spatial resolution to permit visualization of the temporal artery. PET scanning of the cranial arteries is also obscured by high fluorodeoxyglucose (FDG) uptake in the brain. However, higher-resolution PET/CT scanners employing time-of-flight (TOF) imaging could have improved detectability, but further study is needed … When the diagnosis of GCA is still suspected in a patient who has had a negative temporal artery biopsy and/or CDUS, the possibility of large vessel involvement should be considered. The diagnostic procedure of choice for suspected large vessel GCA is an advanced imaging study of the aorta and/or its branches. CT or CT with angiography (CTA), MRI or MR angiography (MRA), and positron emission tomography (PET) or PET with CT are useful for the identification of large vessel GCA”.

11C-Metomidate PET for Predicting Treatment Response in Primary Aldosteronism

Lu et al (2022) stated that appropriate treatment of primary aldosteronism (PA) depends on accurate lateralization. 11C-metomidate (MTO) is a tracer used in PET that provides functional information regarding the adrenal cortex. These researchers carried out MTO PET for patients with PA who are managed according to the guideline and verified its correlation with other lateralization modalities and usefulness in outcome prediction. A total of 17 patients with PA who underwent MTO PET and had 1 or more lateralization modality (adrenal venous sampling and/or NP-59 adrenal scintigraphy) were included. SUVmax of each adrenal gland (higher uptake side, HSUVmax; lower uptake side, LSUVmax) and the ratio of HSUVmax to LSUVmax (contrast) were compared with lateralization modalities, post-surgical outcomes, and medical treatment outcomes. Cut-off values were used as outcome predictors. HSUVmax and LSUVmax

increased in the order of bilateral, unilateral, and negative findings of CT, with opposite order of contrast. High discordant rate between MTO PET and other lateralization modalities was noted. Biochemical responders (n = 8) had significantly lower HSUVmax and LSUVmax than non-responders, and clinical responders (n = 6) had borderline lower HSUVmax than non-responders. By optimal cut-off values of HSUVmax and LSUVmax, MTO PET was able to predict biochemical and clinical outcomes in patients with medical treatment. The authors concluded that according to adrenal CT findings, MTO PET presented different uptake patterns. Patients with PA under medical treatment showed significantly lower tracer uptake in responders. These investigators stated that MTO PET may be a useful imaging biomarker to predict medical treatment outcome; however, prospective, large, multi-center study is needed for further validation.

Pre-Operative Setting, and Re-Staging of Metastatic Cholangiocarcinoma

National Comprehensive Cancer Network's clinical practice guideline on "Hepatobiliary cancers" (Version 2.2022) states that "The routine use of PET/CT in the preoperative setting has not been established in prospective trials".

Treatment Planning of Sweat Gland Tumor

Singh et al (2013) stated that primary apocrine sweat gland carcinomas (PASGCs) are rare tumors, commonly located in the axilla. Metastases are common and confer poor prognosis. Given the rarity of these tumors, there is limited knowledge regarding its diagnosis and management. These investigators showed 18F-FDG PET/CT images of a 61-year-old man with PASGCs of the left axilla. PET/CT confirmed the diagnosis as primary axillary malignancy with nodal, pulmonary, and skeletal metastases. Another interesting finding in this case was the presence of FDG-avid calcified metastatic lymph nodes during the initial evaluation. Follow-up PET/CT showed progression of the disease. The authors concluded that FDG PET/CT appeared to be a promising tool in the management of PASGCs.

Golemi et al (2014) reported on the case of a 61-year-old man with known poroma of right lower abdomen, and malignant transformation to poro-carcinoma was suspected and confirmed by biopsy. PET/CT was requested for tumor staging, which revealed high FDG uptake in

the known skin nodules located in the right side of abdominal and chest wall and identified further some adenopathy in the right axillary. All nodules and axillary lymph nodes were removed, and diagnosis of eccrine poro-carcinoma was confirmed. The primary tumor and secondary lesions of poro-carcinoma showed a high glucose metabolism; therefore, PET/CT could be useful for staging, follow-up, and detection of recurrence of patients with eccrine poro-carcinoma.

Kim et al (2017) stated that eccrine carcinoma (EC) is an extremely rare malignant skin cancer arising from eccrine sweat glands with a high metastatic potential. It mainly occurs in the elderly, with equal incidence in both sexes. It usually spreads to regional lymph nodes, with liver, lungs, and bones being the most common sites of distant metastasis. Because of tumor rarity, little is known regarding the value of F-18 FDG PET/CT in evaluating this disease. This case report aimed to increase current knowledge of F-18 FDG PET/CT in eccrine sweat gland carcinoma as a non-invasive imaging tool for evaluating the extension of the disease and detecting distant metastases. They reported the case of a 96-year-old man who presented with lowly progressive, ill-margined erythematous papules and nodules with a crusted and eroded involving multiple sites of groin, scrotum, penis, left pelvic wall, left hip and left thigh for more than 3 years, which became extensive in the past 2 months. The histologic investigation confirmed the diagnosis of an EC. He received F-18 FDG PET/CT to further evaluate the lesions. The authors concluded that FDG PET/CT imaging revealed FDG uptake at the extensive skin lesion, involvement of lymph nodes, and multiple FDG-avid of liver, skeletal and lung metastases.

18F-Flotufolastat (Posluma)-PET/CT Imaging in Patients with Prostate Cancer

Jani et al (2023) stated that the SPOTLIGHT Trial examined diagnostic performance and safety of radio-hybrid 18F-rhPSMA-7.3, a novel high-affinity positron emission tomography radio-pharmaceutical. In this phase-III clinical trial (the SPOTLIGHT Trial), men with prostate cancer (PCa) recurrence underwent PET/CT 50 to 70 min after IV administration of 296 ± 20 % MBq 18F-rhPSMA-7.3. To evaluate the co-primary endpoints (verified detection rate and combined region-level PPV), 3 blinded, independent central readers evaluated the scans. Verified detection rate was equivalent to the overall detection rate × PPV. Standard of truth was established for each patient using

histopathology or confirmatory imaging. Statistical thresholds (lower bounds of the CIs) of 36.5 % and 62.5 % were pre-specified for verified detection rate and combined region-level PPV, respectively. Additional endpoints included detection rate, verified detection rate, and combined region-level PPV in patients with histopathology standard of truth, and safety. The overall 18F-rhPSMA-7.3 detection rate among all 389 patients with an evaluable scan was 83 % (majority read). Among the 366 patients (median PSA of 1.27 ng/ml) for whom a standard of truth (histopathology [n = 69]/confirmatory imaging only [n = 297]) was available, verified detection rate ranged from 51 % (95 % CI: 46.1 to 56.6) to 54 % (95 % CI: 48.8 to 59.3), exceeding the pre-specified statistical threshold. Combined region-level PPV ranged from 46 % (95 % CI: 42.0 to 50.3) to 60 % (95 % CI: 55.1 to 65.5) across the readers, not meeting the threshold. In the subset of patients with histopathology standard of truth, the verified detection rate and combined region-level PPV were both above the pre-specified thresholds (majority read, 81 % [95 % CI: 69.9 to 89.6] and 72 % [95 % CI: 62.5 to 80.7], respectively). No significant safety concerns were identified. The authors concluded that 18F-rhPSMA-7.3 offered a clinically meaningful verified detection rate for localization of recurrent PCa. Despite missing the co-primary endpoint of combined region-level PPV, the totality of the data supported the potential use of 18F-rhPSMA-7.3.

Surasi et al (2023) noted that radio-hybrid (rh) 18F-rhPSMA-7.3 is a novel high-affinity PSMA-targeting radio-pharmaceutical for PCa imaging. In a prospective, multi-center, phase-III clinical trial (the LIGHTHOUSE Trial), these researchers examined the diagnostic performance and safety of 18F-rhPSMA-7.3 in newly diagnosed PCa patients planned for prostatectomy. Patients underwent PET/CT 50 to 70 mins after an injection of 296 MBq 18F-rhPSMA-7.3. Images were interpreted locally and by 3 blinded independent readers. The co-primary endpoints were patient-level sensitivity and specificity for the detection of pelvic lymph node (PLN) metastases, validated using histopathology at PLN dissection. Pre-specified statistical thresholds (lower bounds of 95 % CI) were set at 22.5 % for sensitivity and 82.5 % for specificity. Of 372 patients screened, 352 had evaluable 18F-rhPSMA-7.3-PET/CT, and 296 (99 [33 %] with unfavorable intermediate-risk [UIR] and 197 [67 %] with high-/very-high-risk [VHR] PCa) subsequently underwent surgery. As per the independent reads, 23 to 37 (7.8 % to 13 %) patients had 18F-rhPSMA-7.3-positive PLN; 70 (24 %) patients had 1 or more positive PLNs on histopathology. The

sensitivity for PLN detection was 30 % (95 % CI: 19.6 % to 42.1 %) for reader 1, 27 % (95 % CI: 17.2 % to 39.1 %) for reader 2, and 23 % (95 % CI: 13.7 % to 34.4 %) for reader 3, not meeting the pre-specified threshold. Specificity was 93 % (95 % CI: 88.8 % to 95.9 %), 94 % (95 % CI: 89.8 % to 96.6 %), and 97 % (95 % CI: 93.7 % to 98.7 %), respectively, exceeding the threshold for all readers. Specificity was high (92 % or greater) across both risk stratifications. Sensitivity was higher among high-risk/VHR (24 % to 33 %) than among UIR (16 % to 21 %) patients. Extra-pelvic (M1) lesions were reported for 56 to 98/352 (16 % to 28 %) patients who underwent 18F-rhPSMA-7.3-PET/CT irrespective of surgery. Verification of these (predominantly by conventional imaging) gave a verified detection rate of 9.9 % to 14 % (PPV, 51 % to 63 %). No serious AEs were observed. The authors concluded that across all risk stratifications, 18F-rhPSMA-7.3-PET/CT had high specificity, meeting the specificity endpoint. The sensitivity endpoint was not met, although higher sensitivity was noted among high-risk/VHR than among UIR patients. Overall, 18F-rhPSMA-7.3-PET/CT was well-tolerated, and identified N1 and M1 disease before surgery in newly diagnosed PCa patients.

Kuo et al (2024) examined the impact of urinary activity on interpretation of 18F-flotufolastat (18F-rhPSMA-7.3) PET/CT. These researchers carried out a post-hoc qualitative and quantitative analysis of scans acquired in 2 phase-III clinical trials of 18F-flotufolastat (the LIGHTHOUSE Trial and the SPOTLIGHT Trial). Newly diagnosed or recurrent PCa patients enrolled in the LIGHTHOUSE Trial or the SPOTLIGHT Trial, respectively, underwent PET/CT 50 to 70 mins after IV administration of 296 MBq 18F-flotufolastat. For the present analysis, 718 18F-flotufolastat scans (352 from LIGHTHOUSE and 366 from SPOTLIGHT) were re-evaluated by 3 board-certified nuclear medicine physicians. Reader 1 carried out a quantitative assessment (SUVmax and SUVmean) of bladder activity in a circular region-of-interest over the maximum diameter of bladder activity in the transverse plane. All 3 readers qualitatively assessed the impact of any urinary activity in the bladder on image interpretation using a 3-point scale (0 = no/minimal visible urinary activity, 1 = urinary activity visible but distinction between urine and disease possible, and 2 = assessment inhibited by urinary activity) and the presence/absence of ureteric activity and halo artifacts. A total of 712/718 scans were evaluable. Reasons for exclusion were cystectomy, renal failure, or urinary catheter in-situ (n = 2 each). The median bladder SUVmax and

SUVmean were 17.1 and 12.5, respectively. By majority read, 682/712 (96 %) patients had either no urinary activity (score = 0) or visible activity that could be distinguished from disease uptake (score = 1). In the minority of patients (24, 3.4 %) where urinary activity did impact assessment (score = 2), the median bladder SUVmean was higher (20.5) than those scored 0 (3.8) or 1 (14.0). Ureteric activity was absent in 401 (56 %) patients. Halo artifacts were observed in only 2 (0.3 %) patients (majority read). The authors concluded that 18F-flotufolastat urinary activity did not influence disease assessment for the majority of patients. While this study was not designed as a head-to-head comparison, the median bladder SUVs were lower than previously reported values for other renally cleared PSMA-PET radiopharmaceuticals.

National Comprehensive Cancer Network's clinical practice guideline on "Prostate Cancer" (Version 4.2023) states that PSMA-PET refers to a growing body of radiopharmaceuticals that target PSMA on the surface of prostate cells. There are multiple PSMA radiopharmaceuticals at various stages of investigation. At this time, the NCCN Guidelines only recommend the currently FDA-approved PSMA agents: F-18 piflufolastat PSMA (also known as F-18 DCFPyL), F-18 flotufolastat PSMA (also known as rh-PSMA-7.3), and Ga-68 PSMA-11. Ga-68 PSMA-11, F-18 piflufolastat PSMA, or F-18 flotufolastat PSMA PET/CT or PET/MRI (full body imaging) can be considered as an alternative to bone scan.

Dotatate PET for Radiation Planning for Meningiomas

Perloe et al (2022) stated that radiation treatment planning for meningiomas traditionally entails MRI contrast enhanced images to define residual tumor. However, the gross tumor volume may be difficult to delineate for patients with a meningioma in the skull base sagittal sinus or after resection. Advanced PET imaging using 68Ga-DOTATATE, which has been shown to be more sensitive and specific than MRI imaging, can be used for target volume delineation in these circumstances. These researchers hypothesized that 68Ga-DOTATATE PET scan-based treatment planning would lead to smaller radiation volumes and would detect additional areas of disease compared with standard MRI alone. The data examined retrospective, de-identified, and blinded gross tumor volume contour delineation with 7 CNS specialists (4 CNS radiation oncologists and 3 neuro-radiologists) for 25 patients with a meningioma diagnosis who received

both a 68Ga-DOTATATE PET and an MRI for radiation treatment planning. Both the MRI and the PET were non-sequentially contoured by each physician for each patient. The median MRI volume for each physician ranged from 16.94 to 25.53 cm3. The median PET volume for each physician ranged from 2.09 to 8.36 cm3. The median PET volume was smaller for each physician. Furthermore, 7 of the 25 patients (28 %) had new non-adjacent areas contoured on PET by at least 6 of the 7 physicians that were not contoured by these physicians on the corresponding MRI. These new areas would not have been in the traditional MRI-based volumes. The authors concluded that the findings of this study supported that 68Ga-DOTATATE PET imaging may help radiation oncologists create more precise radiation treatment volumes via finding undetected areas of disease not observed on MRI. Moreover, these researchers stated that treatment planning guided by 68Ga-DOTATATE PET should be studied prospectively.

Gallium-68 Dotatate PET/CT for Monitoring SDHB Mutation (Pathogenic) Consistent with Hereditary Paraganglioma/Pheochromocytoma Syndrome

Muth et al (2019) noted that pheochromocytomas and paragangliomas (PPGLs) are rare tumors and at least 30 % are part of hereditary syndromes. Approximately 20 % of hereditary PPGL are caused by pathogenic germ-line variants in genes of the succinate dehydrogenase complex (SDHx), TMEM127 or MAX. These investigators presented guidelines regarding genetic testing of family members and their surveillance based on a thorough literature review. All cases of PPGL are recommended genetic testing for germ-line variants regardless of patient and family characteristics. At minimum, FH, NF1, RET, SDHB, SDHD and VHL should be tested. Furthermore, testing of MEN1, SDHA, SDHAF2, SDHC, TMEM127 and MAX is recommended. Healthy 1st-degree relatives (and 2nd-degree relatives in the case of SDHD and SDHAF2 which are maternally imprinted) should be offered carrier testing. Carriers of pathogenic variants should be offered surveillance with annual biochemical measurements of methoxy-catecholamines and bi-annual rapid whole-body MRI and clinical examination. Surveillance should start 5 years before the earliest age of onset in the family; therefore, only children eligible for surveillance should be offered pre-symptomatic genetic testing. The surveillance of children younger than 15 years needs to be individually designed. These guidelines provided a framework for patient management with the possibility to follow outcome via national

registries and/or follow-up studies. Together with improved insights into the disease, this may enable optimization of the surveillance scheme in order to minimize both anxiety and medical complications while ensuring early disease detection. It does not mention PET/CT as a monitoring tool.

Garcia-Carbonero et al (2021) stated that PPGLs are rare neuroendocrine tumors that arise from chromaffin cells of the adrenal medulla and the sympathetic/parasympathetic neural ganglia, respectively. The heterogeneity in its etiology makes PPGL diagnosis and treatment very complex. These researchers provided practical clinical guidelines for the diagnosis and treatment of PPGLs from a multi-disciplinary perspective, with the involvement of the Spanish Societies of Endocrinology and Nutrition (SEEN), Medical Oncology (SEOM), Medical Radiology (SERAM), Nuclear Medicine and Molecular Imaging (SEMNIM), Otorhinolaryngology (SEORL), Pathology (SEAP), Radiation Oncology (SEOR), Surgery (AEC) and the Spanish National Cancer Research Center (CNIO). They reviewed the following topics: epidemiology; anatomy, pathology and molecular pathways; clinical presentation; hereditary predisposition syndromes and genetic counseling and testing; diagnostic procedures, including biochemical testing and imaging studies; treatment including catecholamine blockade, surgery, radiotherapy and radiometabolic therapy, systemic therapy, local ablative therapy and supportive care. Finally, they provided follow-up recommendations. These researchers noted that gallium (68Ga)-DOTA peptide PET showed an overall detection rate of 98.6 % in patients with metastatic SDHB[Mut] PPGLs. In head and neck region PGLs (HNPGLs), this is considered the functional image of choice. In polycythemia-related PPGLs and in FH[Mut] or MAX[Mut] PPGLs, 18F-DOPA PET is the functional imaging of choice, and if not available, (I)-123−MIBG-SPECT/CT is recommended. It does not mention PET/CT as a monitoring tool.

PET as Part of Selective Internal Radiation Therapy (SIRT) for Hepatocellular Carcinoma

An UpToDate review on “Overview of treatment approaches for hepatocellular carcinoma” (Abdalla et al, 2023) does not mention the use of PET scan as a part of SIRT.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Hepatocellular Cancer" (Version 2.2023) states that "PET/CT has limited sensitivity but high specificity, and may be considered when there is an equivocal finding. When an HCC is detected by CT or MRI and has increased metabolic activity on PET/CT, higher intralesional standardized uptake value is a marker of biologic aggressiveness and might predict less optimal response to locoregional therapies".

PET for Disseminated Histoplasmosis

An UpToDate review on "Diagnosis and treatment of disseminated histoplasmosis in HIV-uninfected patients" (Kauffman, 2023) does not mention PET scanning as a management option.

PET for Hepatoblastoma

StatPearls; webpage on "Hepatoblastoma" (Musick et al, 2022) stated that US and either CT or MRI are the imaging modalities used to define the extent of tumor involvement of the liver and aid in pre-surgical planning. A chest CT can help detect lung metastasis; the lung is the most common location of metastases and up to 20 % of cases present with metastases. After imaging, a biopsy, alpha-fetoprotein level (AFP), liver function tests (LFTs), and a hepatitis panel are performed as needed. Consent for participation in biologic studies should be requested before the biopsy is performed so the specimen can be properly allocated and handled; PET is not mentioned as a management option.

Furthermore, an UpToDate review on "Overview of hepatoblastoma" (Malagolowkin, 2023) states that "Radiologic imaging is the basis for designating the Pretreatment Extent of Disease (PRETEXT) stage, which denotes the extent of liver involvement, an important aspect of selecting initial treatment. Cross sectional imaging of the chest should be obtained in all patients to evaluate for lung metastasis and is preferable to obtain prior to surgery in a patient with a liver mass, as post-sedation atelectasis can be difficult to distinguish from pulmonary metastatic disease. Bone metastases are uncommon, and thus, routine radionuclide bone scanning (or fluorodeoxyglucose positron emission tomography) is unwarranted in the absence of suspicion for bone metastases".

PET for Evaluation of Abnormal CT Findings (e.g., Possible Carcinomatosis) Status Post Resection of Dermatofibrosarcoma Protuberans

An UpToDate review on review on “Dermatofibrosarcoma protuberans: Treatment” (Mendenhall e al, 2023) states that “Role of diagnostic imaging -- Given the characteristic histologic appearance of most DFSP on core needle or incisional biopsy, imaging techniques are not routinely used in the diagnostic process. However, because of the infiltrative nature of DFSP, simple clinical palpation is not entirely reliable for ascertaining the depth of tumor invasion. Magnetic resonance imaging (MRI) is more sensitive than palpation for assessing depth of infiltration and may be useful for certain DFSPs, particularly large or recurrent tumors. Although MRI is better than clinical palpation, the extent of local invasion may be difficult to appreciate preoperatively, even with optimal MRI evaluation, particularly in sites in the head and neck and upper thorax. The utility of MRI was shown in a study of 43 DFSPs, which included 22 primary tumors, 9 recurrent tumors, and 12 tumors that were extirpated with positive margins and referred for Mohs micrographic surgery (MMS). Clinical palpation correctly predicted the depth of infiltration in the primary DFSP cases with a sensitivity of 58 %t, a specificity of 90 %, a positive predictive value (PPV) of 88 %, and a negative predictive value (NPV) of 64 %. The corresponding values for MRI were 67, 100, 100, and 71 %, respectively. MRI was of little value in determining lateral tumor extension and lost precision for determining the depth of infiltration in tumors located in the head and neck and upper thorax; in 7 of 8 cases, MRI reported the affected plane to be more superficial than it actually was histologically. This finding was attributed to the lower thickness of the skin at these locations and the close proximity of the muscle to the cutaneous surface, which lessened the ability of MRI to detect spread to the fascia and muscle. For recurrent DFSP, the sensitivity, specificity, PPV and NPV of MRI were 60, 100, 100, and 67 %, respectively. On the other hand, MRI was not accurate in cases with margin-positive excisions. Among 12 such patients, MRI suggested tumor persistence in 5, tumor absence in 2, and was uncertain in 5. Histologic study revealed tumor persistence in 10 cases (only 4 of which were defined positive by MRI), and only 2 cases in which no tumor remained, neither of which were negative by MRI. These data suggest that MRI is beneficial for recurrent tumors but not for detecting tumor persistence in cases of incompletely extirpated tumors. Computed tomography (CT) is only indicated in rare cases in

which underlying bone involvement is suspected … The majority of local recurrences become evident within 3 years, but 25 to 30 % develop after 5 years. Thus, posttreatment follow-up must be carried out lifelong. The time to recurrence may be shorter in transformed DFSP (median 11 versus 33.5 months in one large series). The disease-free interval between recurrences shortens progressively. In one report, the median time to relapse for first and second recurrences was approximately 3.5 years, but it was 1 year for the third relapse. Metastases preferentially localize to the lung, but are reported at other sites including brain, bone, and heart. As only 2 to 5 % of cases metastasize, extensive radiographic surveillance with CT, blood counts, and liver function tests are not warranted, unless there are symptoms or signs to suggest metastatic spread. Although it is difficult to determine which cases are at risk for metastasis, they generally involve recurrent lesions that have progressed for many years and when a fibrosarcomatous component is present (i.e., DFSP-FS). This has led some to recommend that follow-up imaging examinations be limited to those with recurrent DFSP or a DFSP with fibrosarcomatous transformation".

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Dermatofibrosarcoma protuberans" (Version 1.2023) does not mention PET imaging as a management option.

PET/CT for Monitoring Recurrence of Littoral Cell Splenic Angioma

StatPearls' webpage on "Littoral cell splenic angioma" (Rai and Gupta, 2023) stated that radiological imaging studies such as MRI and CT scan have not proven adequate in diagnosing littoral cell angioma, given that it is difficult to differentiate from other splenic neoplasms like angiosarcomas, lymphomas, metastatic tumors on imaging. Owing to the hemosiderin deposits in the tumor cells, hypodense lesions are visible on T1 and T2 weighted MRI images. Sonography evaluation is not helpful because of the variable findings of heterogeneous echotexture and no definite lesions. PET/CT is not mentioned as a management option.

Furthermore, an UpToDate review on "Splenomegaly and other splenic disorders in adults" (Bona, 2023) states that "Several imaging methods can be used to assess splenic size and intrasplenic lesions: these include ultrasound, computed tomography (CT), positron emission

tomography with CT (PET-CT), and magnetic resonance imaging (MRI). The choice of modality depends on the reason for testing and patient characteristics. Ultrasound is the best validated modality for assessing splenomegaly, with a standardized measurement of the longest dimension in an oblique cranial-caudal axis. Spleen size may also be estimated by other methods such as CT or MRI performed for other reasons; however, criteria for spleen length are not as well validated by these methods as for ultrasound … Solid masses include benign and malignant tumors and inflammatory lesions such as sarcoidosis. PET with 18F-fluorodeoxyglucose combined with CT (PET-CT) may be used to determine the likelihood of malignancy; PET-avid lesions are most likely to be malignant or to contain granulomas. In most types of lymphoma, PET-CT is typically used for staging and monitoring of disease response".

Angiomatoid Fibrous Histiocytoma

Colangeli et al (2019) stated that angiomatoid fibrous histiocytoma (AFH) is a rarely metastasizing soft-tissue tumor of low-grade malignancy. These investigators reported a case of AFH located in the leg of a 15-year-old female. This case was of particular interest due to its radiological features that raised 2 questions regarding the nature of the disease (is it reactive or tumoral?) and its site of origin (within soft tissues or the tibial periosteum?). These researchers described US, MRI, CT scan, and PET findings that helped answer these questions, understand the real nature of the disease, and its appropriate treatment. This case showed that a single type of imaging technique may not be sufficient to understand the real nature of a musculoskeletal lesion, and that it is necessary to combine all information derived from various imaging techniques in order to correctly diagnose and treat the disease.

Kobayashi (2021) noted that AFH is a rare intermediate malignant tumor that arises mainly in soft tissues, especially in the superficial extremities of patients younger than 30 years. There have been a few reports of AFH arising from sites other than soft tissue, including bone, and unusual site; and age make it difficult to diagnose this rare tumor. These investigators presented a case of a 54-year-old man who was examined for chest pain, and CT incidentally detected a bone tumor at the scapula with destruction of cortical bone and invasion into soft tissue; MRI revealed multiple cystic components with fluid-fluid levels. FDG-PET showed uptake at the axillary lymph node. The CT-guided

needle biopsy revealed spindle cell sarcoma on histopathology. After neoadjuvant chemotherapy, a scapulectomy was carried out. The final post-resection histopathological diagnosis was the same as the pre-operative diagnosis, and no obvious chemotherapeutic effect was observed. Next-generation sequencing (NGS) of RNA isolated from paraffin-embedded tumor tissue revealed that these lesions harbored the EWSR1-CREB1 fusion gene, and the tumor was diagnosed as AFH. CRP level, which was elevated pre-operatively, decreased after the operation, and there was no recurrence or metastasis 5 years after the treatment. The authors concluded that the diagnosis of AFH is difficult when the tumor has an atypical presentation. Comprehensive genomic analysis, especially RNA sequencing, is efficient in diagnosing this rare tumor. Moreover, MRI findings identical to AFH in soft tissue, the presence of paraneoplastic symptoms such as elevated inflammatory markers, and lymph node swelling were clues towards suspecting this tumor.

Furthermore, National Comprehensive Cancer Network's clinical practice guideline on "Soft Tissue Sarcoma" (Version 2.2024) notes that "FDG-PET/CT scan may be useful in staging, prognostication, grading, and determining response to neoadjuvant therapy".

Gallium-68 Dotatate PET-CT for Grade-I Meningioma Treatment Planning

National Comprehensive Cancer Network's Clinical practice guideline on "Central Nervous System Cancers" (Version 3.2024) notes that dotatate PET/CT brain imaging is indicated for suspected recurrence of meningiomas or progressive disease with inconclusive findings on MRI.

Evaluation of Joint Infection, Joint Inflammation Associated with Osteoarthritis

Kamasaki et al (2020) examined the capability of fluorine-FDG (F-FDG)-PET/CT in screening for musculoskeletal inflammation and injury in the knee region. The study included 12 subjects (6 men and 6 women) who complained of knee pain at rest and 94 controls who did not. SUVmax was calculated for the right and left knees and compared via a 4-point numerical rating scale (NRS) for subjective knee pain. Correlations between SUV values were also evaluated. SUVmax was significantly higher in knee joints with resting and/or walking pain than

in pain-free knee joints. SUVmax in knee joints with walking pain were significantly correlated with NRS category ($\beta = 0.129$, $p < 0.001$). The authors concluded that F-FDG-PET/CT may be useful in screening for musculoskeletal inflammation and injury in the knee region. As knee pain is common, especially among the elderly, these researchers stated that clinicians should consider conducting further examinations when F-FDG uptake is identified in knee joints. Moreover, these researchers stated that further studies are needed to clarify the mechanism of 18F-FDG uptake in knee joints.

The authors stated that this study had several drawbacks. First, because these investigators enrolled individuals who participated in cancer screening by PET/CT, selection bias might have occurred at this stage. Second, they did not obtain the final diagnosis of knee pain by conducting further examinations such as knee X-ray and MRI. Third, as these researchers employed NRS values to evaluate the severity of knee pain, the data were subject to differing perceptions between individuals. Fourth, they did not evaluate inflammation markers (e.g., CRP, matrix metalloproteinase-3 [MMP-3], and tumor necrosis factor-alpha [TNF-α]).

Sandstrom et al (2024) stated that although inflammation is known to play a role in knee osteoarthritis (KOA), inflammation-specific imaging is not routinely performed. These investigators examined the role of joint inflammation, measured using [11C]-PBR28, a radioligand for the inflammatory marker 18-kDa translocator protein (TSPO), in KOA. A total of 21 KOA patients and 11 healthy controls (HC) underwent PET/MRI knee imaging with the TSPO ligand [11C]-PBR28. Standardized uptake values were extracted from regions-of-interest (ROIs) semi-automatically segmented from MRI data, and compared across groups (HC, KOA) and subgroups (unilateral/bilateral KOA symptoms), across knees (most versus least painful), and against clinical variables (e.g., pain and Kellgren-Lawrence [KL] grades). Overall, KOA patients showed elevated [11C]-PBR28 binding across all knee ROIs, compared with HC (all $p < 0.005$). Specifically, PET signal was significantly elevated in both knees in patients with bilateral KOA symptoms (both $p < 0.01$), and in the symptomatic knee ($p < 0.05$), but not the asymptomatic knee ($p = 0.95$) of patients with unilateral KOA symptoms. PET signal was higher in the most versus least painful knee ($p < 0.001$), and the difference in pain ratings across knees was proportional to the difference in PET signal ($r = 0.74$, $p < 0.001$); and K-

L grades neither correlated with PET signal (left knee r = 0.32, p = 0.19; right knee r = 0.18, p = 0.45) nor pain (r = 0.39, p = 0.07). The authors concluded that these findings supported further examination of [11C]-PBR28 PET signal as an imaging marker candidate for KOA and a link between joint inflammation and OA-related pain severity.

Furthermore, an UpToDate review on “Imaging techniques for evaluation of the painful joint” (Baccei et al, 2024) states that “The most common clinical application of PET/CT is in oncology (e.g., for staging known malignancies, monitoring the effect of treatment, and evaluation of tumor recurrence), followed by cardiology (e.g., evaluation of ischemia, viability) and neurology (e.g. differential diagnosis of dementia, Parkinson syndrome). While PET is not routinely used in the clinical evaluation of arthritis, it has been employed in the research setting”.

Initial Staging of Primary Clear Eccrine Hidradenocarcinoma

Patel and Oldan (2018) stated that malignant tumors of the sweat glands are exceedingly rare and aggressive tumors. These investigators presented the case of a 60-year-old man with a malignant hidradenocarcinoma that developed in a background of pre-existing benign hidradenoma on the lateral aspect of the knee that was initially resected, but rapidly recurred with associated inguinal lymphadenopathy. F-FDG PET/CT was carried out as part of pre-operative staging, which showed abnormal inguinal lymph nodes and metastatic disease to the lungs. The authors concluded that FDG PET/CT could play a role in the initial staging and follow-up of this rare malignancy.

Wang et al (2023) reported on a case of hidradenocarcinoma, which showed only slight 18 F-FDG uptake; however, the rare sweat gland tumor showed intense tracer uptake on 68 Ga-FAPI PET/CT. The authors stated that this case showed the potential value of 68 Ga-FAPI PET/CT for the evaluation of hidradenocarcinoma.

Ding et al (2024) noted that hidradenocarcinoma is quite rare in clinical practice. These researchers described the 68Ga-FAPI and 18F-FDG PET/CT findings of hidradenocarcinoma of the head and neck in a 75-year-old man. The primary tumor and secondary lesions showed

intense accumulation of 68Ga-FAPI; but only slight 18F-FDG uptake. The authors concluded that this case showed that 68Ga-FAPI PET/CT might be used as a helpful tool for evaluating hidradenocarcinoma.

Monitoring of Cortico-Basal Syndrome

Palleis et al (2024) noted that cortico-basal syndrome (CBS) with underlying 4-repeat tauopathy is a progressive neurodegenerative disease characterized by declining cognitive and motor functions. Biomarkers for evaluating pathologic brain changes in CBS including tau-PET, 18 kDa translocator protein (TSPO)-PET, structural MRI, neurofilament light chain (NfL), or glial fibrillary acidic protein (GFAP) have recently been examined for differential diagnosis and disease staging, yet their association with disease trajectories remains unclear. These researchers carried out a head-to-head comparison of neuroimaging (tau-PET, TSPO-PET, structural MRI) and plasma biomarkers (NfL, GFAP) as prognostic tools for longitudinal clinical trajectories in β-amyloid (Aβ)-negative CBS. This trial enrolled patients with clinically diagnosed Aβ-negative CBS with clinical follow-up data who underwent baseline structural MRI and plasma-NfL analysis for evaluating neurodegeneration, [18F]PI-2620-PET for assessing tau pathology, [18F]GE-180-PET for assessing microglia activation, and plasma-GFAP analysis for assessing astrocytosis. To quantify tau and microglia load, these researchers assessed summary scores of whole-brain, cortical, and sub-cortical PET signal. For structural MRI analysis, these investigators quantified sub-cortical and cortical gray matter volume. Plasma NfL and GFAP values were assessed using Simoa-based immunoassays. Symptom progression was determined using a battery of cognitive and motor tests (i.e., Progressive Supranuclear Palsy Rating Scale [PSPRS]). Using linear mixed models, these researchers examined if the assessed biomarkers at baseline were associated with faster symptom progression over time (i.e., time × biomarker interaction). A total of 21 patients with Aβ-negative CBS with approximately 2-year clinical follow-up data were included. Patients with CBS with more widespread global tau-PET signal showed faster clinical progression (PSPRS: B/SE = 0.001/0.0005, $p = 0.025$), driven by cortical rather than sub-cortical tau-PET. In contrast, patients with higher global [18F]GE-180-PET readouts showed slower clinical progression (PSPRS: B/SE = -0.056/0.023, $p = 0.019$). No association was found between gray matter volume and clinical progression. Concerning fluid biomarkers, only higher plasma-NfL (PSPRS: B/SE = 0.176/0.046, $p < 0.001$) but

not GFAP was associated with faster clinical deterioration. In a subsequent sensitivity analysis, these investigators found that tau-PET, TSPO-PET, and plasma-NfL showed significant interaction effects with time on clinical trajectories when tested in the same model. The authors concluded that tau-PET, TSPO-PET, and plasma NfL showed effect on future clinical progression in Aβ-negative CBS; therefore, this could be useful for future clinical decision-making and for stratifying patients for clinical trials. Moreover, the power calculation data showed that clinical trials in patients with Aβ-negative CBS are feasible and that these baseline biomarkers have potential to reduce sample sizes when designing interventional trials in CBS.

Monitoring of Neurodegenerative Progression of in Parkinson Disease

Beauchamp et al (2023) noted that there are limited validated biomarkers in Parkinson disease (PD) that substantially hinders the ability to monitor disease progression and consequently measure the effectiveness of disease-modifying treatments. Imaging biomarkers, such as vesicular monoamine transporter type 2 (VMAT2) PET, enable enhanced diagnostic accuracy and detect early neurodegenerative changes associated with prodromal PD. These researchers examined if 18F-AV-133 VMAT2 PET is sensitive enough to monitor and quantify disease progression over a 2-year period. 18F-AV-133 PET scans were carried out on patients with PD and REM sleep behavior disorder (RBD) and neurologic controls (NC). All subjects were scanned twice, approximately 26 months apart. Regional tracer retention was calculated with a primary visual cortex reference region and expressed as the SUV ratio. ROIs included caudate, anterior, and posterior putamen. At the time of scanning, subjects underwent clinical evaluation including UPDRSMOTOR test, Sniffin' Sticks, and Hospital Anxiety and Depression Score. Over the 26-month interval, a significant decline in PET signal was observed in all 3 regions in patients with PD (n = 26) compared with NC (n = 12), consistent with a decrease in VMAT2 level and ongoing neurodegeneration. Imaging trajectory calculations suggested that the neurodegeneration in PD occurs over approximately 3 years (CI: 27.2 to 39.5), with approximately 10.5 years (CI: 9.1 to 11.3) of degeneration in the posterior putamen before it became detectable on a VMAT2 PET scan, a further approximately 6.5 years (CI: 1.6 to 12.7) until symptom onset, and a further approximately 3 years (CI: 0.3 to 8.7) until clinical diagnosis. The authors concluded that assessment of nigrostriatal

integrity with 18F-AV-133 VMAT2 PET showed promise as a tool to monitor therapeutic response in disease-modifying trials. These findings showed that VMAT2 PET provided a sensitive measure to monitor neurodegenerative progression of PD, which has implications for PD diagnostics and subsequently clinical trial patient stratification and monitoring. Classification of Evidence = IV.

Planning of Radiotherapy for Glioblastoma

Jeske et al (1989) noted that glioma are often histologically heterogenous. As many of these tumors could not be fully removed, due to their localization, the most malignant part of the tumor may be missed and information for optimum therapeutic management is incomplete. In addition, low-grade gliomas tend to become more malignant in their development; additional surgical intervention is often not possible. Non-invasive measurement of tumor glucose metabolism with FDG-PET may be used to evaluate tumor malignancy. Malignant gliomas (astrocytoma grade III and glioblastoma) often showed increased peak metabolic rates (in comparison with normal white matter) and uncoupling of FDG transport and phosphorylation. Preliminary experiences with image-guided localized phosphorus-31 MR spectroscopy (P-31 MRS) showed a decrease of phosphodiesters in malignant gliomas, whereas the phosphomonoesters showed an increase in several cases. The phosphocreatine peak was often reduced. A more active therapy of low-grade gliomas might be indicated when signs of hyper-metabolism in FDG-PET and alteration of energy-rich phosphates or membrane-phosphates in P-31 MRS were found.

Gross et al (1998) examined the impact of PET using FDG-PET on the delineation of the target volume in three-dimensional (3D) radiation treatment planning of primary brain tumors. A total of 18 patients with histologically proven (8x biopsy, 10x sub-total resection) primary brain tumors (8 astrocytomas grade III, 1 mixed glioma grade III, and 9 glioblastomas), MRI with gadolinium-DTPA, and FDG-PET were carried out in radiation treatment position within the same week. A computer program was developed for fusion of the PET and MR images. On corresponding axial slices, FDG uptake was compared to contrast enhancement in T1-weighted and to signal hyperintensity in T2-weighted MR images. Based on PET and MRI data, 3D treatment planning was performed. All patients underwent linear accelerator (LINAC) radiotherapy. In MRI, all tumors and the surrounding edema

were visible as hyperintense lesions in the T2-weighted images; 17/18 tumors showed contrast enhancement. In FDG-PET, 16 tumors showed hyper-metabolism compared to normal white matter, whereas only 8/18 tumors showed hyper-metabolism compared to normal gray matter. White matter edema was associated with decreased FDG uptake in all patients. The area of increased FDG uptake correlated closely with contrast enhancement, only in 1 case the volume of increased FDG uptake was larger than the area of contrast enhancement. Mean tumor volumes obtained by MRI T1 + Gd, T2, and PET were 30, 106, and 10 ml, respectively. Survival was comparable to data in the literature with a 1-year survival of 39 % and a median survival of 310 days. The authors concluded that in only in a minority of patients did FDG-PET provide additional information for radiation treatment planning. This was mainly caused by the high intensity of FDG uptake in normal brain tissue. These researchers stated that PET may be of greater value in the definition of regions that should obtain a radiation dose boost.

Douglas et al (2006) stated that FDG-PET imaging for brain tumors has been shown to identify areas of active disease. Radiation dose escalation in the treatment of glioblastoma multiforme (GBM) may result in improved disease control. Based on these premises, these researchers initiated a prospective study of FDG-PET for the treatment planning of radiation dose escalation for the treatment of GBM. A total of 40 patients were enrolled. Subjects were treated with standard conformal fractionated RT with volumes defined by MRI imaging. When patients reached a dose of 45 to 50.4 Gy, they underwent FDG-PET imaging for boost target delineation, for an additional 20 Gy (2 Gy per fraction) to a total dose of 79.4 Gy (n = 30). The estimated 1-year and 2-year OS for the entire group was 70 % and 17 %, respectively, with a median OS of 70 weeks. The estimated 1-year and 2-year PFS was 18 % and 3 %, respectively, with a median of 24 weeks. No significant improvements in OS or PFS were observed for the study group in comparison to institutional historical controls. The authors concluded that radiation dose escalation to 79.4 Gy based on FDG-PET imaging showed no improvement in OS or PFS. These investigators stated that this study established the feasibility of integrating PET metabolic imaging into RT treatment planning.

Stelzer et al (2008) noted that GBM continues to be a difficult therapeutic challenge. These investigators examined if improved survival and tumor control could be achieved with modern delivery of fast neutron radiation using 3D treatment planning; 10 patients were enrolled. Eligibility criteria included pathologic diagnosis of GBM, age of 18 years or older, and KPS of 60 or higher. Subjects underwent MRI and FDG-PET as part of initial 3D treatment planning. Sequential targets were treated with non-coplanar fields to a total dose of 18 Gy in 16 fractions over 4 weeks. Median and 1-year OS were 55 weeks and 60 %, respectively; 1 patient remained alive at last follow-up 255 weeks after diagnosis. Median PFS was 16 weeks, and all patients had tumor progression by 39 weeks. Treatment was clinically well-tolerated; however, evidence of mild-to-moderate gliosis and microvascular sclerosis consistent with radiation injury was observed at autopsy in specimens taken from regions of contralateral brain that received approximately 6 to 10 Gy. Fast neutron radiation using modern imaging, treatment planning, and beam delivery was feasible to a total dose of 18 Gy; however, tumor control probability was poor in comparison to that predicted from a dose-response model based on older studies. The authors concluded that steep dose-response curves for both tumor control and neurotoxicity continued to present a challenge to establishing a therapeutic window for fast neutron radiation in GBM, even with modern techniques.

Galldiks et al (2021) stated that the management of patients with glioma usually requires multi-modality treatment including surgery, RT, and systemic therapy. Accurate neuroimaging plays a central role for RT planning and follow-up after RT completion. In order to maximize the radiation dose to the tumor and to minimize toxic effects on the surrounding brain parenchyma, reliable identification of tumor extent and target volume delineation is crucial. The use of PET for RT planning and monitoring in gliomas has gained considerable interest over the last several years; however, Class I data were not yet available. In addition, PET has been used after RT for response assessment and to distinguish tumor progression from pseudo-progression or radiation necrosis. The authors concluded that in order to improve existing treatment paradigms and to develop novel approaches for the personalization of RT for gliomas, biological information regarding inter- and intra-individual glioma heterogeneity available via metabolic imaging could prove to be crucial. The biological and imaging characterization of individual gliomas may

potentially enable personalized parametrization of mathematical models for tumor control and normal tissue complication probabilities, thereby evolving, evaluating, and bench-marking these models with the objective of implementation into the treatment planning process for biology-dependent dose painting.

UpToDate reviews on “Initial treatment and prognosis of IDH-wildtype glioblastoma in adults” (Batchelor, 2024), “Management of glioblastoma in older adults” (Batchelor and Shih, 2024), and “Radiation therapy for high-grade gliomas” (Shih, 2024) do not mention FDG-PET as a management option.

Furthermore, National Comprehensive Cancer Network’s clinical practice guideline on “Central Nervous System Cancers” (Version 2.2024) provided the following information:

Brain FDG-PET/CT scanning: Assess metabolism within tumor and normal tissue by using radiolabeled tracers:

- May be useful in differentiating tumor from radiation necrosis but has some limitations; may also correlate with tumor grade or provide the optimal area for biopsy
- Limitations: Accuracy of interpretations

SCINTIX Biology-Guided Radiotherapy for the Treatment of Liver Metastases

Liu et al (2024) noted that SCINTIX biology-guided RT (BgRT) is a novel radiotherapy modality using FDG activity on PET imaging performed in real-time to guide RT beamlets. Two PET metrics, activity concentration (AC) and normalized target signal (NTS), need to meet specific thresholds for a tumor to be considered amenable to SCINTIX therapy. In a feasibility study, these researchers examined various characteristics of liver metastases (size, volume, ACmean, ACmax) that meet these thresholds to generate a SCINTIX treatment plan. PET/CT image datasets from 10 patients with liver metastases were acquired on a diagnostic PET/CT system. The PET/CT images of the liver lesions were collected on the MIRADA Medical Imaging Software, and the PET was imported on the RefleXion Treatment Planning System (TPS). In the RefleXion TPS, a simulation tool was used to convert the diagnostic PET images into simulated SCINTIX planning

PET images accounting for differences in system sensitivity, acquisition time, reconstruction method and geometry between the diagnostic PET/CT system and the PET-linac. The gross tumor volumes (GTVs) for the liver metastases were contoured using MIRADA Medical Imaging System. Plan tumor volumes (PTVs) and Biology-Tracking Zones (BTZs) which encompass the full range of motion of the GTVs plus a margin were generated by 5- and 10-mm isotropic GTV expansion, respectively. Furthermore, a background structure BTZ shell created via a 5-mm expanded shell from the BTZ. The mean AC (ACmean) was calculated for each liver metastasis GTV, PTV, BTZ and BTZ shell, and the maximum AC (ACmax) was measured for GTV and BTZ shell on the diagnostic PET images in MIRADA. AC and NTS values were also measured on the simulated RefleXion planning PET images in the RefleXion TPS. The GTV volume, GTV superior-inferior length (SIL), and the PTV's distance to nearest organ at risk (OAR) were recorded in MIRADA for all cases. A total of 5 patients with 8 measurable lesions passed the RefleXion TPS AC and NTS metrics upon conversion to a simulated RefleXion planning PET and were included in the final analysis. The ACmean for GTVs, PTVs, BTZs and BTZ shells was 19.2 +/- 5.8 kBq/ml, 15.3 +/- 4.7 kBq/ml, 12.7 +/- 3.8 kBq/ml, and 9.5 +/- 3.9 kBq/ml, respectively. The ACmax for the GTVs and BTZ shells was 28.4 +/- 7.9.7 kBq/ml, and 21.1 +/- 6.3 kBq/ml, respectively. The GTV volume, GTV SIL and distance from PTV to nearest OAR was 11.2 +/- 8.5 cc, 2.6 +/- 1.0 cm, and 5.1 +/- 2.4 cm, respectively. The TPS AC and NTS were 12.3 +/- 3.4 kBq/ml and 7.8 +/- 4.1, respectively. The authors concluded that this study characterized liver metastases tumor volumes, lengths, ACmax, and ACmean on diagnostic PET/CT scans for cases that met the AC and NTS thresholds for SCINTIX BgRT planning on the RefleXion TPS. These investigators stated that these findings may guide patient selection for future studies on SCINTIX treatment for patients with liver metastases.

Surucu et al (2024) stated that BgRT is a novel RT delivery technique that employs the tumor itself to guide dynamic delivery of treatment dose to the tumor. The RefleXion X1 system is the 1st RT system developed to deliver SCINTIX BgRT. The X1 is characterized by its split arc design, employing two 90-deg PET arcs to guide therapeutic radiation beams in real-time, currently cleared by FDA to treat bone and lung tumors. These investigators examined the capabilities of the SCINTIX RTdelivery system by assessing its sensitivity to changes in

PET contrast, its adaptability in the context of patient motion, and its performance across a spectrum of prescription doses. In a series of experimental scenarios, both static and dynamic, were designed to evaluate the SCINTIX BgRT system's performance, including an end-to-end test. These experiments entailed a range of factors, including changes in PET contrast, motion, and prescription doses. Measurements were carried out using a custom-made ArcCHECK insert, which included a 2.2-cm spherical target and a c-shape structure that can be filled with a PET tracer with varying concentrations. Sinusoidal and cosine4 motion patterns, simulating patient breathing, was used to test the SCINTIX system's ability to deliver BgRT during motion-induced challenges. Each experiment was evaluated against specific metrics, including AC, NTS, and BTZ bounded dose-volume histogram (bDVH) pass rates. The accuracy of the delivered BgRT doses on ArcCHECK and EBT-XD film were evaluated using gamma 3%/2 mm and 3%/3 mm analysis. In static scenarios, the X1 system consistently showed precision and robustness in SCINTIX dose delivery. The end-to-end delivery to the spherical target yielded good results, with AC and NTS values surpassing the critical thresholds of 5 kBq/ml and 2, respectively. In addition, bDVH analysis consistently confirmed 100 % pass rates. These results were re-affirmed in scenarios involving changes in PET contrast, emphasizing the system's ability to adapt to varying PET avidities. Gamma analysis with 3%/2 mm (10 % dose threshold) criteria consistently achieved pass rates of greater than 91.5 % for the static tests. In dynamic SCINTIX delivery scenarios, the X1 system exhibited adaptability under conditions of motion. Sinusoidal and cosine4 motion patterns resulted in 3%/3 mm gamma pass rates of greater than 87 %. Moreover, the comparison with gated stereotactic body radiotherapy (SBRT) delivery on a conventional c-arm Linac resulted in 93.9 % gamma pass rates and used as comparison to examine the inter-play effect. The 1-cm step shift tests showed low overall gamma pass rates of 60.3 % in ArcCHECK measurements, while the doses in the PTV agreed with the plan with 99.9 % for 3%/3 mm measured with film. The authors concluded that the comprehensive evaluation of the X1 RT delivery system for SCINTIX BgRT showed good agreement for the static tests. The system consistently achieved critical metrics and delivered the BgRT doses per plan. The motion tests demonstrated its ability to co-localize the dose where the PET signal was and delivered acceptable BgRT dose distributions. This study did not provide any patient outcomes data.

Oderinde et al (2024) noted that BgRT is a novel technology that uses PET data to direct RT delivery in real-time. BgRT enables the precise delivery of radiation doses based on the PET signals emanating from PET-avid tumors on the fly. In this way, BgRT uniquely utilizes radio-tracer uptake as a biological beacon for controlling and adjusting dose delivery in real-time to account for target motion. These researchers demonstrated using real-time PET for BgRT delivery on the RefleXion X1 RT machine. The X1 RT machine is a rotating ring-gantry RT system that generates a nominal 6MV photon beam, PET, and CT components. The system uses emitted photons from PET-avid targets to deliver effective radiation beamlets or pulses to the tumor in real-time. These researchers demonstrated a real-time PET BgRT delivery experiment under 3 scenarios. These scenarios included BgRT delivering to (S1) a static target in a homogeneous and heterogeneous environment, (S2) a static target with a hot avoidance structure and partial PET-avid target, and (S3) a moving target. The 1st step was to create SBRT and BgRT plans (off-line PET data supported) using RefleXion's custom-built TPS. In addition, to create a BgRT plan using PET-guided delivery, the targets were filled with 18F-FDG, which represents a tumor/target, that is, PET-avid. The background materials were created in the insert with homogeneous water medium (for S1) and heterogeneous water with styrofoam mesh medium. A heterogeneous background medium simulated soft tissue surrounding the tumor. The treatment plan was then delivered to the experimental setups using a pre-commercial version of the X1 machine. As a final step, the dosimetric accuracy for S1 and S2 was assessed using the ArcCheck analysis tool-the gamma criteria of 3%/3 mm. For S3, the delivery dose was quantified using EBT-XD radio-chromic film. The accuracy criteria were based on coverage, where 100 % of the clinical target volume (CTV) receives at least 97 % of the prescription dose, and the maximum dose in the CTV was 130 % or less of the maximum planned dose (97 % ≤ CTV ≤ 130 %). For the S1, both SBRT and BgRT deliveries had gamma pass rates greater than 95 % (SBRT range: 96.9 % to 100 %, BgRT range: 95.2 % to 98.9 %), while in S2, the gamma pass rate was 98 % for SBRT and between 95.2 % and 98.9 % for BgRT plan delivering. For S3, both SBRT and BgRT motion deliveries met CTV dose coverage requirements, with BgRT plans delivering a very high dose to the target. The CTV dose ranges were (a) SBRT:100.4 % to 120.4 %, and (b) BgRT: 121.3 % to 139.9 %. The authors concluded that this phantom-based study demonstrated that PET signals from PET-avid tumors can be utilized to direct real-time

dose delivery to the tumor accurately, which is comparable to the dosimetric accuracy of SBRT. Furthermore, BgRT delivered a PET-signal controlled dose to the moving target, equivalent to the dose distribution to the static target. These researchers stated that a future study will compare the performance of BgRT with conventional image-guided RT.

Surgical Planning of Perivascular Epithelioid Cell Tumors (PEComas)

Perivascular epithelioid cell tumors (PEComas) are a rare variety of mesenchymal tumors composed of distinctive cells. While there is some evidence regarding the use of PET scan for diagnosis/staging of this tumor; there is very limited evidence regarding its use in surgical planning.

Handa et al (2016) noted that PEComa encompasses a group of mesenchymal tumors composed of histologically and immunohistochemically distinctive PECs. A subset of PEComa that typically arises from the falciform ligament and/or ligamentum teres is termed clear cell myo-melanocytic tumor of the falciform ligament/ligamentum teres. To-date, its imaging findings have not been described. These investigators reported the 1st radiological description of a pathologically confirmed tumor. The patient was a 5-year-old girl with a palpable abdominal mass. US, CT, MR and FDG-PET showed a mid-line, well-defined, solid anterior abdominal wall tumor below the rectus abdominis and contiguous with the umbilicus that was hyper-vascular and FDG avid. Awareness of these imaging findings facilitated the diagnosis of this distinctive tumor.

Tripathy et al (2020) stated that PEComas show a focal association with blood vessel walls and usually express melanocytic and smooth muscle markers. These investigators presented a case of 38-year-old man, who was diagnosed with PEComa of the pelvic cavity, and underwent serial 18F-FDG PET/CT scans for the assessment of response to the chemotherapeutic combination of irinotecan and temozolomide.

Zhang et al (2022) reported on the case of a 10-year-old girl with a right renal mass who underwent 18 F-FDG PET/CT and 68 Ga-FAPI-04 PET/MRI for pre-surgical assessment of tumor invasion and malignant potential. The mass showed low 18 F-FDG uptake and intense 68 Ga-

FAPI-04 uptake. Nephrectomy was carried out, and the histopathologic diagnosis was aggressive PEComa. This case showed that 68 Ga-FAPI-04 PET out-performed 18 F-FDG PET in detecting aggressive PEComa.

Li et al (2023) stated that the prevalence of hepatic PEComa is extremely low, and imaging diagnosis is very difficult. These investigators described the use of 18 F-FDG and 68 Ga-FAPI PET/CT in a 58-year-old woman with pathologically confirmed hepatic PEComa. The tumor showed intense 68 Ga-FAPI uptake but only mild 18 F-FDG activity on PET/CT studies. The authors concluded that this case suggested that 68 Ga-FAPI PET/CT might be a useful tool for evaluation of hepatic PEComa with remarkable tumor-to-liver ratios.

Yang et al (2023) noted that hepatic PEComas are rare interstitial tumors that are often mis-diagnosed as HCC due to their unique vascular enhancement patterns. These researchers presented a case of a 61-year-old man who was incidentally found to have a lesion in the left medial segment of the liver during a chest CT examination performed 4 days before his presentation for chest discomfort. Imaging showed solid components with density similar to that of normal liver tissue and areas of low-density adipose tissue within the lesion. The solid components exhibited increased uptake of F-18FDG on PET/CT; MRI showed areas with unevenly high signal intensity in both T1-weighted imaging (T1WI) in-phase and T2-weighted imaging (T2WI) sequences, while T2WI in the opposite phase displayed areas with unevenly low signal intensity, indicating the presence of fatty components. Contrast-enhanced T1WI displayed a "fast in and fast out" enhancement pattern. These distinct imaging features contributed to the diagnosis of hepatic PEComas and distinguish it from HCC.

11C-Choline PET/CT for Hyperparathyroidism

Saha et al (2023) noted that focused parathyroidectomy in primary hyperparathyroidism is possible with accurate pre-operative localization. A growing body of data exists regarding the role of radio-labeled C11 choline positron emission tomography/computed tomography (PET/CT). In cases of non-localized disease, it may be a useful adjunct to ultrasound (US), (123)I/(99)Tc-sestamibi (I-123 sestamibi), or 4-dimensional (4D)-CT imaging. In a retrospective, single-center study, patients who received a neck and chest limited coverage C11 choline PET/CT for evaluation of primary

hyperparathyroidism from 2017 to 2021 were reviewed. These investigators examined the sensitivity, positive predictive value (PPV), and false negative rate. They also compared these rates to the standard modalities of US, I-123 sestamibi, 4D-CT, and assessed concordance rates. These researchers identified 43 patients, of whom 33 had a positive C11 choline PET/CT finding. This cohort of patients had failed to localize on multiple standard imaging modalities. A total of 25 patients proceeded to surgery, 72 % of whom were re-operative cases; 20 patients (80 %) achieved an intra-operative cure. Analysis showed that C11 choline PET/CT achieved a sensitivity of 64 % (95 % confidence interval [CI]: 47 % to 82 %) and PPV of 72 % (95 % CI: 54 % to 90 %). There were 5/25 (20 %) false positive PET C11 choline results found to be lymph nodes, normal parathyroid, and 1 recurrent laryngeal nerve neuroma. The authors concluded that C11 choline PET/CT was a useful adjunct for parathyroid localization in a complex population of patients who have failed standard localization techniques including US, I-123 sestamibi, or 4D-CT and/or prior operations. Although routine inclusion of C11 choline PET/CT imaging may not be necessary, it may aid in pre-operative localization in the re-operative setting.

In a retrospective, single-center study, Van den Bruel et al (2023) examined how accurately pre-operative imaging localized parathyroid adenoma in superior versus inferior parathyroids. Over 6 years, a total of 104 patients with primary hyperparathyroidism underwent parathyroid surgery. Of these, 103 underwent US, 97 [99mTc]pertechnetate/MIBI SPECT/CT and 30 [18F]fluorocholine (FCH) PET/CT; 1 patient with a unilateral double adenoma was excluded from the analysis. Surgical findings with histopathologic confirmation of adenoma were used as the standard. US mis-judged 5 of 48 detected lower adenomas as upper, but 14 of 29 upper adenomas as lower (error rate 10 % versus 48 %, p = 0.0002). The corresponding error rates for [99mTc]pertechnetate/MIBI SPECT/CT were 3 % versus 55 % (p = 0.000014), and for [18F]FCH PET/CT 17 % versus 36 % (p = 0.26). The authors concluded that US, [99mTc]pertechnetate/MIBI SPECT/CT and [18F]FCH PET/CT often mistook superior parathyroid adenomas for inferior ones. These researchers stated that awareness that a significant proportion of upper parathyroid adenoma may be adjacent to the lower third of the thyroid lobe may contribute to more precise pre-operative localization of the parathyroid adenoma.

The authors stated that the retrospective design was the main limitation of this trial. In this real-life study, the analysis relied on the imaging reports retrieved from the patient's record; however, these had all been produced by board-certified imaging specialists consistently judging the relative position of the parathyroid lesion to the thyroid. Apart from mistakes, it appeared unlikely that re-reading by a central reviewer would have changed much. In a small number of patients, imaging reports were ambiguous as to the upper or lower origin of the enlarged parathyroid glands. However, in those patients, these researchers assigned the enlarged glands to the upper location, going against the trial hypothesis; therefore, strengthening these findings. A 2nd limitation was this trial's low detection rate of [99mTc]pertechnetate/MIBI SPECT/CT. Meta-analyses by Treglia and Wong found an estimated pooled sensitivity of 0.86 and 0.88, respectively, on a per-patient basis (and based on publications between 2003 and 2015). Some recent large series with patient characteristics similar to the cohorts in this study showed similarly lower sensitivities. An evolution towards series with lower mean calcium, with a higher proportion of normo-calcemic hyperparathyroidism, might contribute to lower sensitivities. Similarly, the addition of [18F]FCH PET/CT facilitated surgery in the patients in this trial with negative [99mTc]pertechnetate/MIBI SPECT/CT and might somewhat lower the [99mTc]pertechnetate/MIBI SPECT/CT detection rate in a surgical group enriched with [99mTc]pertechnetate/MIBI SPECT/CT-negative patients as compared to patient series 2003 to 2015.

Fang et al (2024) examined the usefulness of US for the localization of ectopic hyperparathyroidism and compared it with 99mTc-sestamibi (99mTc-MIBI), 4D-CT, and 11C-choline PET/CT. Of the 527 patients with surgically confirmed primary hyperparathyroidism, 79 patients with ectopic hyperparathyroidism were enrolled. The diagnostic performance of US, 99mTc-MIBI, US + MIBI, 4D-CT, and 11C-choline PET/CT was calculated, and the factors affecting the sensitivity of US and 99mTc-MIBI were analyzed. A total of 83 ectopic parathyroid lesions were found in 79 patients. The sensitivity was 75.9 %, 81.7 %, 95.1 %, 83.3 %, and 100 % for US, 99mTc-MIBI, US + MIBI, 4D-CT, and 11C-choline PET/CT, respectively. The difference in sensitivity among these different modalities did not achieve statistical significance ($p > 0.05$). The US sensitivity was significantly higher for ectopic lesions in the neck region than for those in the anterior

mediastinum/chest wall (85.9 % versus 42.1 %, p < 0.001). The 99mTc-MIBI and 4D-CT sensitivity was not significantly different between these 2 groups (84.1 % versus 94.6 %, p = 0.193 and 81.3 % versus 85.7 %, p = 1). The 11C-choline PET/CT sensitivity was 100 % in both groups. The authors concluded that US was a valuable tool for the localization of ectopic hyperparathyroidism, especially for ectopic lesions in the neck region.

Calderon et al (2024) noted that in recent years, nuclear medicine imaging methods have proven to be of paramount importance in a wide variety of diseases, especially in oncology, where they are essential for examining the extent of disease when conventional methods fall short. Moreover, nuclear imaging modalities could better characterize lesions using target agents related to specific pathways (e.g., glucose metabolism, cellular proliferation, amino acid transport, lipid metabolism, specific receptor ligands). The clinical presentation of endocrine diseases entails a broad spectrum of sign and symptoms. Furthermore, endocrine tumors show varying degrees of aggressiveness from well differentiated and indolent to highly aggressive cancers, respectively. With the use of new medicinal radio-compounds and increasingly advanced tomographic imaging technology, the application of PET/CT in the field of endocrine diseases is expanding. These researchers analyzed the primary indications of PET/CT, providing a practical approach for clinicians. They carried out a comprehensive literature search on PubMed to provide an updated overview of the available evidence regarding the use of PET/CT in endocrinology. These investigators compared different radio-pharmaceuticals and highlighted the uptake mechanism, excluded neuroendocrine carcinomas from discussion. The authors discussed the emerging opportunities of PET/CT and appropriateness of use in endocrinological diseases.

Thes researchers stated that [18F]F-choline PET/CT is now considered the imaging method of choice in patients with tertiary hyperparathyroidism (tHPT) due to its superior detection capability, also in cases of multi-glandular disease and a low rate of localization failure of parathyroid adenoma. Excellent sensitivity values of [11C]-methionine PET/CT were also reported in the tertiary HPT setting, where multiple lesions were visualized in 57.9 % of patients and ectopic lesions in 21.1 % of patients. According to Bijnens et al (2022), [18F]F-choline PET/CT has proven to be a highly accurate method and

obviates the need for extensive exploration in most cases where US and scintigraphy results do not align or both show negative results. Nonetheless, its use in this context might face limitations because of cost and accessibility factors. These findings highlighted the promising role of nuclear imaging techniques, especially [11C]-choline and [11C]-methionine PET/CT, in optimizing the precision and effectiveness of parathyroid surgery, ultimately benefiting patients via reduced invasiveness and improved treatment outcomes. Moreover, these investigators stated that further research and clinical evaluation are needed to fully harness the capabilities of these advanced imaging modalities for enhanced patient care in the field of endocrinology.

Furthermore, an UpToDate review on “Primary hyperparathyroidism: Diagnosis, differential diagnosis, and evaluation” (El-Hajj Fuleihan and Silverberg, 2024) does not mention C11 choline PET/CT as a management tool.

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